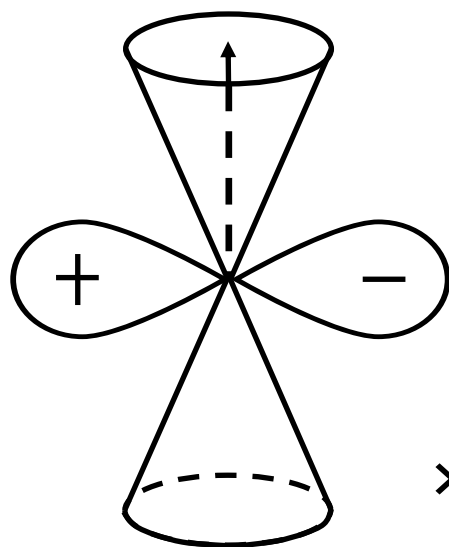


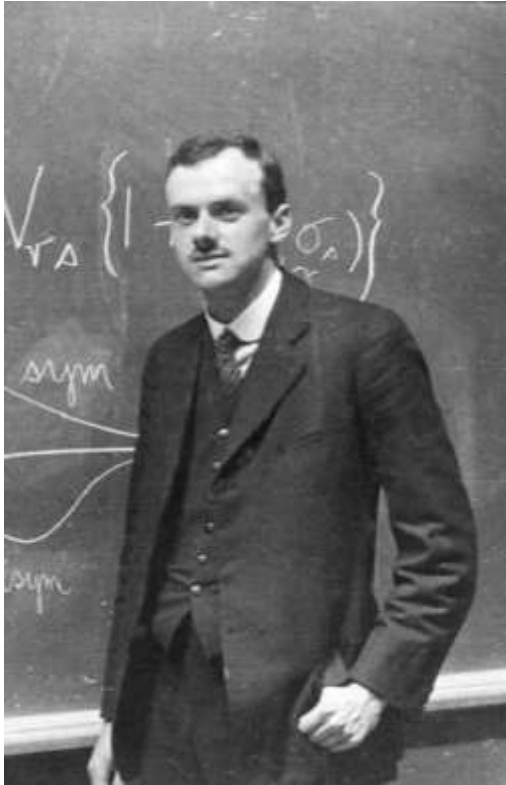
Thomas动力学与 自旋赝偶极矢量(张量)



$$\vec{\mu}_M = \frac{e}{mc} \frac{\hbar}{2} \vec{\sigma}$$

$$\vec{\mu}_E = \frac{1}{2} \frac{e}{mc} \frac{\hbar}{2} (\vec{\sigma} \times \vec{\beta}_p)$$

$$\times \vec{\beta}_p$$



Whatever you want Dirac
1902-1984

“

*It is more important to have beauty
in one's equation than to have them
fit experiment.*

”

方程式之美：Lorentz对称性(协变性)、Poincaré对称性、
规范对称性.....

方程式之美是主观追求还是客观要求？

是否需要，或者什么情况下需要追求方程式的美？

矢量Thomas动力学

区分受Thomas进动修正的物理量与不受修正的物理量必须从动力学办法出发。文献指出^[1,2]，对给定4-vector X^μ ，在任意参考系(lab系)变换到静止系 x^ν 演化服从：

$$\begin{aligned}\dot{x}^\nu|_{rest} &= L_\mu^\nu(-\vec{\beta}) \dot{X}^\mu|_{\vec{\beta}} + \dot{L}_\mu^\nu(-\vec{\beta}) X^\mu|_{X^\mu} \dots\dots \textcircled{1} \\ &= L_\mu^\nu(-\vec{\beta}) \dot{L}_\chi^\mu(\vec{\beta}) \dot{x}^\chi|_{\vec{\beta}} + \dot{L}_\mu^\nu(-\vec{\beta}) \dot{X}^\mu|_{X^\mu} = \dot{x}^\nu|_{\vec{\beta}} + \dot{x}^\nu|_{X^\mu}\end{aligned}$$

其中 $\dot{}$ 表示对固有时求导。结果中第一项为动力学项，可由BMT方程描述；第二项为Thomas进动项，其本质是两次相连帧boost，关注其空间分量Lorentz变换展开有

$$\begin{aligned}\dot{\vec{x}}|_{X^\mu} &= \left(\frac{\dot{\gamma}^2}{\gamma+1}\right)(\vec{\beta} \cdot \vec{X})\vec{\beta} + \left(\frac{\gamma^2}{\gamma+1}\right)(\vec{\beta} \cdot \dot{\vec{X}})\vec{\beta} + \left(\frac{\gamma^2}{\gamma+1}\right)(\vec{\beta} \cdot \vec{X})\dot{\vec{\beta}} - (\gamma\dot{\vec{\beta}}X_0)|_{X_0} \\ &= \frac{(\gamma+2)\gamma^4(\vec{\beta} \cdot \dot{\vec{\beta}})}{(\gamma+1)^2}(\vec{\beta} \cdot \vec{X})\vec{\beta} + \left(\frac{\gamma^2}{\gamma+1}\right)(\dot{\vec{\beta}} \cdot \vec{X})\vec{\beta} + \left(\frac{\gamma^2}{\gamma+1}\right)(\vec{\beta} \cdot \vec{X})\dot{\vec{\beta}} - \gamma^3(\vec{\beta} \cdot \dot{\vec{\beta}})\vec{\beta}X_0 - \gamma\dot{\vec{\beta}}X_0 \\ &= \gamma^3(\vec{\beta} \cdot \dot{\vec{\beta}})\vec{\beta} \left[\frac{(\gamma+2)\gamma(\vec{\beta} \cdot \vec{X})}{(\gamma+1)^2} - X_0 \right] + \left(\frac{\gamma^2}{\gamma+1}\right)(\dot{\vec{\beta}} \cdot \vec{X})\vec{\beta} + \left(\frac{\gamma^2}{\gamma+1}\right)(\vec{\beta} \cdot \vec{X})\dot{\vec{\beta}} - \gamma\dot{\vec{\beta}}X_0\end{aligned}$$

[1] Rębilas, Krzysztof. "Thomas precession and the Bargmann-Michel-Telegdi equation." Foundations of Physics 41 (2011): 1800-1809.

[2] Jackson, John David. "Classical electrodynamics." (1999): 561-565.

矢量Thomas动力学

$$\begin{aligned}
 &= \gamma^3 (\vec{\beta} \cdot \dot{\vec{\beta}}) \vec{\beta} \left[\frac{(\gamma+2)\gamma \left[\vec{\beta} \cdot \left(\vec{x} + \frac{\gamma^2}{\gamma+1} (\vec{\beta} \cdot \vec{x}) \vec{\beta} + \gamma \vec{\beta} x_0 \right) \right]}{(\gamma+1)^2} - \gamma (\vec{x}_0 + \vec{\beta} \cdot \vec{x}) \right] \\
 &+ \left(\frac{\gamma^2}{\gamma+1} \right) \left[\dot{\vec{\beta}} \cdot \left(\vec{x} + \frac{\gamma^2}{\gamma+1} (\vec{\beta} \cdot \vec{x}) \vec{\beta} + \gamma \vec{\beta} x_0 \right) \right] \vec{\beta} + \left(\frac{\gamma^2}{\gamma+1} \right) \left[\vec{\beta} \cdot \left(\vec{x} + \frac{\gamma^2}{\gamma+1} (\vec{\beta} \cdot \vec{x}) \vec{\beta} + \gamma \vec{\beta} x_0 \right) \right] \dot{\vec{\beta}} - \gamma^2 \dot{\vec{\beta}} (x_0 + \vec{\beta} \cdot \vec{x}) \\
 &= \gamma^3 (\vec{\beta} \cdot \dot{\vec{\beta}}) \vec{\beta} \left(1 - \frac{1}{(\gamma+1)^2} \right) (\vec{\beta} \cdot \vec{x}) + \gamma^3 (\vec{\beta} \cdot \dot{\vec{\beta}}) \vec{\beta} \left(1 - \frac{1}{(\gamma+1)^2} \right) (\gamma-1) (\vec{\beta} \cdot \vec{x}) + \gamma^3 (\vec{\beta} \cdot \dot{\vec{\beta}}) \vec{\beta} \left(1 - \frac{1}{(\gamma+1)^2} \right) \gamma \beta^2 x_0 \\
 &- \gamma^4 (\vec{\beta} \cdot \dot{\vec{\beta}}) \vec{\beta} (\vec{x}_0 + \vec{\beta} \cdot \vec{x}) + \frac{\gamma^2}{\gamma+1} (\dot{\vec{\beta}} \cdot \vec{x}) \vec{\beta} + \frac{\gamma^4}{(\gamma+1)^2} (\dot{\vec{\beta}} \cdot \vec{\beta}) (\vec{\beta} \cdot \vec{x}) \vec{\beta} + \frac{\gamma^3}{\gamma+1} (\dot{\vec{\beta}} \cdot \vec{\beta}) \vec{\beta} x_0 + \frac{\gamma^2}{\gamma+1} (\vec{\beta} \cdot \vec{x}) \dot{\vec{\beta}} \\
 &+ \frac{\gamma^4}{(\gamma+1)^2} \beta^2 (\vec{\beta} \cdot \vec{x}) \dot{\vec{\beta}} + \frac{\gamma^3}{\gamma+1} \beta^2 \dot{\vec{\beta}} x_0 - \gamma^2 \dot{\vec{\beta}} x_0 - \gamma^2 \dot{\vec{\beta}} (\vec{\beta} \cdot \vec{x}) \\
 &= \gamma^4 (\vec{\beta} \cdot \dot{\vec{\beta}}) \vec{\beta} \left(1 - \frac{1}{(\gamma+1)^2} \right) [(\vec{\beta} \cdot \vec{x}) + \beta^2 x_0] - \gamma^4 (\vec{\beta} \cdot \dot{\vec{\beta}}) \vec{\beta} (\vec{x}_0 + \vec{\beta} \cdot \vec{x}) + \frac{\gamma^2}{\gamma+1} (\dot{\vec{\beta}} \cdot \vec{x}) \vec{\beta} + \frac{\gamma^2}{\gamma+1} (\vec{\beta} \cdot \vec{x}) \dot{\vec{\beta}} \\
 &+ \left[\frac{\gamma^2}{\gamma+1} (\gamma-1) - \gamma^2 \right] (\vec{\beta} \cdot \vec{x}) \dot{\vec{\beta}} + (\gamma^2 - \gamma) \dot{\vec{\beta}} x_0 - \gamma^2 \dot{\vec{\beta}} x_0 + \frac{\gamma^3}{\gamma+1} (\dot{\vec{\beta}} \cdot \vec{\beta}) \vec{\beta} x_0 + \frac{\gamma^4}{(\gamma+1)^2} (\dot{\vec{\beta}} \cdot \vec{\beta}) (\vec{\beta} \cdot \vec{x}) \vec{\beta} \\
 &= \left[\gamma^4 \left(1 - \frac{1}{(\gamma+1)^2} \right) + \frac{\gamma^4}{(\gamma+1)^2} - \gamma^4 \right] (\vec{\beta} \cdot \dot{\vec{\beta}}) (\vec{\beta} \cdot \vec{x}) \vec{\beta} - \gamma^4 (\vec{\beta} \cdot \dot{\vec{\beta}}) \vec{\beta} \left(\frac{1}{\gamma^2} + \frac{\beta^2}{(\gamma+1)^2} \right) x_0 \\
 &+ \frac{\gamma^2}{\gamma+1} (\dot{\vec{\beta}} \cdot \vec{x}) \vec{\beta} - \frac{\gamma^2}{\gamma+1} (\vec{\beta} \cdot \vec{x}) \dot{\vec{\beta}} - \gamma \dot{\vec{\beta}} x_0 + \frac{\gamma^3}{\gamma+1} (\dot{\vec{\beta}} \cdot \vec{\beta}) \vec{\beta} x_0 \\
 &\underline{\underline{\dot{\vec{\beta}}' = \frac{d}{dt} \vec{\beta} = \frac{1}{\gamma} \frac{d}{d\tau} \vec{\beta} = \frac{1}{\gamma} \dot{\vec{\beta}}}} \quad \gamma \frac{\gamma^2}{\gamma+1} \left[\vec{x} \times (\vec{\beta} \times \dot{\vec{\beta}}') \right] - \gamma \dot{\vec{\beta}} x_0 - \frac{\gamma^3}{\gamma+1} (\dot{\vec{\beta}} \cdot \vec{\beta}) \vec{\beta} x_0 \quad \dots\dots ②
 \end{aligned}$$

blue: $(\vec{\beta} \cdot \dot{\vec{\beta}}) (\vec{\beta} \cdot \vec{x}) \vec{\beta}$ 项

red: $(\dot{\vec{\beta}} \cdot \vec{\beta}) \vec{\beta} x_0$ 项

green: Thomas进动项

备注

4

矢量Thomas动力学

以上推导表明Thomas动力学修正对任意物理量演化是普适的，但由①式可知满足

$$X^\mu \propto U^\mu \dots\dots \textcircled{3}$$

的4-vector比较特殊，有

$$\dot{x}^\nu|_{X^\mu} = \left(L_\mu^\nu(-\vec{\beta}) X^\mu \right) \Big|_{X^\mu} = \dot{L}_\mu^\nu(-\vec{\beta}) X^\mu \Big|_{X^\mu} = \dot{L}_\mu^\nu(-\vec{\beta}) X^\mu \Big|_{U^\mu} \stackrel{\vec{\beta} \neq 0}{=} \dot{L}_\mu^\nu(-\vec{\beta}) X^\mu \Big|_{\vec{\beta}} = 0$$

故全微分中不存在②式，即不受Thomas动力学修正；对②式的结果代入

$$\dot{\vec{\beta}} = 0$$

可得到相同结论。常见的满足这一关系的4-vector有

$$J^\mu = (c\rho, \vec{J}) = (c\gamma\rho_0, \gamma\rho_0 \vec{u}) = \rho_0 U^\mu$$

$$p^\mu = \left(\frac{W}{c}, \vec{p} \right) = (c\gamma m_0, \gamma m_0 \vec{u}) = m_0 U^\mu$$

从Thomas运动学角度也能得到Thomas进动不修正动量/动量速度：所有4-vector都满

足③式的 (即仅轨道运动)的基本粒子是各向同性的，粒子静止参考系的自转不具备观测效

应，Thomas进动消失*。值得注意的是四维电流密度也不受进动影响，说明利用Biot-

Savart定律，正比于电流密度的源激发场场强也不受进动影响，Thomas进动角速度不激

发磁场。 Thomas进动对Mag SOC的影响源于对自旋的影响。

矢量Thomas动力学

从②式结果亦可可知若4-vector满足

$$X^\mu \propto (ct, \vec{x})^\mu \dots\dots \textcircled{4}$$

则按照Thomas动力学有

$$\dot{x}^\nu|_{X^\mu} = -\gamma\dot{\beta}x_0$$

不满足③、④式的一般4-vector如构造自旋4-vector S^μ 、偶极矢量(张量)等, 虽可能包含速度相关项, 但相关速度与Lorentz变换速度 β 无关, 故满足②式结果包含Thomas进动动力学项的完整表达。

总结一下Thomas运动学与Thomas动力学的不同视角和研究办法。

Thomas动力学办法

- Thomas动力学是变系效应, 认为协变物理量的演化只有在静止系中才有(比较)意义。
- 只有不满足③、④式的4-vector在静止系的演化才受Thomas动力学修正。
- 利用动力学办法研究对de Broglie波的修正通过将 $(ct, \vec{x})$ Fourier变换 $(\omega/c, \vec{k})$, 按照de Broglie假设化为 $\frac{1}{\hbar}(W/c, \vec{p}) = \frac{1}{\hbar}p^\mu \propto U^\mu$, 故不受Thomas动力学修正。对Doppler效应的处理就是基于这一观点*, 不受修正的结果也和Doppler效应Thomas运动学结果一致。

Thomas运动学办法

- ▶ Thomas运动学是基于连续帧的非变系效应，认为曲线运动的对象经历“多余”的运动模式。
- ▶ 任何曲线运动的对象都可观测到Thomas运动，当对象有静止自旋时表现为Thomas进动^[1]。利用Wigner转动、高阶Wigner转动导出的Thomas进动、高阶Thomas进动都属于Thomas运动学的研究范畴。
- ▶ 对de Broglie波的修正通过de Broglie波角速度与曲线运动角速度关联，而 ω_T 是运动系自转角速度，与曲线运动角速度无关，故无法修正 ω (或 $\vec{k}$)，与动力学结果一致。

Thomas动力学与Thomas运动学是统一、互补的。例如它们都不会影响动量速度、群速度、波速度，也不会修正(延拓)复色关系。对于类氢原子的自旋-轨道耦合，从核静止系(lab系)观测电子运动在运动学上存在Thomas进动^[1]，而若不变换到电子静止系则自旋4-vector演化不受Thomas动力学影响，这两个方面是理解互补统一的关键。

[1] Rębilas, Krzysztof. "Subtleties of the Thomas precession." European Journal of Physics 36.4 (2015): 045007.

二阶反对称张量Thomas动力学

从②式结果亦可知若4-vector满足

$$X^\mu \propto (ct, \vec{x})^\mu \dots\dots ④$$

则按照Thomas动力学有

$$\dot{x}^\nu|_{X^\mu} = -\gamma\dot{\vec{\beta}}x_0$$

不满足③、④式的一般4-vector如构造自旋4-vector $\mathbf{S}^\mu$ 、偶极矢量(张量)等, 虽可能包含速度相关项, 但相关速度与Lorentz变换速度 $\vec{\beta}$ 无关, 故满足②式结果包含Thomas进动动力学项的完整表达。

考虑二阶张量 $X^{\mu\nu}$ 演化的动力学规律:

$$\begin{aligned}\dot{x}^{\mu\nu}|_{rest} &= L_\lambda^\mu(-\vec{\beta})L_\sigma^\nu(-\vec{\beta})\dot{X}^{\lambda\sigma}\Big|_{\vec{\beta}} + \left[L_\lambda^\mu(-\vec{\beta})L_\sigma^\nu(-\vec{\beta}) \right] X^{\lambda\sigma}\Big|_{X^{\lambda\sigma}} \\ &= L_\lambda^\mu(-\vec{\beta})L_\sigma^\nu(-\vec{\beta}) \left[L_\chi^\sigma(\vec{\beta})L_\eta^\lambda(\vec{\beta})x^{\chi\eta} \right] \Big|_{\vec{\beta}} + \left[L_\lambda^\mu(-\vec{\beta})L_\sigma^\nu(-\vec{\beta}) X^{\lambda\sigma} \right] \Big|_{X^{\lambda\sigma}} \\ &= L_\lambda^\mu(-\vec{\beta})L_\sigma^\nu(-\vec{\beta})L_\chi^\sigma(\vec{\beta})L_\eta^\lambda(\vec{\beta})\dot{x}^{\chi\eta}\Big|_{\vec{\beta}} + \dot{x}^{\mu\nu}\Big|_{X^{\lambda\sigma}} \\ &= \dot{x}^{\chi\eta}\Big|_{\vec{\beta}} + \dot{x}^{\mu\nu}\Big|_{X^{\lambda\sigma}}\end{aligned}$$

二阶反对称张量Thomas动力学

同理，结果中第一项为动力学项，可由动力学方程描述；第二项为Thomas进动项，其本质是两次相连帧boost，设被研究二阶张量满足两个矢量组合的协变形式

$$x^{\mu\nu}|_{rest} = \begin{pmatrix} 0 & -m^1 & -m^2 & -m^3 \\ m^1 & 0 & -n^3 & n^2 \\ m^2 & n^3 & 0 & -n^1 \\ m^3 & -n^2 & n^1 & 0 \end{pmatrix}$$

则其矢量分量Lorentz变换展开有

$$\begin{aligned} \dot{m}|_{X^{\lambda\sigma}} &= \dot{\gamma}(\vec{M} + \vec{\beta} \times \vec{N})|_{X^{\lambda\sigma}} + \gamma(\vec{M} + \dot{\vec{\beta}} \times \vec{N})|_{X^{\lambda\sigma}} \\ &\quad - \left(\frac{\dot{\gamma}^2}{\gamma+1} \right) \vec{\beta}(\vec{\beta} \cdot \vec{M})|_{X^{\lambda\sigma}} - \frac{\gamma^2}{\gamma+1} \dot{\vec{\beta}}(\vec{\beta} \cdot \vec{M})|_{X^{\lambda\sigma}} - \frac{\gamma^2}{\gamma+1} \vec{\beta}(\vec{\beta} \cdot \dot{\vec{M}})|_{X^{\lambda\sigma}} \\ &= \gamma^3(\vec{\beta} \cdot \dot{\vec{\beta}})(\vec{M} + \vec{\beta} \times \vec{N})|_{X^{\lambda\sigma}} + \gamma(\dot{\vec{\beta}} \times \vec{N})|_{X^{\lambda\sigma}} \\ &\quad - \left[\frac{(\gamma+2)\gamma^4(\vec{\beta} \cdot \dot{\vec{\beta}})}{(\gamma+1)^2}(\vec{\beta} \cdot \vec{M})\vec{\beta} + \left(\frac{\gamma^2}{\gamma+1} \right)(\dot{\vec{\beta}} \cdot \vec{M})\vec{\beta} + \left(\frac{\gamma^2}{\gamma+1} \right)(\vec{\beta} \cdot \vec{M})\dot{\vec{\beta}} \right] \end{aligned}$$

二阶反对称张量Thomas动力学

$$\begin{aligned}
 & \gamma^3 \left(\vec{\beta} \cdot \dot{\vec{\beta}} \right) \left(\vec{M} + \vec{\beta} \times \vec{N} \right) \Big|_{X^{\lambda\sigma}} + \gamma \left(\dot{\vec{\beta}} \times \vec{N} \right) \Big|_{X^{\lambda\sigma}} \\
 & - \left[\frac{(\gamma+2)\gamma^4 (\vec{\beta} \cdot \dot{\vec{\beta}})}{(\gamma+1)^2} (\vec{\beta} \cdot \vec{M}) \vec{\beta} + \left(\frac{\gamma^2}{\gamma+1} \right) (\dot{\vec{\beta}} \cdot \vec{M}) \vec{\beta} + \left(\frac{\gamma^2}{\gamma+1} \right) (\vec{\beta} \cdot \vec{M}) \dot{\vec{\beta}} \right] \\
 & = \gamma^3 \left(\vec{\beta} \cdot \dot{\vec{\beta}} \right) \left[\gamma (\vec{m} - \vec{\beta} \times \vec{n}) - \frac{\gamma^2}{\gamma+1} (\vec{\beta} (\vec{\beta} \cdot \vec{m})) + \vec{\beta} \times \left(\gamma (\vec{n} + \vec{\beta} \times \vec{m}) - \frac{\gamma^2}{\gamma+1} (\vec{\beta} (\vec{\beta} \cdot \vec{n})) \right) \right] \Big|_{X^{\lambda\sigma}} \\
 & + \gamma \left[\dot{\vec{\beta}} \times \left(\gamma (\vec{n} + \vec{\beta} \times \vec{m}) - \frac{\gamma^2}{\gamma+1} (\vec{\beta} (\vec{\beta} \cdot \vec{n})) \right) \right] \Big|_{X^{\lambda\sigma}} \\
 & - \frac{(\gamma+2)\gamma^4 (\vec{\beta} \cdot \dot{\vec{\beta}})}{(\gamma+1)^2} \left[\vec{\beta} \cdot \left(\gamma (\vec{m} - \vec{\beta} \times \vec{n}) - \frac{\gamma^2}{\gamma+1} (\vec{\beta} (\vec{\beta} \cdot \vec{m})) \right) \right] \vec{\beta} \\
 & - \left(\frac{\gamma^2}{\gamma+1} \right) \left[\dot{\vec{\beta}} \cdot \left(\gamma (\vec{m} - \vec{\beta} \times \vec{n}) - \frac{\gamma^2}{\gamma+1} (\vec{\beta} (\vec{\beta} \cdot \vec{m})) \right) \right] \vec{\beta} - \left(\frac{\gamma^2}{\gamma+1} \right) \left[\vec{\beta} \cdot \left(\gamma (\vec{m} - \vec{\beta} \times \vec{n}) - \frac{\gamma^2}{\gamma+1} (\vec{\beta} (\vec{\beta} \cdot \vec{m})) \right) \right] \dot{\vec{\beta}} \\
 & = \gamma^4 \left(\vec{\beta} \cdot \dot{\vec{\beta}} \right) (\vec{m} - \vec{\beta} \times \vec{n}) - \frac{\gamma^5 (\vec{\beta} \cdot \dot{\vec{\beta}})}{\gamma+1} \vec{\beta} (\vec{\beta} \cdot \vec{m}) + \gamma^4 \left(\vec{\beta} \cdot \dot{\vec{\beta}} \right) \vec{\beta} \times (\vec{n} + \vec{\beta} \times \vec{m}) - \frac{\gamma^5 (\vec{\beta} \cdot \dot{\vec{\beta}})}{\gamma+1} \vec{\beta} \times [\vec{\beta} (\vec{\beta} \cdot \vec{n})] \\
 & \gamma^2 \dot{\vec{\beta}} \times (\vec{n} + \vec{\beta} \times \vec{m}) - \frac{\gamma^3}{\gamma+1} \dot{\vec{\beta}} \times [\vec{\beta} (\vec{\beta} \cdot \vec{n})]
 \end{aligned}$$

二阶反对称张量Thomas动力学

$$\begin{aligned}
 & -\frac{(\gamma+2)\gamma^5(\vec{\beta} \cdot \dot{\vec{\beta}})}{(\gamma+1)^2} \vec{\beta} \cdot (\vec{m} - \vec{\beta} \times \vec{n}) \vec{\beta} + \frac{(\gamma+2)\gamma^6(\vec{\beta} \cdot \dot{\vec{\beta}})}{(\gamma+1)^3} \vec{\beta} \cdot [\vec{\beta}(\vec{\beta} \cdot \vec{m})] \vec{\beta} \\
 & -\frac{\gamma^3}{\gamma+1} \dot{\vec{\beta}} \cdot (\vec{m} - \vec{\beta} \times \vec{n}) \vec{\beta} + \frac{\gamma^4}{(\gamma+1)^2} \dot{\vec{\beta}} \cdot [\vec{\beta}(\vec{\beta} \cdot \vec{m})] \vec{\beta} - \frac{\gamma^3}{\gamma+1} \vec{\beta} \cdot (\vec{m} - \vec{\beta} \times \vec{n}) \dot{\vec{\beta}} + \frac{\gamma^4}{(\gamma+1)^2} \vec{\beta} \cdot [\vec{\beta}(\vec{\beta} \cdot \vec{m})] \dot{\vec{\beta}} \\
 & = \gamma^4(\vec{\beta} \cdot \dot{\vec{\beta}}) [(\vec{m} - \vec{\beta} \times \vec{n}) + \vec{\beta} \times (\vec{n} + \vec{\beta} \times \vec{m})] - \frac{\gamma^5(\vec{\beta} \cdot \dot{\vec{\beta}})}{\gamma+1} [\vec{\beta}(\vec{\beta} \cdot \vec{m}) + \vec{\beta} \times [\vec{\beta}(\vec{\beta} \cdot \vec{n})]] \\
 & + \gamma^2 [\dot{\vec{\beta}} \times (\vec{n} + \vec{\beta} \times \vec{m})] - \frac{\gamma^3}{\gamma+1} [\dot{\vec{\beta}} \times [\vec{\beta}(\vec{\beta} \cdot \vec{n})]] \\
 & - \frac{(\gamma+2)\gamma^5(\vec{\beta} \cdot \dot{\vec{\beta}})}{(\gamma+1)^2} \left[\vec{\beta} \cdot (\vec{m} - \vec{\beta} \times \vec{n}) \vec{\beta} - \frac{\gamma}{\gamma+1} \vec{\beta} \cdot [\vec{\beta}(\vec{\beta} \cdot \vec{m})] \vec{\beta} \right] \\
 & - \frac{\gamma^3}{\gamma+1} \left[\dot{\vec{\beta}} \cdot (\vec{m} - \vec{\beta} \times \vec{n}) \vec{\beta} + \vec{\beta} \cdot (\vec{m} - \vec{\beta} \times \vec{n}) \dot{\vec{\beta}} \right] + \frac{\gamma^4}{(\gamma+1)^2} \left[\dot{\vec{\beta}} \cdot [\vec{\beta}(\vec{\beta} \cdot \vec{m})] \vec{\beta} + \vec{\beta} \cdot [\vec{\beta}(\vec{\beta} \cdot \vec{m})] \dot{\vec{\beta}} \right] \\
 & = \gamma^4(\vec{\beta} \cdot \dot{\vec{\beta}}) [\vec{m} + \vec{\beta} \times (\vec{\beta} \times \vec{m})] - \frac{\gamma^5(\vec{\beta} \cdot \dot{\vec{\beta}})}{\gamma+1} [\vec{\beta}(\vec{\beta} \cdot \vec{m}) + \vec{\beta} \times [\vec{\beta}(\vec{\beta} \cdot \vec{n})]] \\
 & + \gamma^2 [\dot{\vec{\beta}} \times \vec{n} + \vec{\beta} \times (\vec{\beta} \times \vec{m})] - \frac{\gamma^3}{\gamma+1} [\dot{\vec{\beta}} \times [\vec{\beta}(\vec{\beta} \cdot \vec{n})]]
 \end{aligned}$$

二阶反对称张量Thomas动力学

$$\begin{aligned}
 & -\frac{(\gamma+2)\gamma^5(\vec{\beta} \cdot \dot{\vec{\beta}})}{(\gamma+1)^2} \left[(\vec{\beta} \cdot \vec{m})\vec{\beta} - \vec{\beta} \cdot (\vec{\beta} \times \vec{n})\vec{\beta} - \frac{\gamma}{\gamma+1}\beta^2(\vec{\beta} \cdot \vec{m})\vec{\beta} \right] \\
 & -\frac{\gamma^3}{\gamma+1} \left[\vec{\beta}(\dot{\vec{\beta}} \cdot \vec{m}) - \dot{\vec{\beta}} \cdot (\vec{\beta} \times \vec{n})\vec{\beta} + \dot{\vec{\beta}}(\vec{\beta} \cdot \vec{m}) - \vec{\beta} \cdot (\vec{\beta} \times \vec{n})\dot{\vec{\beta}} \right] + \frac{\gamma^4}{(\gamma+1)^2} \left[(\vec{\beta} \cdot \vec{m})(\dot{\vec{\beta}} \cdot \vec{\beta})\vec{\beta} + \beta^2(\vec{\beta} \cdot \vec{m})\dot{\vec{\beta}} \right] \\
 & = \gamma^4(\vec{\beta} \cdot \dot{\vec{\beta}}) \left[\frac{1}{\gamma^2}\vec{m} + \vec{\beta}(\vec{\beta} \cdot \vec{m}) \right] - \frac{\gamma^5(\vec{\beta} \cdot \dot{\vec{\beta}})}{\gamma+1} [\vec{\beta}(\vec{\beta} \cdot \vec{m})] \\
 & + \gamma^2 \left[\dot{\vec{\beta}} \times \vec{n} + \vec{\beta}(\dot{\vec{\beta}} \cdot \vec{m}) - \vec{m}(\dot{\vec{\beta}} \cdot \vec{\beta}) \right] - \frac{\gamma^3}{\gamma+1} [(\vec{\beta} \cdot \vec{n})(\dot{\vec{\beta}} \times \vec{\beta})] \\
 & -\frac{(\gamma+2)\gamma^5(\vec{\beta} \cdot \dot{\vec{\beta}})}{(\gamma+1)^2} \left[\left(1 - \frac{\gamma}{\gamma+1}\beta^2\right)(\vec{\beta} \cdot \vec{m})\vec{\beta} \right] \\
 & -\frac{\gamma^3}{\gamma+1} \left[\vec{\beta}(\dot{\vec{\beta}} \cdot \vec{m}) - \dot{\vec{\beta}} \cdot (\vec{\beta} \times \vec{n})\vec{\beta} + \dot{\vec{\beta}}(\vec{\beta} \cdot \vec{m}) \right] + \frac{\gamma^4}{(\gamma+1)^2} \left[(\vec{\beta} \cdot \vec{m})(\dot{\vec{\beta}} \cdot \vec{\beta})\vec{\beta} + \beta^2(\vec{\beta} \cdot \vec{m})\dot{\vec{\beta}} \right]
 \end{aligned}$$

$$\begin{aligned}
 1 - \frac{\gamma}{\gamma+1}\beta^2 &= 1 - \left(1 - \frac{1}{\gamma+1}\right)\beta^2 = 1 - \beta^2 + \frac{1}{\gamma+1}\beta^2 = \frac{1}{\gamma^2} - \frac{1}{\gamma+1}\beta^2 = \frac{1}{\gamma^2} + \frac{1}{\gamma+1}\left(1 - \frac{1}{\gamma^2}\right) \\
 &= \frac{\gamma+1}{\gamma^2(\gamma+1)} + \frac{\gamma^2}{\gamma^2(\gamma+1)} - \frac{1}{\gamma^2(\gamma+1)} = \frac{\gamma(\gamma+1)}{\gamma^2(\gamma+1)} = \frac{1}{\gamma}
 \end{aligned}$$

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$$\begin{aligned}
 \therefore \dot{\vec{m}}|_{x^{\wedge\sigma}} &= \gamma^2 \left[\dot{\vec{\beta}} \times \vec{n} + \vec{\beta} \left(\dot{\vec{\beta}} \cdot \vec{m} \right) \right] - \left[\frac{\gamma^5}{\gamma+1} + \frac{(\gamma+2)\gamma^4}{(\gamma+1)^2} - \gamma^4 - \frac{\gamma^4}{(\gamma+1)^2} \right] \left[\vec{\beta} \left(\vec{\beta} \cdot \dot{\vec{\beta}} \right) \left(\vec{\beta} \cdot \vec{m} \right) \right] \\
 &- \frac{\gamma^3}{\gamma+1} \left[\left(\vec{\beta} \cdot \vec{n} \right) \left(\dot{\vec{\beta}} \times \vec{\beta} \right) + \vec{\beta} \left(\dot{\vec{\beta}} \cdot \vec{m} \right) - \dot{\vec{\beta}} \cdot \left(\vec{\beta} \times \vec{n} \right) \vec{\beta} + \dot{\vec{\beta}} \left(\vec{\beta} \cdot \vec{m} \right) \right] + \frac{\gamma^4}{(\gamma+1)^2} \left[\beta^2 \left(\vec{\beta} \cdot \vec{m} \right) \dot{\vec{\beta}} \right] \\
 &= \gamma^2 \left[\dot{\vec{\beta}} \times \vec{n} \right] - \frac{\gamma^3}{\gamma+1} \left[\left(\vec{\beta} \cdot \vec{n} \right) \left(\dot{\vec{\beta}} \times \vec{\beta} \right) - \vec{n} \cdot \left(\dot{\vec{\beta}} \times \vec{\beta} \right) \vec{\beta} \right] + \frac{\gamma^2}{\gamma+1} \left[\vec{m} \times \left(\vec{\beta} \times \dot{\vec{\beta}} \right) \right] \\
 &= -\gamma\gamma^2 \left[\vec{n} \times \dot{\vec{\beta}}' \right] - \gamma \frac{\gamma^3}{\gamma+1} \left\{ \vec{n} \times \left[\vec{\beta} \times \left(\vec{\beta} \times \dot{\vec{\beta}}' \right) \right] \right\} + \gamma \frac{\gamma^2}{\gamma+1} \left[\vec{m} \times \left(\vec{\beta} \times \dot{\vec{\beta}}' \right) \right] \quad \dots\dots \textcircled{5}
 \end{aligned}$$

上面的推导中有五点考虑：

$$\textcircled{1}: \frac{\gamma^5}{\gamma+1} + \frac{(\gamma+2)\gamma^4}{(\gamma+1)^2} - \gamma^4 - \frac{\gamma^4}{(\gamma+1)^2} = \frac{\gamma^5}{\gamma+1} + \frac{\gamma^4}{\gamma+1} - \frac{\gamma^5 + \gamma^4}{\gamma+1} = 0$$

$$\textcircled{2}: \frac{\gamma^3}{\gamma+1} \left[\frac{\gamma}{\gamma+1} \beta^2 - 1 \right] \dot{\vec{\beta}} \left(\vec{\beta} \cdot \vec{m} \right) = - \frac{\gamma^2}{\gamma+1} \dot{\vec{\beta}} \left(\vec{\beta} \cdot \vec{m} \right)$$

$$\textcircled{3}: \gamma^2 \vec{\beta} \left(\dot{\vec{\beta}} \cdot \vec{m} \right) - \frac{\gamma^3}{\gamma+1} \vec{\beta} \left(\dot{\vec{\beta}} \cdot \vec{m} \right) = \left[\frac{\gamma^3 + \gamma^2}{\gamma+1} - \frac{\gamma^3}{\gamma+1} \right] \vec{\beta} \left(\dot{\vec{\beta}} \cdot \vec{m} \right) = \frac{\gamma^2}{\gamma+1} \vec{\beta} \left(\dot{\vec{\beta}} \cdot \vec{m} \right)$$

$$\textcircled{4}: \frac{\gamma^2}{\gamma+1} \left[\vec{\beta} \left(\dot{\vec{\beta}} \cdot \vec{m} \right) - \dot{\vec{\beta}} \left(\vec{\beta} \cdot \vec{m} \right) \right] = \frac{\gamma^2}{\gamma+1} \left[\vec{\beta} \left(\vec{m} \cdot \dot{\vec{\beta}} \right) - \dot{\vec{\beta}} \left(\vec{m} \cdot \vec{\beta} \right) \right] = \frac{\gamma^2}{\gamma+1} \left[\vec{m} \times \left(\vec{\beta} \times \dot{\vec{\beta}} \right) \right]$$

$$\textcircled{5}: \left(\vec{\beta} \cdot \vec{n} \right) \left(\dot{\vec{\beta}} \times \vec{\beta} \right) - \vec{n} \cdot \left(\dot{\vec{\beta}} \times \vec{\beta} \right) \vec{\beta} = \left(\dot{\vec{\beta}} \times \vec{\beta} \right) \left(\vec{n} \cdot \vec{\beta} \right) - \vec{\beta} \left[\vec{n} \cdot \left(\dot{\vec{\beta}} \times \vec{\beta} \right) \right] = \vec{n} \times \left[\left(\dot{\vec{\beta}} \times \vec{\beta} \right) \times \vec{\beta} \right] \quad 13$$

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同理可得另一矢量组分的Thomas动力学满足：

$$\begin{aligned} \dot{\vec{n}}|_{X^{\lambda\sigma}} &= \dot{\gamma}(\vec{N} - \vec{\beta} \times \vec{M})|_{X^{\lambda\sigma}} + \gamma(\vec{N} - \vec{\beta} \times \vec{M})|_{X^{\lambda\sigma}} - \left(\frac{\dot{\gamma}^2}{\gamma+1}\right)\vec{\beta}(\vec{\beta} \cdot \vec{N})|_{X^{\lambda\sigma}} - \frac{\gamma^2}{\gamma+1}\dot{\vec{\beta}}(\vec{\beta} \cdot \vec{N})|_{X^{\lambda\sigma}} - \frac{\gamma^2}{\gamma+1}\vec{\beta}(\vec{\beta} \cdot \dot{\vec{N}})|_{X^{\lambda\sigma}} \\ &= \underbrace{\gamma^2[\vec{m} \times \dot{\vec{\beta}}]}_{\beta^0 \text{项}} + \underbrace{\frac{\gamma^3}{\gamma+1}\left\{\vec{m} \times [\vec{\beta} \times (\vec{\beta} \times \dot{\vec{\beta}})]\right\}}_{\beta^2 \text{项}} + \underbrace{\frac{\gamma^2}{\gamma+1}[\vec{n} \times (\vec{\beta} \times \dot{\vec{\beta}})]}_{\beta^1 \text{项}} \end{aligned}$$

短交叉项

时间分量正值，空
间分量负值

长交叉项

进动项

↓

始终正值

短交叉项： 被研究矢量组分乘以速度 involve到另一矢量组分的Lorentz变换，进而影响被研究矢量组分Thomas动力学

长交叉项： 另一矢量组分乘以速度 involve到被研究矢量组分的Lorentz变换，进而影响被研究矢量组分Thomas动力学

Thomas动力学的一个核心观点是，研究人员认为实验室系观测的动力学结果必须经过变系修正到静止系，**只有**静止系的动力学结果是有普遍意义的，这也是Thomas动力学与Thomas运动学的观点差别(Thomas运动学是只有实验室系才能观测的运动，认为研究人员应当以实验室系结果为准)。

对于场强张量的Thomas动力学，也可以通过4-potential的Thomas动力学结果的旋度、协变散度得到，在此不再重复。对于更广义的二阶张量Thomas动力学，不能用矢量分解的思路处理，而必须对Lorentz变换矩阵直接做固有时求导处理。

一般二阶张量Thomas动力学

$$\dot{X}^{\mu\nu}\Big|_{x^{\lambda\sigma}} = \left[L_{\lambda}^{\mu}(\vec{\beta}) L_{\sigma}^{\nu}(\vec{\beta}) \right] x^{\lambda\sigma}\Big|_{x^{\lambda\sigma}} \dots\dots \textcircled{6}$$

$$= \begin{pmatrix} \gamma & -\gamma\beta_1 & -\gamma\beta_2 & -\gamma\beta_3 \\ -\gamma\beta_1 & 1 + \frac{(\gamma-1)\beta_1^2}{\beta^2} & \frac{(\gamma-1)\beta_1\beta_2}{\beta^2} & \frac{(\gamma-1)\beta_1\beta_3}{\beta^2} \\ -\gamma\beta_2 & \frac{(\gamma-1)\beta_1\beta_2}{\beta^2} & 1 + \frac{(\gamma-1)\beta_2^2}{\beta^2} & \frac{(\gamma-1)\beta_2\beta_3}{\beta^2} \\ -\gamma\beta_3 & \frac{(\gamma-1)\beta_1\beta_3}{\beta^2} & \frac{(\gamma-1)\beta_2\beta_3}{\beta^2} & 1 + \frac{(\gamma-1)\beta_3^2}{\beta^2} \end{pmatrix} \begin{pmatrix} \gamma & -\gamma\beta_1 & -\gamma\beta_2 & -\gamma\beta_3 \\ -\gamma\beta_1 & 1 + \frac{(\gamma-1)\beta_1^2}{\beta^2} & \frac{(\gamma-1)\beta_1\beta_2}{\beta^2} & \frac{(\gamma-1)\beta_1\beta_3}{\beta^2} \\ -\gamma\beta_2 & \frac{(\gamma-1)\beta_1\beta_2}{\beta^2} & 1 + \frac{(\gamma-1)\beta_2^2}{\beta^2} & \frac{(\gamma-1)\beta_2\beta_3}{\beta^2} \\ -\gamma\beta_3 & \frac{(\gamma-1)\beta_1\beta_3}{\beta^2} & \frac{(\gamma-1)\beta_2\beta_3}{\beta^2} & 1 + \frac{(\gamma-1)\beta_3^2}{\beta^2} \end{pmatrix} x^{\lambda\sigma}$$

$$+ \begin{pmatrix} \gamma & -\gamma\beta_1 & -\gamma\beta_2 & -\gamma\beta_3 \\ -\gamma\beta_1 & 1 + \frac{(\gamma-1)\beta_1^2}{\beta^2} & \frac{(\gamma-1)\beta_1\beta_2}{\beta^2} & \frac{(\gamma-1)\beta_1\beta_3}{\beta^2} \\ -\gamma\beta_2 & \frac{(\gamma-1)\beta_1\beta_2}{\beta^2} & 1 + \frac{(\gamma-1)\beta_2^2}{\beta^2} & \frac{(\gamma-1)\beta_2\beta_3}{\beta^2} \\ -\gamma\beta_3 & \frac{(\gamma-1)\beta_1\beta_3}{\beta^2} & \frac{(\gamma-1)\beta_2\beta_3}{\beta^2} & 1 + \frac{(\gamma-1)\beta_3^2}{\beta^2} \end{pmatrix} \begin{pmatrix} \gamma & -\gamma\beta_1 & -\gamma\beta_2 & -\gamma\beta_3 \\ -\gamma\beta_1 & 1 + \frac{(\gamma-1)\beta_1^2}{\beta^2} & \frac{(\gamma-1)\beta_1\beta_2}{\beta^2} & \frac{(\gamma-1)\beta_1\beta_3}{\beta^2} \\ -\gamma\beta_2 & \frac{(\gamma-1)\beta_1\beta_2}{\beta^2} & 1 + \frac{(\gamma-1)\beta_2^2}{\beta^2} & \frac{(\gamma-1)\beta_2\beta_3}{\beta^2} \\ -\gamma\beta_3 & \frac{(\gamma-1)\beta_1\beta_3}{\beta^2} & \frac{(\gamma-1)\beta_2\beta_3}{\beta^2} & 1 + \frac{(\gamma-1)\beta_3^2}{\beta^2} \end{pmatrix} x^{\lambda\sigma}$$

$$\hat{\mathbf{x}}^{\text{new}}|_{\mathbf{x}^-} = [\mathbf{L}_x^* (-\hat{\beta}) \mathbf{L}_w^* (-\hat{\beta})] \mathbf{x}^{\text{old}}|_{\mathbf{x}^-}$$

[illegible]

一般二阶张量Thomas动力学

令 $\alpha = \gamma^3 (\vec{\beta} \cdot \dot{\vec{\beta}})$, 化简得

[illegible]

其中

其中

$$\dot{L}(\vec{\beta}) = \frac{1}{\beta^3} \begin{pmatrix} \alpha\beta^3 & -(\alpha\beta_1\beta^3 + \gamma\dot{\beta}_1\beta^3) & -(\alpha\beta_2\beta^3 + \gamma\dot{\beta}_2\beta^3) & -(\alpha\beta_3\beta^3 + \gamma\dot{\beta}_3\beta^3) \\ -(\alpha\beta_1\beta^3 + \gamma\dot{\beta}_1\beta^3) & \alpha\beta_1\beta_1\beta + (\gamma-1)[(\beta\dot{\beta}_1\beta_1) - 3\dot{\beta}\beta_1\beta_1] & \alpha\beta_1\beta_2\beta + (\gamma-1)[(\beta\dot{\beta}_1\beta_2) - 3\dot{\beta}\beta_1\beta_2] & \alpha\beta_1\beta_3\beta + (\gamma-1)[(\beta\dot{\beta}_1\beta_3) - 3\dot{\beta}\beta_1\beta_3] \\ -(\alpha\beta_2\beta^3 + \gamma\dot{\beta}_2\beta^3) & \alpha\beta_1\beta_2\beta + (\gamma-1)[(\beta\dot{\beta}_1\beta_2) - 3\dot{\beta}\beta_1\beta_2] & \alpha\beta_1\beta_1\beta + (\gamma-1)[(\beta\dot{\beta}_2\beta_2) - 3\dot{\beta}\beta_2\beta_2] & \alpha\beta_1\beta_2\beta + (\gamma-1)[(\beta\dot{\beta}_2\beta_3) - 3\dot{\beta}\beta_2\beta_3] \\ -(\alpha\beta_3\beta^3 + \gamma\dot{\beta}_3\beta^3) & \alpha\beta_1\beta_3\beta + (\gamma-1)[(\beta\dot{\beta}_1\beta_3) - 3\dot{\beta}\beta_1\beta_3] & \alpha\beta_1\beta_2\beta + (\gamma-1)[(\beta\dot{\beta}_2\beta_3) - 3\dot{\beta}\beta_2\beta_3] & \alpha\beta_3\beta_3\beta + (\gamma-1)[(\beta\dot{\beta}_3\beta_3) - 3\dot{\beta}\beta_3\beta_3] \end{pmatrix}$$

$$L(\vec{\beta}) = \frac{1}{\beta^2} \begin{pmatrix} \left(\gamma - \frac{1}{\gamma}\right) & -\beta_1\left(\gamma - \frac{1}{\gamma}\right) & -\beta_2\left(\gamma - \frac{1}{\gamma}\right) & -\beta_3\left(\gamma - \frac{1}{\gamma}\right) \\ -\beta_1\left(\gamma - \frac{1}{\gamma}\right) & \beta^2 + (\gamma-1)\beta_1^2 & (\gamma-1)\beta_1\beta_2 & (\gamma-1)\beta_1\beta_3 \\ -\beta_2\left(\gamma - \frac{1}{\gamma}\right) & (\gamma-1)\beta_1\beta_2 & \beta^2 + (\gamma-1)\beta_2^2 & (\gamma-1)\beta_2\beta_3 \\ -\beta_3\left(\gamma - \frac{1}{\gamma}\right) & (\gamma-1)\beta_1\beta_3 & (\gamma-1)\beta_2\beta_3 & \beta^2 + (\gamma-1)\beta_3^2 \end{pmatrix}$$

$$= \frac{1}{\beta^2} \begin{pmatrix} \frac{\gamma+1}{\gamma}(\gamma-1) & -\beta_1\left(\frac{\gamma+1}{\gamma}(\gamma-1)\right) & -\beta_2\left(\frac{\gamma+1}{\gamma}(\gamma-1)\right) & -\beta_3\left(\frac{\gamma+1}{\gamma}(\gamma-1)\right) \\ -\beta_1\left(\frac{\gamma+1}{\gamma}(\gamma-1)\right) & \frac{\gamma+1}{\gamma^2}(\gamma-1) + (\gamma-1)\beta_1^2 & (\gamma-1)\beta_1\beta_2 & (\gamma-1)\beta_1\beta_3 \\ -\beta_2\left(\frac{\gamma+1}{\gamma}(\gamma-1)\right) & (\gamma-1)\beta_1\beta_2 & \frac{\gamma+1}{\gamma^2}(\gamma-1) + (\gamma-1)\beta_2^2 & (\gamma-1)\beta_2\beta_3 \\ -\beta_3\left(\frac{\gamma+1}{\gamma}(\gamma-1)\right) & (\gamma-1)\beta_1\beta_3 & (\gamma-1)\beta_2\beta_3 & \frac{\gamma+1}{\gamma^2}(\gamma-1) + (\gamma-1)\beta_3^2 \end{pmatrix}$$

一般二阶张量Thomas动力学

其中蓝色矩阵元是可以写出通项的，只要考虑到变换

$$\begin{aligned} \alpha\beta_0\beta_0\beta + (\gamma - 1) \left[(\beta\dot{\beta}_0\beta_0) - 3\dot{\beta}\beta_0\beta_0 \right] &= \alpha\beta^3 + (\gamma - 1) \left[(\dot{\beta}^3) - 3\beta^2\dot{\beta} \right] \\ &= \alpha\beta^3 + (\gamma - 1) \left[3\beta^2\dot{\beta} - 3\beta^2\dot{\beta} \right] = \alpha\beta^3 \end{aligned}$$

为进一步写出通项，考虑 $\beta^2 = 1 - \frac{1}{\gamma^2}$ ，得到

$$\begin{aligned} \vec{L}(\vec{\beta}) &= \frac{1}{\beta^3} \begin{pmatrix} \alpha\beta^3 & -\left(\alpha\beta_1\beta^3 + \left(\gamma - \frac{1}{\gamma}\right)\dot{\beta}_1\beta\right) & -\left(\alpha\beta_2\beta^3 + \left(\gamma - \frac{1}{\gamma}\right)\dot{\beta}_2\beta\right) & -\left(\alpha\beta_3\beta^3 + \left(\gamma - \frac{1}{\gamma}\right)\dot{\beta}_3\beta\right) \\ -\left(\alpha\beta_1\beta^3 + \left(\gamma - \frac{1}{\gamma}\right)\dot{\beta}_1\beta\right) & \alpha\beta_1\beta_1\beta + (\gamma - 1) \left[(\beta\dot{\beta}_1\beta_1) - 3\dot{\beta}\beta_1\beta_1 \right] & \alpha\beta_1\beta_2\beta + (\gamma - 1) \left[(\beta\dot{\beta}_1\beta_2) - 3\dot{\beta}\beta_1\beta_2 \right] & \alpha\beta_1\beta_3\beta + (\gamma - 1) \left[(\beta\dot{\beta}_1\beta_3) - 3\dot{\beta}\beta_1\beta_3 \right] \\ -\left(\alpha\beta_2\beta^3 + \left(\gamma - \frac{1}{\gamma}\right)\dot{\beta}_2\beta\right) & \alpha\beta_1\beta_2\beta + (\gamma - 1) \left[(\beta\dot{\beta}_1\beta_2) - 3\dot{\beta}\beta_1\beta_2 \right] & \alpha\beta_1\beta_1\beta + (\gamma - 1) \left[(\beta\dot{\beta}_2\beta_1) - 3\dot{\beta}\beta_2\beta_1 \right] & \alpha\beta_1\beta_2\beta + (\gamma - 1) \left[(\beta\dot{\beta}_2\beta_2) - 3\dot{\beta}\beta_2\beta_2 \right] \\ -\left(\alpha\beta_3\beta^3 + \left(\gamma - \frac{1}{\gamma}\right)\dot{\beta}_3\beta\right) & \alpha\beta_1\beta_3\beta + (\gamma - 1) \left[(\beta\dot{\beta}_1\beta_3) - 3\dot{\beta}\beta_1\beta_3 \right] & \alpha\beta_1\beta_2\beta + (\gamma - 1) \left[(\beta\dot{\beta}_2\beta_3) - 3\dot{\beta}\beta_2\beta_3 \right] & \alpha\beta_3\beta_3\beta + (\gamma - 1) \left[(\beta\dot{\beta}_3\beta_3) - 3\dot{\beta}\beta_3\beta_3 \right] \end{pmatrix} \\ &= \frac{1}{\beta^3} \begin{pmatrix} \alpha\beta^3 & -\left[\alpha\beta_1\beta^3 + \beta^{-1}\left(\gamma - \frac{1}{\gamma}\right)\left((\beta\dot{\beta}_1\beta_0) - 2\dot{\beta}\beta_1\beta_0\right)\right] & -\left[\alpha\beta_2\beta^3 + \beta^{-1}\left(\gamma - \frac{1}{\gamma}\right)\left((\beta\dot{\beta}_2\beta_0) - 2\dot{\beta}\beta_2\beta_0\right)\right] & -\left[\alpha\beta_3\beta^3 + \beta^{-1}\left(\gamma - \frac{1}{\gamma}\right)\left((\beta\dot{\beta}_3\beta_0) - 2\dot{\beta}\beta_3\beta_0\right)\right] \\ -\left[\alpha\beta_1\beta^3 + \beta^{-1}\left(\gamma - \frac{1}{\gamma}\right)\left((\beta\dot{\beta}_1\beta_0) - 2\dot{\beta}\beta_1\beta_0\right)\right] & \alpha\beta_1\beta_1\beta + (\gamma - 1) \left[(\beta\dot{\beta}_1\beta_1) - 3\dot{\beta}\beta_1\beta_1 \right] & \alpha\beta_1\beta_2\beta + (\gamma - 1) \left[(\beta\dot{\beta}_1\beta_2) - 3\dot{\beta}\beta_1\beta_2 \right] & \alpha\beta_1\beta_3\beta + (\gamma - 1) \left[(\beta\dot{\beta}_1\beta_3) - 3\dot{\beta}\beta_1\beta_3 \right] \\ -\left[\alpha\beta_2\beta^3 + \beta^{-1}\left(\gamma - \frac{1}{\gamma}\right)\left((\beta\dot{\beta}_2\beta_0) - 2\dot{\beta}\beta_2\beta_0\right)\right] & \alpha\beta_1\beta_2\beta + (\gamma - 1) \left[(\beta\dot{\beta}_1\beta_2) - 3\dot{\beta}\beta_1\beta_2 \right] & \alpha\beta_1\beta_1\beta + (\gamma - 1) \left[(\beta\dot{\beta}_2\beta_1) - 3\dot{\beta}\beta_2\beta_1 \right] & \alpha\beta_1\beta_2\beta + (\gamma - 1) \left[(\beta\dot{\beta}_2\beta_2) - 3\dot{\beta}\beta_2\beta_2 \right] \\ -\left[\alpha\beta_3\beta^3 + \beta^{-1}\left(\gamma - \frac{1}{\gamma}\right)\left((\beta\dot{\beta}_3\beta_0) - 2\dot{\beta}\beta_3\beta_0\right)\right] & \alpha\beta_1\beta_3\beta + (\gamma - 1) \left[(\beta\dot{\beta}_1\beta_3) - 3\dot{\beta}\beta_1\beta_3 \right] & \alpha\beta_1\beta_2\beta + (\gamma - 1) \left[(\beta\dot{\beta}_2\beta_3) - 3\dot{\beta}\beta_2\beta_3 \right] & \alpha\beta_3\beta_3\beta + (\gamma - 1) \left[(\beta\dot{\beta}_3\beta_3) - 3\dot{\beta}\beta_3\beta_3 \right] \end{pmatrix} \\ L(\vec{\beta}) &= \frac{\gamma - 1}{\beta^2} \begin{pmatrix} 1 - \beta^2 + \beta_0^2 + \frac{1}{\gamma} & -\beta_1\left(1 + \frac{1}{\gamma}\right) & -\beta_2\left(1 + \frac{1}{\gamma}\right) & -\beta_3\left(1 + \frac{1}{\gamma}\right) \\ -\beta_1\left(1 + \frac{1}{\gamma}\right) & 1 - \beta^2 + \beta_1^2 + \frac{1}{\gamma} & \beta_1\beta_2 & \beta_1\beta_3 \\ -\beta_2\left(1 + \frac{1}{\gamma}\right) & \beta_1\beta_2 & 1 - \beta^2 + \beta_2^2 + \frac{1}{\gamma} & \beta_2\beta_3 \\ -\beta_3\left(1 + \frac{1}{\gamma}\right) & \beta_1\beta_3 & \beta_2\beta_3 & 1 - \beta^2 + \beta_3^2 + \frac{1}{\gamma} \end{pmatrix} \end{aligned}$$

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令 $\beta_0 = \beta$ (β [4]不是协变矢量), 则Lorentz变换矩阵、固有时一阶导的Lorentz变换矩阵通项公式分别为:

$$L_{\nu}^{\mu}(\vec{\beta}) = \frac{\gamma-1}{\beta^2} \left[g_{\mu\mu} g_{\nu\nu} \beta^{-\theta(-g_{\mu\mu} g_{\nu\nu})} \beta_{\mu} \beta_{\nu} \left(\frac{\gamma + \theta(-g_{\mu\mu} g_{\nu\nu})}{\gamma} \right) + \delta_{\mu\nu} \left(\frac{\gamma+1}{\gamma^2} \right) \right]$$

$$\dot{L}_{\nu}^{\mu}(\vec{\beta}) = \frac{1}{\beta^3} g_{\mu\mu} g_{\nu\nu} \left\{ \beta^{\theta(-g_{\mu\mu} g_{\nu\nu})} \alpha \beta_{\mu} \beta_{\nu} \beta + \beta^{-\theta(-g_{\mu\mu} g_{\nu\nu})} \left(\gamma - \frac{1}{\gamma^{\theta(-g_{\mu\mu} g_{\nu\nu})}} \right) \left[(\beta_{\mu} \dot{\beta}_{\nu} \beta) - (2 + \theta(g_{\mu\mu} g_{\nu\nu})) \dot{\beta} \beta_{\mu} \beta_{\nu} \right] \right\}$$

用Mathematica将上述结果 $\mu=\nu=1$ 代入⑥式, 化简后的结果依然十分复杂:

$$\text{Out[31]= } \frac{1}{\left(1 - \frac{1}{\gamma^2}\right)^3 \gamma^5} \\ (\gamma-1)(\gamma+1) \left((\gamma-1) \sqrt{1 - \frac{1}{\gamma^2}} \gamma^2 \text{X11} (\gamma^2 \text{dot}\beta\beta\beta - 3 \beta \text{dot}(\gamma^2-1)) + \beta x \text{X12} \left(-\alpha(\gamma^2-1)^2 - \sqrt{1 - \frac{1}{\gamma^2}} (\gamma-1) \gamma^2 (\gamma^2 \text{dot}\beta\beta\beta - 3 \beta \text{dot}(\gamma^2-1)) \right) + \beta y \text{X13} \left(-\alpha(\gamma^2-1)^2 - \sqrt{1 - \frac{1}{\gamma^2}} (\gamma-1) \gamma^2 (\gamma^2 \text{dot}\beta\beta\beta - 3 \beta \text{dot}(\gamma^2-1)) \right) + \right. \\ \left. \beta z \text{X14} \left(-\alpha(\gamma^2-1)^2 - \sqrt{1 - \frac{1}{\gamma^2}} (\gamma-1) \gamma^2 (\gamma^2 \text{dot}\beta\beta\beta - 3 \beta \text{dot}(\gamma^2-1)) \right) - (\gamma^2-1) \text{X21} \left(\alpha \beta x (\gamma^2-1) + \gamma^3 \left(\text{dot}\beta\beta\beta x - 2 \beta \text{dot} \beta x \sqrt{1 - \frac{1}{\gamma^2}} \right) \right) + \right. \\ \left. \beta x (\gamma^2-1) \text{X22} \left(\alpha \beta x (\gamma^2-1) + \gamma^3 \left(\text{dot}\beta\beta\beta x - 2 \beta \text{dot} \beta x \sqrt{1 - \frac{1}{\gamma^2}} \right) \right) + \beta y (\gamma^2-1) \text{X23} \left(\alpha \beta x (\gamma^2-1) + \gamma^3 \left(\text{dot}\beta\beta\beta x - 2 \beta \text{dot} \beta x \sqrt{1 - \frac{1}{\gamma^2}} \right) \right) + \beta z (\gamma^2-1) \text{X24} \left(\alpha \beta x (\gamma^2-1) + \gamma^3 \left(\text{dot}\beta\beta\beta x - 2 \beta \text{dot} \beta x \sqrt{1 - \frac{1}{\gamma^2}} \right) \right) - \right. \\ (\gamma^2-1) \text{X31} \left(\alpha \beta y (\gamma^2-1) + \gamma^3 \left(\text{dot}\beta\beta\beta y - 2 \beta \text{dot} \beta y \sqrt{1 - \frac{1}{\gamma^2}} \right) \right) + \beta x (\gamma^2-1) \text{X32} \left(\alpha \beta y (\gamma^2-1) + \gamma^3 \left(\text{dot}\beta\beta\beta y - 2 \beta \text{dot} \beta y \sqrt{1 - \frac{1}{\gamma^2}} \right) \right) + \beta y (\gamma^2-1) \text{X33} \left(\alpha \beta y (\gamma^2-1) + \gamma^3 \left(\text{dot}\beta\beta\beta y - 2 \beta \text{dot} \beta y \sqrt{1 - \frac{1}{\gamma^2}} \right) \right) + \\ \left. \beta z (\gamma^2-1) \text{X34} \left(\alpha \beta y (\gamma^2-1) + \gamma^3 \left(\text{dot}\beta\beta\beta y - 2 \beta \text{dot} \beta y \sqrt{1 - \frac{1}{\gamma^2}} \right) \right) - (\gamma^2-1) \text{X41} \left(\alpha \beta z (\gamma^2-1) + \gamma^3 \left(\text{dot}\beta\beta\beta z - 2 \beta \text{dot} \beta z \sqrt{1 - \frac{1}{\gamma^2}} \right) \right) + \beta x (\gamma^2-1) \text{X42} \left(\alpha \beta z (\gamma^2-1) + \gamma^3 \left(\text{dot}\beta\beta\beta z - 2 \beta \text{dot} \beta z \sqrt{1 - \frac{1}{\gamma^2}} \right) \right) + \right. \\ \left. \beta y (\gamma^2-1) \text{X43} \left(\alpha \beta z (\gamma^2-1) + \gamma^3 \left(\text{dot}\beta\beta\beta z - 2 \beta \text{dot} \beta z \sqrt{1 - \frac{1}{\gamma^2}} \right) \right) + \beta z (\gamma^2-1) \text{X44} \left(\alpha \beta z (\gamma^2-1) + \gamma^3 \left(\text{dot}\beta\beta\beta z - 2 \beta \text{dot} \beta z \sqrt{1 - \frac{1}{\gamma^2}} \right) \right) + \alpha (\gamma^2-1)^2 \text{X11} \right)$$

为了方便地总结规律, 改用非矩阵求导的方法, 先给出矩阵变换的结果再求导。

用Mathematica计算二阶张量Lorentz变换结果并总结矩阵元的规律:

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$$X^{\mu\nu}|_{x^{\lambda\sigma}} = [L_\lambda^\mu(\vec{\beta}) L_\sigma^\nu(\vec{\beta})] x^{\lambda\sigma}|_{x^{\lambda\sigma}} = \begin{pmatrix} X_{t-t}^{11} & X_{t-s}^{12} & X_{t-s}^{13} & X_{t-s}^{14} \\ X_{s-t}^{21} & X_{s-s}^{22} & X_{s-s}^{23} & X_{s-s}^{24} \\ X_{s-t}^{31} & X_{s-s}^{32} & X_{s-s}^{33} & X_{s-s}^{34} \\ X_{s-t}^{41} & X_{s-s}^{42} & X_{s-s}^{43} & X_{s-s}^{44} \end{pmatrix}$$

其中不同类矩阵元满足不同的通项：

$$X_{t-t}^{\mu\nu} = \sum_\lambda \sum_\sigma g_{\lambda\lambda} g_{\sigma\sigma} \gamma^2 \beta_\lambda \beta_\sigma x^{\lambda\sigma} \quad (\beta_\lambda = (1, \beta_x, \beta_y, \beta_z))$$

$$X_{s-t}^{\mu\nu} = \sum_\lambda \sum_\sigma -g_{\lambda\lambda} g_{\sigma\sigma} \frac{\gamma}{\gamma+1} \beta_\sigma [\beta_\mu \beta_\lambda (\gamma^2 + \delta_{\lambda 1} \gamma) + \delta_{\lambda\mu} (\gamma+1)] x^{\lambda\sigma} \quad (\beta_\lambda = (1, \beta_x, \beta_y, \beta_z))$$

$$X_{t-s}^{\mu\nu} = \sum_\lambda \sum_\sigma -g_{\lambda\lambda} g_{\sigma\sigma} \frac{\gamma}{\gamma+1} \beta_\lambda [\beta_\nu \beta_\sigma (\gamma^2 + \delta_{\sigma 1} \gamma) + \delta_{\sigma\nu} (\gamma+1)] x^{\lambda\sigma} \quad (\beta_\lambda = (1, \beta_x, \beta_y, \beta_z))$$

$$X_{s-s}^{\mu\nu} = \sum_\lambda \sum_\sigma g_{\lambda\lambda} g_{\sigma\sigma} \frac{1}{(\gamma+1)^2} \beta_\mu \beta_\nu \gamma^2 \left[\beta_\lambda \gamma + \delta_{\lambda 1} + \frac{\gamma+1}{\beta_\lambda \gamma} \delta_{\lambda\mu} \right] \left[\beta_\sigma \gamma + \delta_{\sigma 1} + \frac{\gamma+1}{\beta_\sigma \gamma} \delta_{\sigma\nu} \right] x^{\lambda\sigma}$$

正如一阶张量Thomas动力学一样，我们关注张量的空间部分经Lorentz后的结果。

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$$\begin{aligned}
 \therefore X_{s-s}^{\mu\nu} &= \sum_{\lambda} \sum_{\sigma} g_{\lambda\lambda} g_{\sigma\sigma} \frac{\gamma^2}{(\gamma+1)^2} \beta_{\mu} \beta_{\nu} \left[\beta_{\lambda} \gamma + \delta_{\lambda 1} + \frac{\gamma+1}{\beta_{\lambda} \gamma} \delta_{\lambda\mu} \right] \left[\beta_{\sigma} \gamma + \delta_{\sigma 1} + \frac{\gamma+1}{\beta_{\sigma} \gamma} \delta_{\sigma\nu} \right] x^{\lambda\sigma} \\
 &= \sum_{\lambda=2}^4 \sum_{\sigma=2}^4 \frac{\gamma^2}{(\gamma+1)^2} \beta_{\mu} \beta_{\nu} \left[\beta_{\lambda} \gamma + \frac{\gamma+1}{\beta_{\lambda} \gamma} \delta_{\lambda\mu} \right] \left[\beta_{\sigma} \gamma + \frac{\gamma+1}{\beta_{\sigma} \gamma} \delta_{\sigma\nu} \right] x^{\lambda\sigma} \\
 &\quad - \sum_{\sigma=2}^4 \frac{\gamma^2}{\gamma+1} \beta_{\mu} \beta_{\nu} \left[\beta_{\sigma} \gamma + \frac{\gamma+1}{\beta_{\sigma} \gamma} \delta_{\sigma\nu} \right] x^{1\sigma} \\
 &\quad - \sum_{\lambda=2}^4 \frac{\gamma^2}{\gamma+1} \beta_{\mu} \beta_{\nu} \left[\beta_{\lambda} \gamma + \frac{\gamma+1}{\beta_{\lambda} \gamma} \delta_{\lambda\mu} \right] x^{\lambda 1} \\
 &\quad + \gamma^2 \beta_{\mu} \beta_{\nu} x^{11}
 \end{aligned}$$

$$\begin{aligned}
 \text{其中纯空间项} &\sum_{\lambda=2}^4 \sum_{\sigma=2}^4 \frac{\gamma^2}{(\gamma+1)^2} \beta_{\mu} \beta_{\nu} \left[\beta_{\lambda} \beta_{\sigma} \gamma^2 + \frac{\beta_{\sigma} (\gamma+1)}{\beta_{\lambda}} \delta_{\lambda\mu} + \frac{\beta_{\lambda} (\gamma+1)}{\beta_{\sigma}} \delta_{\sigma\nu} + \frac{(\gamma+1)^2}{\beta_{\lambda} \beta_{\sigma} \gamma^2} \delta_{\lambda\mu} \delta_{\sigma\nu} \right] x^{\lambda\sigma} \\
 &= \frac{\gamma^4}{(\gamma+1)^2} \beta_{\mu} \beta_{\nu} \sum_{\lambda=2}^4 \sum_{\sigma=2}^4 \beta_{\lambda} \beta_{\sigma} x^{\lambda\sigma} + \beta_{\nu} \frac{\gamma^2}{\gamma+1} \sum_{\sigma=2}^4 \beta_{\sigma} x^{\mu\sigma} + \beta_{\mu} \frac{\gamma^2}{\gamma+1} \sum_{\lambda=2}^4 \beta_{\lambda} x^{\lambda\nu} + x^{\mu\nu}
 \end{aligned}$$

定义二阶张量点积 $\vec{M} \cdot \vec{N} = \sum_{\lambda=2}^4 \sum_{\sigma=2}^4 M^{\lambda\sigma} N^{\lambda\sigma} = \text{tr}(\vec{M}^T \vec{N})$ ，不构造速度张量，否则错过好的运动学

$$\therefore \frac{\gamma^4}{(\gamma+1)^2} \beta_{\mu} \beta_{\nu} \sum_{\lambda=2}^4 \sum_{\sigma=2}^4 \beta_{\lambda} \beta_{\sigma} x^{\lambda\sigma} = \frac{\gamma^4}{(\gamma+1)^2} \vec{\beta} \vec{\beta}^T [(\vec{\beta} \vec{\beta}^T) \cdot \vec{x}], \text{ 这和一阶张量Lorentz变换形式类似}$$

$\vec{x}$ 上标表示二阶张量

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$$\begin{aligned} \therefore [\vec{\beta} \cdot \vec{x}]^\mu &= \begin{pmatrix} \vec{\beta} \cdot \vec{x}^{2c} & \vec{\beta} \cdot \vec{x}^{2c} & \vec{\beta} \cdot \vec{x}^{2c} \\ \vec{\beta} \cdot \vec{x}^{3c} & \vec{\beta} \cdot \vec{x}^{3c} & \vec{\beta} \cdot \vec{x}^{3c} \\ \vec{\beta} \cdot \vec{x}^{4c} & \vec{\beta} \cdot \vec{x}^{4c} & \vec{\beta} \cdot \vec{x}^{4c} \end{pmatrix} = \begin{pmatrix} \beta_1 x^{22} + \beta_2 x^{23} + \beta_3 x^{24} & \beta_1 x^{22} + \beta_2 x^{23} + \beta_3 x^{24} & \beta_1 x^{22} + \beta_2 x^{23} + \beta_3 x^{24} \\ \beta_1 x^{32} + \beta_2 x^{33} + \beta_3 x^{34} & \beta_1 x^{32} + \beta_2 x^{33} + \beta_3 x^{34} & \beta_1 x^{32} + \beta_2 x^{33} + \beta_3 x^{34} \\ \beta_1 x^{42} + \beta_2 x^{43} + \beta_3 x^{44} & \beta_1 x^{42} + \beta_2 x^{43} + \beta_3 x^{44} & \beta_1 x^{42} + \beta_2 x^{43} + \beta_3 x^{44} \end{pmatrix} \\ &= \begin{pmatrix} \beta_1 x^{22} & \beta_1 x^{22} & \beta_1 x^{22} \\ \beta_1 x^{32} & \beta_1 x^{32} & \beta_1 x^{32} \\ \beta_1 x^{42} & \beta_1 x^{42} & \beta_1 x^{42} \end{pmatrix} + \begin{pmatrix} \beta_2 x^{23} & \beta_2 x^{23} & \beta_2 x^{23} \\ \beta_2 x^{33} & \beta_2 x^{33} & \beta_2 x^{33} \\ \beta_2 x^{43} & \beta_2 x^{43} & \beta_2 x^{43} \end{pmatrix} + \begin{pmatrix} \beta_3 x^{24} & \beta_3 x^{24} & \beta_3 x^{24} \\ \beta_3 x^{34} & \beta_3 x^{34} & \beta_3 x^{34} \\ \beta_3 x^{44} & \beta_3 x^{44} & \beta_3 x^{44} \end{pmatrix} \\ &= \beta_1 \begin{pmatrix} x^{22} & x^{22} & x^{22} \\ x^{32} & x^{32} & x^{32} \\ x^{42} & x^{42} & x^{42} \end{pmatrix} + \beta_2 \begin{pmatrix} x^{23} & x^{23} & x^{23} \\ x^{33} & x^{33} & x^{33} \\ x^{43} & x^{43} & x^{43} \end{pmatrix} + \beta_3 \begin{pmatrix} x^{24} & x^{24} & x^{24} \\ x^{34} & x^{34} & x^{34} \\ x^{44} & x^{44} & x^{44} \end{pmatrix} = \vec{\beta} \cdot \vec{\vec{x}}^\mu \end{aligned}$$

$$\text{其中 } \vec{\vec{x}}^\mu = (\vec{x}^2, \vec{x}^3, \vec{x}^4), \quad \vec{x}^n = \begin{pmatrix} x^{2n} & x^{2n} & x^{2n} \\ x^{3n} & x^{3n} & x^{3n} \\ x^{4n} & x^{4n} & x^{4n} \end{pmatrix}$$

$$\text{同理 } [\vec{\beta} \cdot \vec{x}]^\nu = \vec{\beta} \cdot \vec{\vec{x}}^\nu, \quad \vec{\vec{x}}^\nu = (\vec{x}^2, \vec{x}^3, \vec{x}^4), \quad \vec{x}^\nu = \begin{pmatrix} x^{n2} & x^{n3} & x^{n4} \\ x^{n2} & x^{n3} & x^{n4} \\ x^{n2} & x^{n3} & x^{n4} \end{pmatrix}$$

$$\begin{aligned} \therefore \frac{\gamma^4}{(\gamma+1)^2} \beta_\mu \beta_\nu \sum_{\lambda=2}^4 \sum_{\sigma=2}^4 \beta_\lambda \beta_\sigma x^{\lambda\sigma} + \beta_\nu \frac{\gamma^2}{\gamma+1} \sum_{\sigma=2}^4 \beta_\sigma x^{\mu\sigma} + \beta_\mu \frac{\gamma^2}{\gamma+1} \sum_{\lambda=2}^4 \beta_\lambda x^{\lambda\nu} + x^{\mu\nu} \\ = \frac{\gamma^4}{(\gamma+1)^2} \vec{\beta} \vec{\beta}^T [(\vec{\beta} \vec{\beta}^T) \cdot \vec{x}] + \frac{\gamma^2}{(\gamma+1)} \beta_\nu (\vec{\beta} \cdot \vec{x}^\mu) + \frac{\gamma^2}{(\gamma+1)} \beta_\mu (\vec{\beta} \cdot \vec{x}^\nu) + x^{\mu\nu} \\ = \frac{\gamma^4}{(\gamma+1)^2} \vec{\beta} \vec{\beta}^T [(\vec{\beta} \vec{\beta}^T) \cdot \vec{x}] + \frac{\gamma^2}{(\gamma+1)} (\vec{x} \vec{\beta}) \vec{\beta}^T + \frac{\gamma^2}{(\gamma+1)} \vec{\beta} (\vec{\beta}^T \vec{x}) + \vec{x} \quad \text{最后一步是凑出来的} \end{aligned}$$

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$$\begin{aligned}
 & \text{含时间项} \left\{ - \sum_{\sigma=2}^4 \frac{\gamma^2}{\gamma+1} \beta_\mu \beta_\nu \left[\beta_\sigma \gamma + \frac{\gamma+1}{\beta_\sigma \gamma} \delta_{\sigma\nu} \right] x^{1\sigma} - \sum_{\lambda=2}^4 \frac{\gamma^2}{\gamma+1} \beta_\mu \beta_\nu \left[\beta_\lambda \gamma + \frac{\gamma+1}{\beta_\lambda \gamma} \delta_{\lambda\mu} \right] x^{\lambda 1} + \gamma^2 \beta_\mu \beta_\nu x^{11} \right\} \\
 &= - \frac{\gamma^3}{\gamma+1} \beta_\mu \beta_\nu \sum_{\sigma=2}^4 \beta_\sigma x^{1\sigma} - \gamma \beta_\mu x^{1\nu} - \frac{\gamma^3}{\gamma+1} \beta_\mu \beta_\nu \sum_{\lambda=2}^4 \beta_\lambda x^{\lambda 1} - \gamma \beta_\nu x^{\mu 1} + \gamma^2 \beta_\mu \beta_\nu x^{11} \\
 &= - \frac{\gamma^3}{(\gamma+1)} \vec{\beta} \vec{\beta}^T (\vec{\beta}^T \vec{x}_{lin1}) - \gamma \vec{\beta} \vec{x}_{lin1}^T - \frac{\gamma^3}{(\gamma+1)} \vec{\beta} \vec{\beta}^T (\vec{x}_{col1}^T \vec{\beta}) - \gamma \vec{x}_{col1} \vec{\beta}^T + \gamma^2 \vec{\beta} \vec{\beta}^T x^{11}
 \end{aligned}$$

$$\text{其中 } \vec{x}_{lin1} = (x^{12} \ x^{13} \ x^{14})^T, \quad \vec{x}_{col1} = (x^{21} \ x^{31} \ x^{41})^T$$

$$\begin{aligned}
 \therefore \vec{X}_{s-s} &= \frac{\gamma^4}{(\gamma+1)^2} \vec{\beta} \vec{\beta}^T [(\vec{\beta} \vec{\beta}^T) \cdot \vec{x}] + \frac{\gamma^2}{(\gamma+1)} (\vec{x} \vec{\beta}) \vec{\beta}^T + \frac{\gamma^2}{(\gamma+1)} \vec{\beta} (\vec{\beta}^T \vec{x}) + \vec{x} \\
 &- \frac{\gamma^3}{(\gamma+1)} \vec{\beta} \vec{\beta}^T (\vec{\beta}^T \vec{x}_{lin1}) - \gamma \vec{\beta} \vec{x}_{lin1}^T - \frac{\gamma^3}{(\gamma+1)} \vec{\beta} \vec{\beta}^T (\vec{\beta}^T \vec{x}_{col1}) - \gamma \vec{\beta} \vec{x}_{col1}^T + \gamma^2 \vec{\beta} \vec{\beta}^T x^{11}
 \end{aligned}$$

Note: 所有矢量为列矢量，两矢量乘积若右边转置则结果为二阶张量，
 左边转置则为内积，矢量本身交换结果不变

$$\text{定义 } \vec{x}_1 = \vec{x}_{lin1} + \vec{x}_{col1} = (x_{12} + x_{21} \ x_{31} + x_{13} \ x_{41} + x_{14})^T;$$

$$\begin{aligned}
 \therefore \vec{X}_{s-s} &= \vec{x} + \frac{\gamma^4}{(\gamma+1)^2} \vec{\beta} \vec{\beta}^T [(\vec{\beta} \vec{\beta}^T) \cdot \vec{x}] - \frac{\gamma^3}{(\gamma+1)} \vec{\beta} \vec{\beta}^T (\vec{\beta}^T \vec{x}_1) + \gamma^2 \vec{\beta} \vec{\beta}^T x^{11} \\
 &+ \frac{\gamma^2}{(\gamma+1)} (\vec{x} \vec{\beta}) \vec{\beta}^T + \frac{\gamma^2}{(\gamma+1)} \vec{\beta} (\vec{\beta}^T \vec{x}) - \gamma \vec{\beta} \vec{x}_{lin1}^T - \gamma \vec{x}_{col1} \vec{\beta}^T
 \end{aligned}$$

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$$\begin{aligned} \because \frac{\gamma^2}{(\gamma+1)} (\vec{x}\vec{\beta})\vec{\beta}^T &= \frac{\gamma^2}{(\gamma+1)} \vec{\beta}(\vec{x}\vec{\beta})^T = \frac{\gamma^2}{(\gamma+1)} \vec{\beta}(\vec{\beta}^T \vec{x}^T) \\ \because \frac{\gamma^2}{(\gamma+1)} (\vec{x}\vec{\beta})\vec{\beta}^T + \frac{\gamma^2}{(\gamma+1)} \vec{\beta}(\vec{\beta}^T \vec{x}) &= \frac{\gamma^2}{(\gamma+1)} [\vec{\beta}(\vec{\beta}^T \vec{x}^T) + \vec{\beta}(\vec{\beta}^T \vec{x})] \\ &= \frac{\gamma^2}{(\gamma+1)} \vec{\beta}[\vec{\beta}^T (\vec{x}^T + \vec{x})] \text{ 可证分配律成立} \end{aligned}$$

$$\text{定义 } \vec{x}^S = \vec{x}^T + \vec{x} = \begin{pmatrix} 2x^{22} & x^{23} + x^{32} & x^{42} + x^{24} \\ x^{23} + x^{32} & 2x^{33} & x^{43} + x^{34} \\ x^{24} + x^{42} & x^{43} + x^{34} & 2x^{44} \end{pmatrix}$$

$$\begin{aligned} \text{则 } \vec{X}_{s-s} = \vec{x} + \frac{\gamma^4}{(\gamma+1)^2} \vec{\beta}\vec{\beta}^T [(\vec{\beta}\vec{\beta}^T) \cdot \vec{x}] - \frac{\gamma^3}{(\gamma+1)} \vec{\beta}\vec{\beta}^T (\vec{\beta}^T \vec{x}_1) + \gamma^2 \vec{\beta}\vec{\beta}^T x^{11} \\ + \frac{\gamma^2}{(\gamma+1)} \vec{\beta}(\vec{\beta}^T \vec{x}^S) - \gamma \vec{\beta} \vec{x}_{lin1}^T - \gamma \vec{x}_{col1} \vec{\beta}^T \end{aligned}$$

同样办法可以得到变换后时间-空间项、空间-时间项与变换前的关系：

$$\begin{aligned} X_{t-s}^{\mu\nu} &= \sum_{\lambda} \sum_{\sigma} -g_{\lambda\lambda} g_{\sigma\sigma} \frac{\gamma}{\gamma+1} \beta_{\lambda} [\beta_{\nu} \beta_{\sigma} (\gamma^2 + \delta_{\sigma 1} \gamma) + \delta_{\sigma\nu} (\gamma+1)] x^{\lambda\sigma} \\ &= \sum_{\lambda=2}^4 \sum_{\sigma=2}^4 -\frac{\gamma}{\gamma+1} \beta_{\lambda} [\beta_{\nu} \beta_{\sigma} \gamma^2 + \delta_{\sigma\nu} (\gamma+1)] x^{\lambda\sigma} + \sum_{\sigma=2}^4 \frac{\gamma}{\gamma+1} [\beta_{\nu} \beta_{\sigma} \gamma^2 + \delta_{\sigma\nu} (\gamma+1)] x^{1\sigma} \\ &+ \sum_{\lambda=2}^4 \gamma \beta_{\lambda} (\beta_{\nu} \gamma + \delta_{1\nu}) x^{\lambda 1} - \gamma (\gamma \beta_{\nu} + \delta_{1\nu}) x^{11} \end{aligned}$$

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$$\begin{aligned}
 &= -\frac{\gamma^3}{\gamma+1}\beta_\nu \sum_{\lambda=2}^4 \sum_{\sigma=2}^4 \beta_\lambda \beta_\sigma x^{\lambda\sigma} - \gamma \sum_{\lambda=2}^4 \beta_\lambda x^{\lambda\nu} + \frac{\gamma^3}{\gamma+1}\beta_\nu \sum_{\sigma=2}^4 \beta_\sigma x^{1\sigma} + \gamma x^{1\nu} \\
 &+ \gamma^2 \beta_\nu \sum_{\lambda=2}^4 \beta_\lambda x^{\lambda 1} + \gamma \delta_{1\nu} \sum_{\lambda=2}^4 \beta_\lambda x^{\lambda 1} - \gamma^2 \beta_\nu x^{11} - \delta_{1\nu} \gamma x^{11} \\
 &= -\frac{\gamma^3}{\gamma+1} \vec{\beta} [(\vec{\beta} \vec{\beta}^T) \cdot \vec{x}] - \gamma \vec{x}^T \vec{\beta} + \frac{\gamma^3}{\gamma+1} \vec{\beta} (\vec{\beta} \cdot \vec{x}_{lin1}) + \gamma \vec{x}_{lin1} + \gamma^2 \vec{\beta} (\vec{\beta} \cdot \vec{x}_{col1} - x^{11}) \\
 X_{s-t}^{\mu\nu} &= \sum_{\lambda} \sum_{\sigma} -g_{\lambda\lambda} g_{\sigma\sigma} \frac{\gamma}{\gamma+1} \beta_\sigma [\beta_\mu \beta_\lambda (\gamma^2 + \delta_{\lambda 1} \gamma) + \delta_{\lambda\mu} (\gamma+1)] x^{\lambda\sigma} \\
 &= \sum_{\lambda=2}^4 \sum_{\sigma=2}^4 -\frac{\gamma}{\gamma+1} \beta_\sigma [\beta_\mu \beta_\lambda \gamma^2 + \delta_{\lambda\mu} (\gamma+1)] x^{\lambda\sigma} + \sum_{\sigma=2}^4 \gamma \beta_\sigma (\gamma \beta_\mu + \delta_{1\mu}) x^{1\sigma} \\
 &+ \sum_{\lambda=2}^4 \frac{\gamma}{\gamma+1} [\beta_\mu \beta_\lambda \gamma^2 + \delta_{\lambda\mu} (\gamma+1)] x^{\lambda 1} - \gamma (\gamma \beta_\mu + \delta_{1\mu}) x^{11} \\
 &= -\frac{\gamma^3}{\gamma+1} \beta_\mu \sum_{\lambda=2}^4 \sum_{\sigma=2}^4 \beta_\sigma \beta_\lambda x^{\lambda\sigma} - \gamma \sum_{\sigma=2}^4 \beta_\sigma x^{\mu\sigma} + \gamma^2 \beta_\mu \sum_{\sigma=2}^4 \beta_\sigma x^{1\sigma} \\
 &+ \frac{\gamma^3}{\gamma+1} \beta_\mu \sum_{\lambda=2}^4 \beta_\lambda x^{\lambda 1} + \gamma x^{\mu 1} - \gamma^2 \beta_\mu x^{11} \\
 &= -\frac{\gamma^3}{\gamma+1} \vec{\beta} [(\vec{\beta} \vec{\beta}^T) \cdot \vec{x}] - \gamma \vec{x} \vec{\beta} + \frac{\gamma^3}{\gamma+1} \vec{\beta} (\vec{\beta} \cdot \vec{x}_{col1}) + \gamma \vec{x}_{col1} + \gamma^2 \vec{\beta} (\vec{\beta} \cdot \vec{x}_{lin1} - x^{11})
 \end{aligned}$$

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同理时间-时间项满足关系：

$$\begin{aligned}
 X_{t-t}^{\mu\nu} &= \sum_{\lambda} \sum_{\sigma} g_{\lambda\lambda} g_{\sigma\sigma} \gamma^2 \beta_{\lambda} \beta_{\sigma} x^{\lambda\sigma} \\
 &= \gamma^2 \sum_{\lambda=2}^4 \sum_{\sigma=2}^4 \beta_{\lambda} \beta_{\sigma} x^{\lambda\sigma} - \gamma^2 \sum_{\lambda=2}^4 \beta_{\lambda} x^{\lambda 1} - \gamma^2 \sum_{\sigma=2}^4 \beta_{\sigma} x^{1\sigma} + \gamma^2 x^{11} \\
 &= \gamma^2 [(\vec{\beta} \vec{\beta}^T) \cdot \vec{x}] - \gamma^2 (\vec{\beta} \cdot \vec{x}_{col1}) - \gamma^2 (\vec{\beta} \cdot \vec{x}_{lin1}) + \gamma^2 x^{11}
 \end{aligned}$$

综上，二阶张量Lorentz变换前后各部分满足

$$\begin{aligned}
 \star \vec{X} &= \vec{x} + \frac{\gamma^4}{(\gamma+1)^2} \vec{\beta} \vec{\beta}^T [(\vec{\beta} \vec{\beta}^T) \cdot \vec{x}] - \frac{\gamma^3}{\gamma+1} \vec{\beta} \vec{\beta}^T (\vec{\beta} \cdot \vec{x}_1) + \gamma^2 \vec{\beta} \vec{\beta}^T x^{11} + \frac{\gamma^2}{\gamma+1} (\vec{\beta} \vec{\beta}^T) \vec{x}^S - \gamma \vec{\beta} \vec{x}_{lin1}^T - \gamma \vec{x}_{col1} \vec{\beta}^T \\
 \vec{X}^T &= \vec{x}^T + \frac{\gamma^4}{(\gamma+1)^2} \vec{\beta} \vec{\beta}^T [(\vec{\beta} \vec{\beta}^T) \cdot \vec{x}^T] - \frac{\gamma^3}{\gamma+1} \vec{\beta} \vec{\beta}^T (\vec{\beta} \cdot \vec{x}_1) + \gamma^2 \vec{\beta} \vec{\beta}^T x^{11} + \frac{\gamma^2}{\gamma+1} \vec{x}^S (\vec{\beta} \vec{\beta}^T) - \gamma \vec{\beta} \vec{x}_{col1}^T - \gamma \vec{x}_{lin1} \vec{\beta}^T \\
 \left\{ \begin{aligned}
 \vec{X}^S &= \vec{x}^S + \frac{\gamma^4}{(\gamma+1)^2} \vec{\beta} \vec{\beta}^T [(\vec{\beta} \vec{\beta}^T) \cdot \vec{x}^S] - \frac{2\gamma^3}{\gamma+1} \vec{\beta} \vec{\beta}^T (\vec{\beta} \cdot \vec{x}_1) + 2\gamma^2 \vec{\beta} \vec{\beta}^T x^{11} + \frac{\gamma^2}{\gamma+1} [\vec{\beta} \vec{\beta}^T, \vec{x}^S]_+ - \gamma (\vec{\beta} \vec{x}_1^T + \vec{x}_1 \vec{\beta}^T) \\
 \vec{X}_1 &= -\frac{2\gamma^3}{\gamma+1} \vec{\beta} [(\vec{\beta} \vec{\beta}^T) \cdot \vec{x}] - \gamma \vec{x}^S \vec{\beta} + \frac{\gamma^3}{\gamma+1} \vec{\beta} (\vec{\beta} \cdot \vec{x}_1) + \gamma \vec{x}_1 + \gamma^2 \vec{\beta} (\vec{\beta} \cdot \vec{x}_1 - 2x^{11}) \\
 X^{11} &= \gamma^2 [(\vec{\beta} \vec{\beta}^T) \cdot \vec{x}] - \gamma^2 (\vec{\beta} \cdot \vec{x}_1) + \gamma^2 x^{11} \\
 \vec{X}_{lin1} &= -\frac{\gamma^3}{\gamma+1} \vec{\beta} [(\vec{\beta} \vec{\beta}^T) \cdot \vec{x}] - \gamma \vec{x}^T \vec{\beta} + \frac{\gamma^3}{\gamma+1} \vec{\beta} (\vec{\beta} \cdot \vec{x}_{lin1}) + \gamma \vec{x}_{lin1} + \gamma^2 \vec{\beta} (\vec{\beta} \cdot \vec{x}_{col1} - x^{11}) \\
 \vec{X}_{col1} &= -\frac{\gamma^3}{\gamma+1} \vec{\beta} [(\vec{\beta} \vec{\beta}^T) \cdot \vec{x}] - \gamma \vec{x} \vec{\beta} + \frac{\gamma^3}{\gamma+1} \vec{\beta} (\vec{\beta} \cdot \vec{x}_{col1}) + \gamma \vec{x}_{col1} + \gamma^2 \vec{\beta} (\vec{\beta} \cdot \vec{x}_{lin1} - x^{11})
 \end{aligned} \right.
 \end{aligned}$$

一般二阶张量Thomas动力学

为探讨一般二阶张量Thomas动力学，首先给出含 γ 系数求导结果：

$$\left(\frac{\dot{\gamma}^4}{(\gamma+1)^2}\right) = \frac{(2\gamma+4)\gamma^6(\vec{\beta} \cdot \dot{\vec{\beta}})}{(\gamma+1)^3} \quad \left(\frac{\dot{\gamma}^3}{\gamma+1}\right) = \frac{(2\gamma+3)\gamma^5(\vec{\beta} \cdot \dot{\vec{\beta}})}{(\gamma+1)^2} \quad \left(\frac{\dot{\gamma}^2}{\gamma+1}\right) = \frac{(\gamma+2)\gamma^4(\vec{\beta} \cdot \dot{\vec{\beta}})}{(\gamma+1)^2}$$

$$\begin{aligned} \text{则 } \dot{\vec{x}}^{\mu\nu}|_{X^{\lambda\sigma}} &= \left(\frac{\dot{\gamma}^4}{(\gamma+1)^2}\right) \vec{\beta} \vec{\beta}^T [(\vec{\beta} \vec{\beta}^T) \cdot \vec{X}] + \frac{\gamma^4}{(\gamma+1)^2} (\vec{\beta} \dot{\vec{\beta}}^T) [(\vec{\beta} \vec{\beta}^T) \cdot \vec{X}] + \frac{\gamma^4}{(\gamma+1)^2} \vec{\beta} \vec{\beta}^T [(\vec{\beta} \dot{\vec{\beta}}^T) \cdot \vec{X}] \\ &+ \left(\frac{\dot{\gamma}^3}{\gamma+1}\right) \vec{\beta} \vec{\beta}^T (\vec{\beta} \cdot \vec{X}_1) + \frac{\gamma^3}{\gamma+1} (\vec{\beta} \dot{\vec{\beta}}^T) (\vec{\beta} \cdot \vec{X}_1) + \frac{\gamma^3}{\gamma+1} \vec{\beta} \vec{\beta}^T (\dot{\vec{\beta}} \cdot \vec{X}_1) \\ &+ (\dot{\gamma}^2) \vec{\beta} \vec{\beta}^T \vec{X}^{11} + \gamma^2 (\vec{\beta} \dot{\vec{\beta}}^T) \vec{X}^{11} + \left(\frac{\dot{\gamma}^2}{\gamma+1}\right) (\vec{\beta} \vec{\beta}^T) \vec{X}^S + \frac{\gamma^2}{\gamma+1} (\vec{\beta} \dot{\vec{\beta}}^T) \vec{X}^S + \dot{\gamma} \vec{\beta} \vec{X}_1^T + \gamma \dot{\vec{\beta}} \vec{X}_1^T \\ &= \left\{ \frac{(2\gamma+4)\gamma^6(\vec{\beta} \cdot \dot{\vec{\beta}})}{(\gamma+1)^3} \vec{\beta} \vec{\beta}^T (\vec{\beta} \vec{\beta}^T)^{ip} + \frac{\gamma^4}{(\gamma+1)^2} (\vec{\beta} \dot{\vec{\beta}}^T) (\vec{\beta} \vec{\beta}^T)^{ip} + \frac{\gamma^4}{(\gamma+1)^2} \vec{\beta} \vec{\beta}^T (\vec{\beta} \dot{\vec{\beta}}^T)^{ip} \right\} \vec{X} \\ &+ \left[\frac{(2\gamma+3)\gamma^5(\vec{\beta} \cdot \dot{\vec{\beta}})}{(\gamma+1)^2} (\vec{\beta} \vec{\beta}^T) \vec{\beta}^T + \frac{\gamma^3}{\gamma+1} (\vec{\beta} \dot{\vec{\beta}}^T) \vec{\beta}^T + \frac{\gamma^3}{\gamma+1} (\vec{\beta} \vec{\beta}^T) \dot{\vec{\beta}}^T \right] \vec{X}_1 + \left[2\gamma^4 (\vec{\beta} \cdot \dot{\vec{\beta}}) \vec{\beta} \vec{\beta}^T + \gamma^2 (\vec{\beta} \dot{\vec{\beta}}^T) \right] \vec{X}^{11} \\ &+ \left[\frac{(\gamma+2)\gamma^4(\vec{\beta} \cdot \dot{\vec{\beta}})}{(\gamma+1)^2} (\vec{\beta} \vec{\beta}^T) + \frac{\gamma^2}{\gamma+1} (\vec{\beta} \dot{\vec{\beta}}^T) \right] \vec{X}^S + \left[\gamma^3 (\vec{\beta} \cdot \dot{\vec{\beta}}) \vec{\beta} + \gamma \dot{\vec{\beta}} \right] \vec{X}_{lin1}^T + \left[\gamma^3 (\vec{\beta} \cdot \dot{\vec{\beta}}) \vec{X}_{col1} \vec{\beta}^T + \gamma \vec{X}_{col1} \dot{\vec{\beta}}^T \right] \end{aligned}$$

一般二阶张量Thomas动力学

其中 $(\vec{\beta}\vec{\beta}^T)^{ip} \vec{X} = (\vec{\beta}\vec{\beta}^T) \cdot \vec{X} = \text{tr}[(\vec{\beta}\vec{\beta}^T)^T \vec{X}]$

$$\begin{aligned}
 \text{展开 } \dot{\vec{x}}^{\mu\nu}|_{x^{\lambda\sigma}} = & \left\{ \frac{(2\gamma+4)\gamma^6(\vec{\beta} \cdot \dot{\vec{\beta}})}{(\gamma+1)^3} \vec{\beta}\vec{\beta}^T (\vec{\beta}\vec{\beta}^T)^{ip} + \frac{\gamma^4}{(\gamma+1)^2} (\vec{\beta}\dot{\vec{\beta}}^T) (\vec{\beta}\vec{\beta}^T)^{ip} + \frac{\gamma^4}{(\gamma+1)^2} \vec{\beta}\vec{\beta}^T (\vec{\beta}\dot{\vec{\beta}}^T)^{ip} \right\} \\
 & \left\{ \vec{x} + \frac{\gamma^4}{(\gamma+1)^2} \vec{\beta}\vec{\beta}^T [(\vec{\beta}\vec{\beta}^T) \cdot \vec{x}] - \frac{\gamma^3}{\gamma+1} \vec{\beta}\vec{\beta}^T (\vec{\beta} \cdot \vec{x}_1) + \gamma^2 \vec{\beta}\vec{\beta}^T x^{11} + \frac{\gamma^2}{\gamma+1} (\vec{\beta}\vec{\beta}^T) \vec{x}^s - \gamma \vec{\beta} \vec{x}_{lin1}^T - \gamma \vec{x}_{col1} \vec{\beta}^T \right\} \\
 & + \left[\frac{(2\gamma+3)\gamma^5(\vec{\beta} \cdot \dot{\vec{\beta}})}{(\gamma+1)^2} (\vec{\beta}\vec{\beta}^T) \vec{\beta}^T + \frac{\gamma^3}{\gamma+1} (\vec{\beta}\dot{\vec{\beta}}^T) \vec{\beta}^T + \frac{\gamma^3}{\gamma+1} (\vec{\beta}\vec{\beta}^T) \dot{\vec{\beta}}^T \right] \\
 & \left\{ -\frac{2\gamma^3}{\gamma+1} \vec{\beta} [(\vec{\beta}\vec{\beta}^T) \cdot \vec{x}] - \gamma \vec{x}^s \vec{\beta} + \frac{\gamma^3}{\gamma+1} \vec{\beta} (\vec{\beta} \cdot \vec{x}_1) + \gamma \vec{x}_1 + \gamma^2 \vec{\beta} (\vec{\beta} \cdot \vec{x}_1 - 2x^{11}) \right\} \\
 & + \left[2\gamma^4 (\vec{\beta} \cdot \dot{\vec{\beta}}) \vec{\beta}\vec{\beta}^T + \gamma^2 (\vec{\beta}\dot{\vec{\beta}}^T) \right] \left\{ \gamma^2 [(\vec{\beta}\vec{\beta}^T) \cdot \vec{x}] - \gamma^2 (\vec{\beta} \cdot \vec{x}_1) + \gamma^2 x^{11} \right\} \\
 & + \left[\frac{(\gamma+2)\gamma^4(\vec{\beta} \cdot \dot{\vec{\beta}})}{(\gamma+1)^2} (\vec{\beta}\vec{\beta}^T) + \frac{\gamma^2}{\gamma+1} (\vec{\beta}\dot{\vec{\beta}}^T) \right] \\
 & \left\{ \vec{x}^s + \frac{\gamma^4}{(\gamma+1)^2} \vec{\beta}\vec{\beta}^T [(\vec{\beta}\vec{\beta}^T) \cdot \vec{x}^s] - \frac{2\gamma^3}{\gamma+1} \vec{\beta}\vec{\beta}^T (\vec{\beta} \cdot \vec{x}_1) + 2\gamma^2 \vec{\beta}\vec{\beta}^T x^{11} + \frac{\gamma^2}{\gamma+1} [\vec{\beta}\vec{\beta}^T, \vec{x}^s]_+ - \gamma (\vec{\beta} \vec{x}_1^T + \vec{x}_1 \vec{\beta}^T) \right\} \\
 & + \left[\gamma^3 (\vec{\beta} \cdot \dot{\vec{\beta}}) \vec{\beta} + \gamma \dot{\vec{\beta}} \right] \left\{ -\frac{\gamma^3}{\gamma+1} \vec{\beta}^T [(\vec{\beta}\vec{\beta}^T) \cdot \vec{x}] - \gamma \vec{\beta}^T \vec{x} + \frac{\gamma^3}{\gamma+1} \vec{\beta}^T (\vec{\beta} \cdot \vec{x}_{lin1}) + \gamma \vec{x}_{lin1}^T + \gamma^2 \vec{\beta}^T (\vec{\beta} \cdot \vec{x}_{col1} - x^{11}) \right\} \\
 & + \gamma^3 (\vec{\beta} \cdot \dot{\vec{\beta}}) \left[-\frac{\gamma^3}{\gamma+1} \vec{\beta} [(\vec{\beta}\vec{\beta}^T) \cdot \vec{x}] - \gamma \vec{x} \vec{\beta} + \frac{\gamma^3}{\gamma+1} \vec{\beta} (\vec{\beta} \cdot \vec{x}_{col1}) + \gamma \vec{x}_{col1} + \gamma^2 \vec{\beta} (\vec{\beta} \cdot \vec{x}_{lin1} - x^{11}) \right] \vec{\beta}^T \\
 & + \gamma \left[-\frac{\gamma^3}{\gamma+1} \vec{\beta} [(\vec{\beta}\vec{\beta}^T) \cdot \vec{x}] - \gamma \vec{x} \vec{\beta} + \frac{\gamma^3}{\gamma+1} \vec{\beta} (\vec{\beta} \cdot \vec{x}_{col1}) + \gamma \vec{x}_{col1} + \gamma^2 \vec{\beta} (\vec{\beta} \cdot \vec{x}_{lin1} - x^{11}) \right] \dot{\vec{\beta}}^T
 \end{aligned}$$

一般二阶张量Thomas动力学

化简后 $\vec{x}$ 演化方程如下：

$$\begin{aligned}
 & \left. \dot{\vec{x}}^{\mu\nu} \right|_{X^{\lambda\sigma}} \\
 = & \frac{\gamma^5}{(\gamma+1)^2} [(\vec{\beta}\vec{\beta}^T) \cdot \vec{x}] (\dot{\vec{\beta}}\vec{\beta}^T) - \frac{\gamma^5}{(\gamma+1)^2} [(\vec{\beta}\vec{\beta}^T) \cdot \vec{x}] (\vec{\beta}\dot{\vec{\beta}}^T) + \gamma^4 (\vec{\beta} \cdot \dot{\vec{\beta}}) (\vec{\beta}\vec{\beta}^T) \vec{x}^T - \gamma^4 (\vec{\beta} \cdot \dot{\vec{\beta}}) \vec{x} (\vec{\beta}\vec{\beta}^T) \\
 & + \frac{\gamma^3}{\gamma+1} (\dot{\vec{\beta}}\vec{\beta}^T) \vec{x}^T - \frac{\gamma^3}{\gamma+1} \vec{x} (\dot{\vec{\beta}}\vec{\beta}^T) \\
 & + \frac{\gamma^4}{\gamma+1} (\vec{\beta} \cdot \vec{x}_1) (\vec{\beta}\dot{\vec{\beta}}^T) - \frac{\gamma^4}{\gamma+1} (\vec{\beta} \cdot \vec{x}_1) (\dot{\vec{\beta}}\vec{\beta}^T) + \frac{\gamma^3}{\gamma+1} (\vec{\beta} \cdot \vec{x}_{col1}) (\dot{\vec{\beta}}\vec{\beta}^T) - \frac{\gamma^3}{\gamma+1} (\vec{\beta} \cdot \vec{x}_{col1}) (\vec{\beta}\dot{\vec{\beta}}^T) \\
 & + \gamma^2 (\gamma-1) x^{11} (\dot{\vec{\beta}}\vec{\beta}^T) - \gamma^2 (\gamma-1) x^{11} (\vec{\beta}\dot{\vec{\beta}}^T) + \gamma^2 (\vec{x}_{col1} \dot{\vec{\beta}}^T) - \gamma^2 (\dot{\vec{\beta}} \vec{x}_{col1}^T) + \gamma^4 (\vec{\beta} \cdot \dot{\vec{\beta}}) (\vec{x}_{col1} \vec{\beta}^T) - \gamma^4 (\vec{\beta} \cdot \dot{\vec{\beta}}) (\vec{\beta} \vec{x}_{col1}^T) \\
 & + \frac{\gamma^2}{\gamma+1} (\dot{\vec{\beta}}\vec{\beta}^T) \vec{x}^T - \frac{\gamma^2}{\gamma+1} (\vec{\beta}\dot{\vec{\beta}}^T) \vec{x} + \frac{\gamma^2}{\gamma+1} (\dot{\vec{\beta}}\vec{\beta}^T) \vec{x} - \frac{\gamma^2}{\gamma+1} \vec{x} (\dot{\vec{\beta}}\vec{\beta}^T) + \frac{\gamma^3 (\vec{\beta} \cdot \dot{\vec{\beta}})}{\gamma+1} (\vec{\beta} \vec{x}_1^T) + \gamma (\dot{\vec{\beta}} \vec{x}_1^T) \quad \dots\dots ⑦
 \end{aligned}$$

化简过程详见末尾附录。上式中灰底色项都可以被表示为含 γ 系数与含 $\vec{\beta}$ 、 $\vec{x}$ 、 $\vec{x}_{col1}$ 、 $\vec{x}_1$ 、 x_{11} 矩阵减去自身转置的乘积，绿底色项与矢量Thomas动力学表达式中除去Thomas进动项以外的项形式一致。比较特殊的是黄底色项(含 γ 系数与 $(\dot{\vec{\beta}}\vec{\beta}^T) \vec{x}^T$ 减去其分别转置的乘积)以及蓝底色项(含 γ 系数与 $[(\dot{\vec{\beta}}\vec{\beta}^T), \vec{x}]_-$ 的乘积)。

若二阶张量是反对称的，则一般二阶张量Thomas动力学结果与矢量分解的二阶反对称张量(⑤式)结果一致。这很容易看出来： $\vec{x}_{col1}$ 、 $\vec{x}$ 项的含 γ 系数是对得上的，并且矢量外

一般二阶张量Thomas动力学

积的表达式与自身转置相减的形式一致。

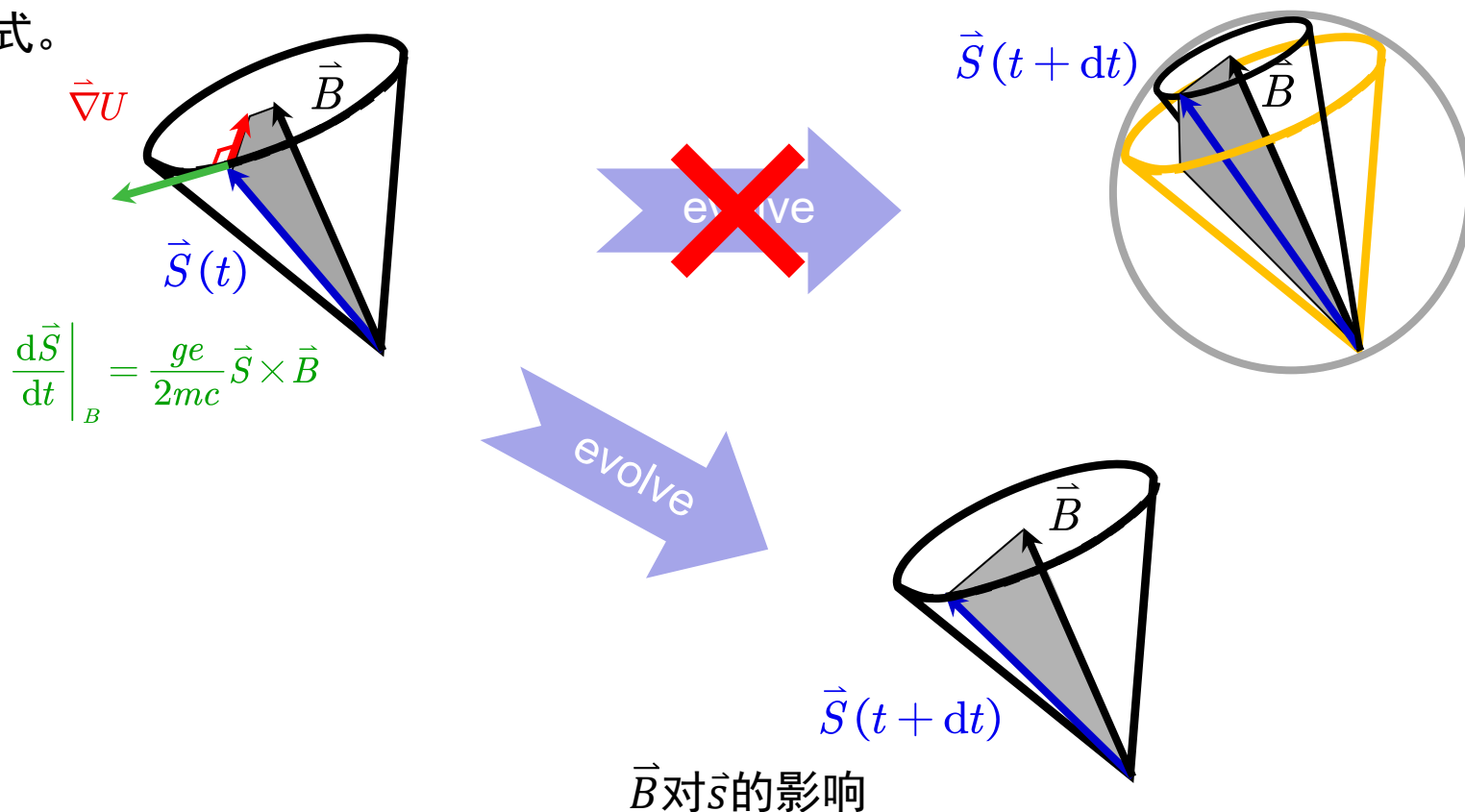
$$\begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{pmatrix} \times \begin{pmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \end{pmatrix} = \begin{pmatrix} \alpha_2 \beta_3 - \beta_2 \alpha_3 \\ \alpha_1 \beta_3 - \beta_1 \alpha_3 \\ \alpha_1 \beta_2 - \beta_1 \alpha_2 \end{pmatrix} \quad \kappa(\text{反对称}) = \begin{pmatrix} 0 & -\alpha_1 & -\alpha_2 & -\alpha_3 \\ \alpha_1 & 0 & -\beta_3 & \beta_2 \\ \alpha_2 & \beta_3 & 0 & -\beta_1 \\ \alpha_3 & -\beta_2 & \beta_1 & 0 \end{pmatrix}$$

$$\omega - \omega^T = \begin{pmatrix} 0 & \omega_{21} - \omega_{12} & \omega_{13} - \omega_{31} & \omega_{14} - \omega_{41} \\ \omega_{21} - \omega_{12} & 0 & \omega_{23} - \omega_{32} & \omega_{24} - \omega_{42} \\ \omega_{31} - \omega_{13} & \omega_{32} - \omega_{23} & 0 & \omega_{34} - \omega_{43} \\ \omega_{41} - \omega_{14} & \omega_{42} - \omega_{24} & \omega_{43} - \omega_{34} & 0 \end{pmatrix}$$

一般二阶张量Thomas动力学的表达式不如矢量Thomas动力学简洁，原因是⑦式并不是最终的Thomas动力学表达式。最终表达式应当与矢量分解的二阶张量Thomas动力学结果类似，即矢量/张量与内积的交换相减。但也不能期望二阶张量Thomas动力学结果与矢量Thomas动力学结果存在显性规律；实际上，追求这种最终解只有结合二阶张量的Lorentz变换探究二阶张量的一阶和高阶Thomas运动学才是有意义的。现在应当意识到，张量对象的阶数与Thomas效应的阶数是研究的两个维度；Thomas运动学与Thomas动力学是另一个角度的两个研究维度。

自旋-进动耦合Hamiltonian

将Thomas动力学应用量子力学或量子场论中必须明确 ω_T 对Hamiltonian及Lagrangian的作用，首先研究对Hamiltonian的作用。在 $\vec{S}$ 的演化方程中存在 $\vec{\omega}_T$ 项是值得讨论的，如果 $\vec{\omega}_T$ 仅作为与 $\vec{S}$ 并列的运动模式，两个运动模式间只能运动上合成，不能在动力学上相互影响。Thomas动力学的结果表明 $\vec{\omega}_T$ 影响 $\vec{S}$ 演化，原因是 $\vec{\omega}_T$ 是参考系转动导致的更底层的运动模式。



自旋-进动耦合Hamiltonian

从能量 $U = -\frac{ge}{2mc} \vec{S} \cdot \vec{B}$ 可以给出 $\frac{d\vec{S}}{dt} = \frac{ge}{2mc} \vec{S} \times \vec{B}$ 的演化关系式。取外加磁场方向为Z轴方

向, 对于固定的固有偶极(如自旋), 其偶极角动量中的 $\vec{r}$ 不随时间演化, 即 $\frac{d\vec{r}}{dt} = 0$, 故球坐标

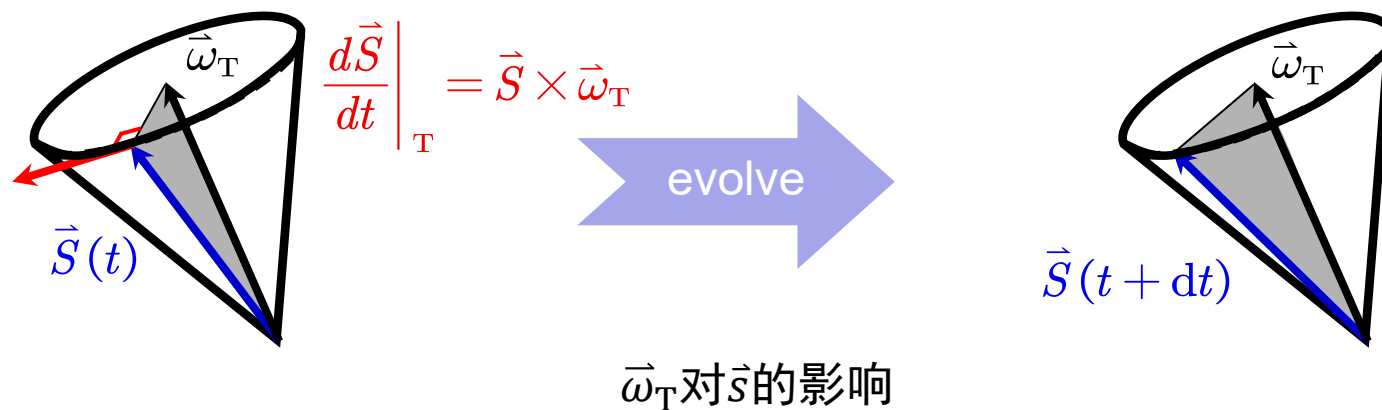
系下演化 “推动力” 满足

$$\frac{d\vec{S}}{dt} = \frac{d(\vec{r} \times m\vec{v})}{dt} = \vec{r} \times \frac{d\vec{p}}{dt} = \vec{r} \times \vec{\nabla}U = \vec{r} \times \frac{1}{r} \frac{\partial U}{\partial \theta} \vec{e}_\theta = \frac{\partial U}{\partial \theta} \vec{e}_\varphi = \frac{ge}{2mc} |\vec{S}| |\vec{B}| \sin \theta \vec{e}_\varphi = \frac{ge}{2mc} \vec{S} \times \vec{B}$$

因此, 固定磁矩在外加磁场下发生进动(Larmor进动), 而非直接沿能量梯度方向演化。

这与Thomas进动动力学一致, 因此Thomas进动与Larmor进动对Hamiltonian作用形式一致

$$U_T = -\vec{\omega}_T \cdot \vec{S} \quad \text{or} \quad \hat{H}_T = -\vec{\omega}_T \cdot \hat{\vec{S}}$$



自旋-进动耦合Hamiltonian

为什么要考虑Thomas动力学(自旋-进动耦合)? 它与Dirac方程经Foldy–Wouthuysen变换解耦的二分量Hamiltonian是什么关系?

连续帧动力学必须有意义

$\vec{S}(\tau)$ 全部时刻都有意义

$\vec{S}(\tau)$ 展开式各项Lorentz线性变换

$$\hat{L}\vec{S}(\tau) = \sum_{i=0}^{\infty} \hat{L} \left[\frac{\vec{S}^{(i)}(0)}{i!} \tau^i \right] \quad \blacktriangle$$

$\vec{S}$ 对 τ 的任意阶导Lorentz线性变换

$\dot{\vec{S}}' \rightarrow \dot{\vec{S}}$ 作为空间矢量参与Lorentz变换

多出的自旋-进动耦合能应考虑进 $\dot{\vec{S}}$
使 $\dot{\vec{S}}' = (\dot{\vec{S}} + \Gamma \vec{\omega}_T \times \vec{S}) = \hat{L} \dot{\vec{S}}'$

单电子Dirac方程 - 四分量

自旋偶极 - 二分量

.....

自旋-自旋耦合
自旋-电磁耦合

$\dot{\vec{S}}$ +

自旋-进动耦合
 $\Gamma \vec{\omega}_T \times \vec{S}$

自旋-进动耦合项是由连续帧动力学引入的对(单帧)Dirac方程解耦项的补充; 是运动(时空)性质对电子(场)性质的补充。

$\blacktriangle$ 核心是: 有物理意义的矢量必循线性Lorentz变换。

自旋-进动耦合Hamiltonian

对于自旋偶极4*4张量模型，自旋磁偶极 $\vec{S}$ 是偶极张量的空间分量， $c\vec{d}$ 是偶极张量的时间分量。除Larmor进动和Thomas进动外， $\vec{S}$ 的演化还会受到 $c\vec{d}$ 影响：

$$\begin{aligned}\mu \frac{d\vec{S}}{dt} &= \gamma^2 \left[\vec{M} \times \dot{\vec{\beta}} \right] + \frac{\gamma^3}{\gamma + 1} \left\{ \vec{M} \times \left[\vec{\beta} \times \left(\vec{\beta} \times \dot{\vec{\beta}} \right) \right] \right\} \\ &= \vec{M} \times \vec{\omega}_{\text{SX}} + \vec{M} \times \vec{\omega}_{\text{LX}} = \vec{M} \times \vec{\omega}_{\text{X}}\end{aligned}$$

则球坐标系下演化“推动力”方程相应添加：

$$\begin{aligned}\mu \left. \frac{d\vec{S}}{dt} \right|_{\text{X}} &= \frac{\partial U_{\text{X}}}{\partial \theta} \vec{e}_{\varphi} = c\vec{D} \times \vec{\omega}_{\text{X}} = |c\vec{D}| |\vec{\omega}_{\text{X}}| \sin\theta \\ \therefore U_{\text{X}} &= -\frac{c}{\mu} \vec{D} \cdot \vec{\omega}_{\text{X}} \quad \text{or} \quad \hat{H}_{\text{X}} = -\frac{c}{\mu} \vec{\omega}_{\text{X}} \cdot \hat{\vec{D}}\end{aligned}$$

值得注意的是，尽管自旋电偶极“不满足演化不变性”，但并不改变自旋磁偶极演化的简洁结果，推导见上页。且电子静止系 $\vec{d} = \frac{1}{2}(\vec{s} \times \vec{\beta}) = 0$ ，所以在静止系自旋偶极张量中并不存在 U_{X} 修正项。

拟向变换自旋赝偶极矢量

(Imitative Transfer Pseudodipole Vector, ITPDV)

自旋偶极的定义与性质与Dirac方程和Thomas动力学关系密切。电子*Mag* SOC固有磁矩是广义坐标Dirac方程正负能态解耦产生的能量附加项，决定自旋的两大特点：

- 在不考虑QED修正的情况下自旋Landé因子 $g_s = 2$ ；
- 自旋角动量在磁场方向的投影大小 $\hbar/2$ 是一个相对论标量，不随参考系变换而变化。

经典的Thomas动力学之所以考虑Thomas进动对动力学演化的影响是因为要建立任意惯性系动力学与静止系动力学的联系。例如下文提到的T-BMT方程认为只在静止系成立

$U = -\frac{g_s e}{2mc} \vec{S} \cdot \vec{B}$ ，必须将4-vector从惯性系的向静止系正反两次变换从而建立与静止系动力学的联系并同时引入Thomas进动^[1]，这与Dirac方程矛盾。

在P32对Thomas动力学进行新的诠释，即连续帧下4-vector必须遵循特定的变换方式(如Lorentz变换或下文的拟向变换)。此诠释下静止系与其他惯性系平等，考虑Thomas进动不是为了联系静止系而是为了构造满足特定变换方式的新4-vector。此诠释下Thomas动力学是非变系效应，与Thomas运动学一致。

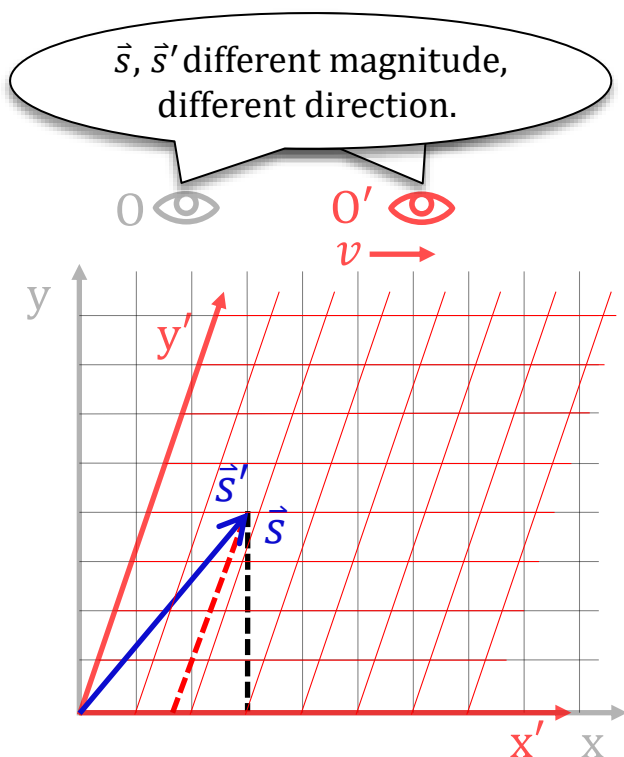
构造自旋偶极经典思路是自旋磁偶极4-vector(T-BMT方程)^[1]，即自旋磁偶极的三个

[1] Jackson, John David. "Classical electrodynamics." (1999): 561-565.

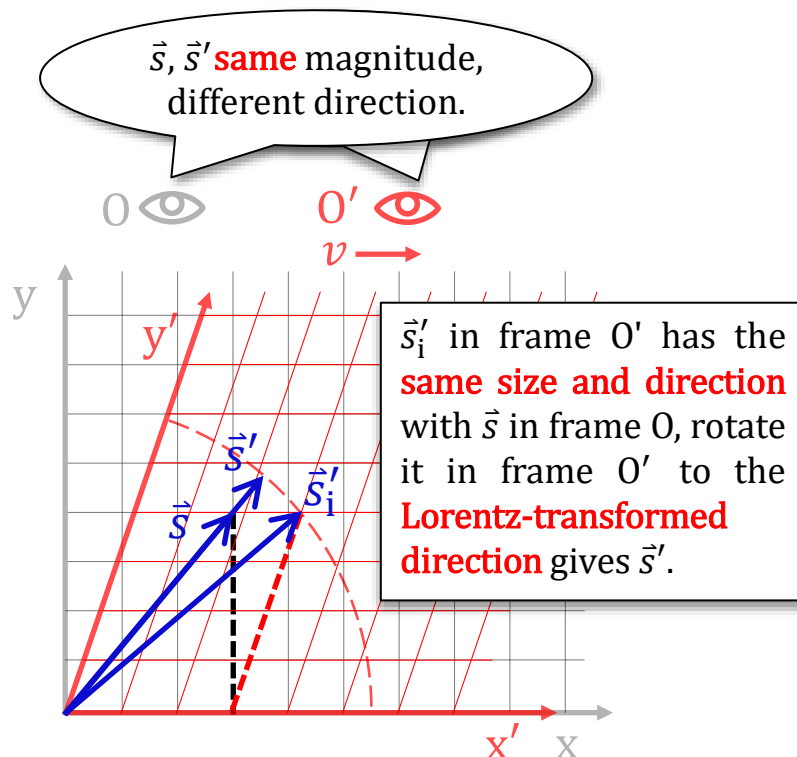
拟向变换自旋赝偶极矢量

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空间分量和一个时间分量。Jackson建议用静止系自旋角动量时间分量为零导出 $U_\alpha S^\alpha = 0$ 的结论。我们继承这一思路，但注意到：Dirac方程F-W变换结果表明自旋角动量大小及其z分量大小是相对论标量，同时角动量确是三维矢量。应当考虑参考系变换中**方向**的变化，即自旋角动量方向按“老中新”坐标轴方向变换，但变换前后自旋角动量大小不变。据此构造新的参考系变换规则——拟向变换(Imitative transformation)，满足如下条件：



Lorentz transformation



Imitative transformation

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在此设定下，y方向自旋本征态经过L.T.和I.T.前后分别满足如下关系：

$$\begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} \xrightarrow{\text{L.T.}} \begin{pmatrix} -\gamma\beta_2 \\ \frac{(\gamma-1)\beta_1\beta_2}{\beta^2} \\ 1 + \frac{(\gamma-1)\beta_2^2}{\beta^2} \\ \frac{(\gamma-1)\beta_2\beta_3}{\beta^2} \end{pmatrix} \quad \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} \xrightarrow{\text{I.T.}} \begin{pmatrix} 0 \\ \frac{(\gamma-1)\beta_1\beta_2}{\beta^2}\zeta_2 \\ \left[1 + \frac{(\gamma-1)\beta_2^2}{\beta^2}\right]\zeta_2 \\ \frac{(\gamma-1)\beta_2\beta_3}{\beta^2}\zeta_2 \end{pmatrix}$$

根据x、y、z方向的对称性，定义I.T.操作矩阵为 L_I ：

$$L_I = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \left[1 + \frac{(\gamma-1)\beta_1^2}{\beta^2}\right]\zeta_1 & \frac{(\gamma-1)\beta_1\beta_2}{\beta^2}\zeta_2 & \frac{(\gamma-1)\beta_1\beta_3}{\beta^2}\zeta_3 \\ 0 & \frac{(\gamma-1)\beta_1\beta_2}{\beta^2}\zeta_1 & \left[1 + \frac{(\gamma-1)\beta_2^2}{\beta^2}\right]\zeta_2 & \frac{(\gamma-1)\beta_2\beta_3}{\beta^2}\zeta_3 \\ 0 & \frac{(\gamma-1)\beta_1\beta_3}{\beta^2}\zeta_1 & \frac{(\gamma-1)\beta_2\beta_3}{\beta^2}\zeta_2 & \left[1 + \frac{(\gamma-1)\beta_3^2}{\beta^2}\right]\zeta_3 \end{pmatrix}$$

满足：

$$[(\gamma-1)\beta_1\beta_2]^2 + [\beta^2 + (\gamma-1)\beta_2^2]^2 + [(\gamma-1)\beta_2\beta_3]^2 = \frac{1}{\kappa_2^2} \quad \text{其中 } \kappa_2^2 = \frac{\zeta_2^2}{\beta^4}$$

$$\therefore \beta^2 = \frac{\gamma^2 - 1}{\gamma^2} = (\gamma - 1) \frac{\gamma + 1}{\gamma^2}$$

$$\therefore \beta_1^2\beta_2^2 + \left[\frac{\gamma+1}{\gamma^2} + \beta_2^2\right]^2 + \beta_2^2\beta_3^2 = \frac{1}{(\gamma-1)^2\kappa_2^2}$$

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$$\therefore \beta_1^2 \beta_2^2 + \frac{(\gamma+1)^2}{\gamma^4} + 2\frac{\gamma+1}{\gamma^2} \beta_2^2 + \beta_2^4 + \beta_2^2 \beta_3^2 = \frac{1}{(\gamma-1)^2 \kappa_2^2}$$

$$\text{又} \because \beta_3^2 = \beta^2 - \beta_1^2 - \beta_2^2$$

$$\therefore \beta_1^2 \beta_2^2 + \frac{(\gamma+1)^2}{\gamma^4} + 2\frac{\gamma+1}{\gamma^2} \beta_2^2 + \beta_2^4 + \beta_2^2 (\beta^2 - \beta_1^2 - \beta_2^2) = \frac{1}{(\gamma-1)^2 \kappa_2^2}$$

$$\therefore \frac{(\gamma+1)^2}{\gamma^4} + 2\frac{\gamma+1}{\gamma^2} \beta_2^2 + \beta_2^2 \beta^2 = \frac{1}{(\gamma-1)^2 \kappa_2^2}$$

$$\therefore \frac{(\gamma+1)^2}{\gamma^4} + 2\frac{\gamma+1}{\gamma^2} \beta_2^2 + \frac{\gamma^2-1}{\gamma^2} \beta_2^2 = \frac{1}{(\gamma-1)^2 \kappa_2^2}$$

$$\therefore \frac{(\gamma+1)^2}{\gamma^4} + \frac{(\gamma+1)^2}{\gamma^2} \beta_2^2 = \frac{1}{(\gamma-1)^2 \kappa_2^2}$$

$$\therefore \left(\frac{\gamma}{\gamma^2-1} \right)^2 \frac{1}{\frac{1}{\gamma^2} + \beta_2^2} = \kappa_2^2 = \frac{\zeta_2^2}{\beta^4} = \left(\frac{\gamma^2}{\gamma^2-1} \right)^2 \zeta_2^2$$

$$\therefore \zeta_2 = \frac{1}{\sqrt{1 + \beta_2^2 \gamma^2}}$$

故缩放系数 $\zeta_2 = (1 + \alpha_2^2)^{-1/2}$ ，其中定义补偿 $\alpha_2 = \gamma \beta_2$ ，意义是y方向时间膨胀补偿；

此结果也可在“Lorentz变换前后矢量四维范数不变”前提下规定时间分量也不变直接给出， ζ_2 表达式分母的加号源于Minkowski度规。与Jackson的处理不同，Dirac方程建议任意参考系自旋角动量时间分量为一非零常数：

$$\gamma^0 = I_4, \langle \sigma_t \rangle = \langle \psi | \frac{\hbar}{2} I_4 | \psi \rangle = \frac{\hbar}{2}$$

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在此设定下，y方向自旋本征态经过L.T.和I.T.前后分别满足如下关系：

$$\begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix} \xrightarrow{\text{L.T.}} \begin{pmatrix} \gamma - \gamma\beta_2 \\ -\gamma\beta_1 + \frac{(\gamma-1)\beta_1\beta_2}{\beta^2} \\ -\gamma\beta_1 + 1 + \frac{(\gamma-1)\beta_2^2}{\beta^2} \\ -\gamma\beta_1 + \frac{(\gamma-1)\beta_2\beta_3}{\beta^2} \end{pmatrix} \quad \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix} \xrightarrow{\text{I.T.}} \begin{pmatrix} 1 \\ \left[-\gamma\beta_1 + \frac{(\gamma-1)\beta_1\beta_2}{\beta^2} \right] \zeta_2 \\ \left[-\gamma\beta_2 + 1 + \frac{(\gamma-1)\beta_2^2}{\beta^2} \right] \zeta_2 \\ \left[-\gamma\beta_3 + \frac{(\gamma-1)\beta_2\beta_3}{\beta^2} \right] \zeta_2 \end{pmatrix}$$

根据x、y、z方向的对称性，定义I.T.操作矩阵为 L_I ：

$$L_I = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \left[-\gamma\beta_1 + 1 + \frac{(\gamma-1)\beta_1^2}{\beta^2} \right] \zeta_1 & \left[-\gamma\beta_1 + \frac{(\gamma-1)\beta_1\beta_2}{\beta^2} \right] \zeta_2 & \left[-\gamma\beta_1 + \frac{(\gamma-1)\beta_1\beta_3}{\beta^2} \right] \zeta_3 \\ 0 & \left[-\gamma\beta_2 + \frac{(\gamma-1)\beta_1\beta_2}{\beta^2} \right] \zeta_1 & \left[-\gamma\beta_2 + 1 + \frac{(\gamma-1)\beta_2^2}{\beta^2} \right] \zeta_2 & \left[-\gamma\beta_2 + \frac{(\gamma-1)\beta_2\beta_3}{\beta^2} \right] \zeta_3 \\ 0 & \left[-\gamma\beta_3 + \frac{(\gamma-1)\beta_1\beta_3}{\beta^2} \right] \zeta_1 & \left[-\gamma\beta_3 + \frac{(\gamma-1)\beta_2\beta_3}{\beta^2} \right] \zeta_2 & \left[-\gamma\beta_3 + 1 + \frac{(\gamma-1)\beta_3^2}{\beta^2} \right] \zeta_3 \end{pmatrix}$$

满足：

$$\left[-\gamma\beta_1\beta^2 + (\gamma-1)\beta_1\beta_2 \right]^2 + \left[-\gamma\beta_2\beta^2 + \beta^2 + (\gamma-1)\beta_2^2 \right]^2 + \left[-\gamma\beta_3\beta^2 + (\gamma-1)\beta_2\beta_3 \right]^2 = \frac{1}{\kappa_2^2} \quad \text{其中 } \kappa_2^2 = \frac{\zeta_2^2}{\beta^4}$$

$$\therefore \beta^2 = \frac{\gamma^2 - 1}{\gamma^2} = (\gamma - 1) \frac{\gamma + 1}{\gamma^2}$$

$$\therefore \left[\beta_1\beta_2 - \beta_1 \frac{\gamma + 1}{\gamma} \right]^2 + \left[\frac{\gamma + 1}{\gamma^2} + \beta_2^2 - \beta_2 \frac{\gamma + 1}{\gamma} \right]^2 + \left[\beta_2\beta_3 - \beta_3 \frac{\gamma + 1}{\gamma} \right]^2 = \frac{1}{(\gamma - 1)^2 \kappa_2^2}$$

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$$\because \beta^2 = \beta_1^2 + \beta_2^2 + \beta_3^2$$

$$\therefore \beta_2^2 \beta^2 - 2\beta^2 \frac{\gamma+1}{\gamma} \beta_2 + \beta^2 \frac{(\gamma+1)^2}{\gamma^2} + 2\beta_2^2 \frac{\gamma+1}{\gamma^2} + \frac{(\gamma+1)^2}{\gamma^4} - 2\beta_2 \frac{(\gamma+1)^2}{\gamma^3} = \frac{1}{(\gamma-1)^2 \kappa_2^2}$$

$$\therefore \beta_2^2 \gamma^2 (\gamma^2 - 1) - 2(\gamma^2 - 1)(\gamma^2 + \gamma)\beta_2 + (\gamma^2 - 1)(\gamma + 1)^2 + 2\beta_2^2 \gamma^2 (\gamma + 1) + (\gamma + 1)^2 - 2\beta_2 \gamma (\gamma + 1)^2$$

$$= \frac{\gamma^4}{(\gamma-1)^2 \kappa_2^2} = \frac{\gamma^4}{(\gamma-1)^2} \frac{(\gamma^2 - 1)^2}{\gamma^4} \frac{1}{\zeta_2^2} = \frac{(\gamma+1)^2}{\zeta_2^2}$$

$$\therefore \beta_2^2 \gamma^2 \frac{\gamma-1}{\gamma+1} - 2\gamma(\gamma-1)\beta_2 + (\gamma^2 - 1) + 2\beta_2^2 \frac{\gamma^2}{\gamma+1} + 1 - 2\beta_2 \gamma = \frac{1}{\zeta_2^2}$$

$$\therefore \beta_2^2 \gamma^2 - 2\gamma^2 \beta_2 + \gamma^2 = \frac{1}{\zeta_2^2}$$

$$\therefore \zeta_2 = \frac{1}{\gamma - \gamma\beta_2} \quad (\beta_2 \leq 1)$$

缩放系数可改写为： $\zeta_2 = (1 + (\alpha_2^t)^2 - (\alpha_2^s)^2)^{-1/2}$ ，其中定义时间膨胀补偿 $\alpha_2^t = \gamma - \gamma\beta_2$ 及空间收缩补偿 $\alpha_2^s = 1$ ，二者符号分别对应Minkowski度规的时间、空间对角元。这里出现空间收缩补偿归因为 L_1 矩阵的空间矩阵元与 L 矩阵的空间矩阵元的非线性关系。

以 $\beta_2 = \beta_3 = 0$ ， $\beta_1 \neq 0$ 为例，我目前认为拟向变换的非对称性使之无法用单一的生成元表达，(就像Lorentz变换或旋转变换那样)。这个想法暂未被严格证明。总之，在找到单一生成元及生成函数之前，无法用不共线生成元对易子研究拟向变换的Wigner转动及Thomas进动，只能用最原始的物理图像研究之。

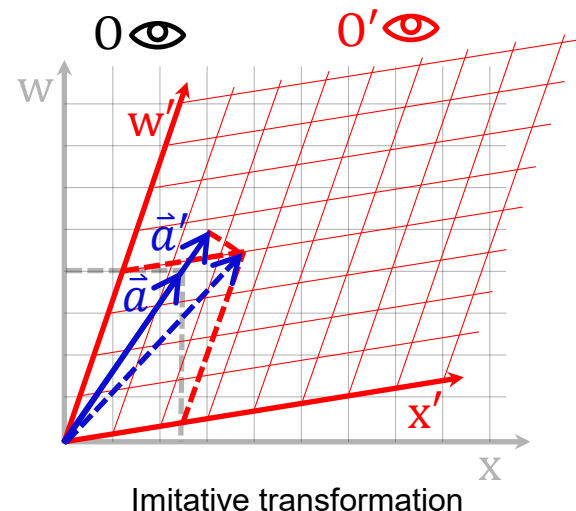
拟向变换自旋赝偶极矢量

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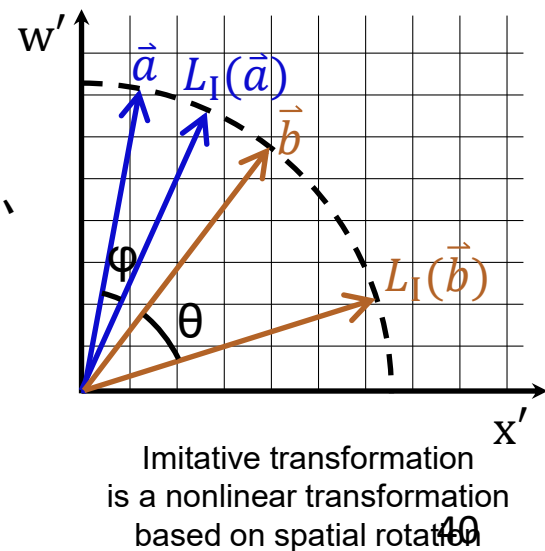
$$L_I(\beta_2 = \beta_3 = 0, \beta_1 \neq 0) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & -\beta_1 & -\beta_1 \\ 0 & 0 & \frac{1}{\gamma} & 0 \\ 0 & 0 & 0 & \frac{1}{\gamma} \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & -\sin \varepsilon & -\sin \varepsilon \\ 0 & 0 & \cos \varepsilon & 0 \\ 0 & 0 & 0 & \cos \varepsilon \end{pmatrix} \quad \text{其中 } \varepsilon = \sin^{-1} \beta_1$$

回到对拟向变换的定义。从物理图像上拟向变换应当是基于Lorentz变换后的坐标系(w' - x' 坐标系)的空间旋转变换基础上的。旋转矩阵是反对称矩阵，但已构造的两个 L_I 变换矩阵都不满足反对称性。实际上，这种物理图像下的拟向变换是基于空间旋转的非线性变换，即不同矢量 $\vec{a}$ 、 $\vec{b}$ 经空间旋转变换的角度 φ 、 θ 是不同的。非线性变换只有在指定的某一部分才有线性近似，之前构造的两个 L_I 变换矩阵是在 $(0/1, 1, 0, 0)^T$ 、 $(0/1, 0, 1, 0)^T$ 、 $(0/1, 0, 0, 1)^T$ 三处的局域线性近似。

巧合的是，弯曲时空的广义Lorentz变换是基于时-空双曲旋转的非线性变换，而拟向变换是基于空间旋转的非线性变换。考虑到后者是人为构造的，我不认为二者有任何关联。



In $w' - x'$ frame



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现在考虑非线性变换。对任意矢量 $(0, a, b, c)^T$ ，满足 $a^2 + b^2 + c^2 = 1$ 。经过拟向变换和Lorentz变换后分别满足：

$$\begin{pmatrix} 0 \\ \left\{ a \left[1 + \frac{(\gamma-1)\beta_1^2}{\beta^2} \right] + b \frac{(\gamma-1)\beta_1\beta_2}{\beta^2} + c \frac{(\gamma-1)\beta_1\beta_3}{\beta^2} \right\} \zeta \\ \left\{ a \frac{(\gamma-1)\beta_1\beta_2}{\beta^2} + b \left[1 + \frac{(\gamma-1)\beta_2^2}{\beta^2} \right] + c \frac{(\gamma-1)\beta_2\beta_3}{\beta^2} \right\} \zeta \\ \left\{ a \frac{(\gamma-1)\beta_1\beta_3}{\beta^2} + b \frac{(\gamma-1)\beta_2\beta_3}{\beta^2} + c \left[1 + \frac{(\gamma-1)\beta_3^2}{\beta^2} \right] \right\} \zeta \end{pmatrix} \xleftarrow{\text{L.T.}} \begin{pmatrix} 0 \\ a \\ b \\ c \end{pmatrix} \xrightarrow{\text{L.T.}} \begin{pmatrix} -a\gamma\beta_1 - b\gamma\beta_2 - c\gamma\beta_3 \\ a \left[1 + \frac{(\gamma-1)\beta_1^2}{\beta^2} \right] + b \frac{(\gamma-1)\beta_1\beta_2}{\beta^2} + c \frac{(\gamma-1)\beta_1\beta_3}{\beta^2} \\ a \frac{(\gamma-1)\beta_1\beta_2}{\beta^2} + b \left[1 + \frac{(\gamma-1)\beta_2^2}{\beta^2} \right] + c \frac{(\gamma-1)\beta_2\beta_3}{\beta^2} \\ a \frac{(\gamma-1)\beta_1\beta_3}{\beta^2} + b \frac{(\gamma-1)\beta_2\beta_3}{\beta^2} + c \left[1 + \frac{(\gamma-1)\beta_3^2}{\beta^2} \right] \end{pmatrix}$$

$$\therefore \vec{y} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \left[1 + \frac{(\gamma-1)\beta_1^2}{\beta^2} \right] \zeta & \frac{(\gamma-1)\beta_1\beta_2}{\beta^2} \zeta & \frac{(\gamma-1)\beta_1\beta_3}{\beta^2} \zeta \\ 0 & \frac{(\gamma-1)\beta_1\beta_2}{\beta^2} \zeta & \left[1 + \frac{(\gamma-1)\beta_2^2}{\beta^2} \right] \zeta & \frac{(\gamma-1)\beta_2\beta_3}{\beta^2} \zeta \\ 0 & \frac{(\gamma-1)\beta_1\beta_3}{\beta^2} \zeta & \frac{(\gamma-1)\beta_2\beta_3}{\beta^2} \zeta & \left[1 + \frac{(\gamma-1)\beta_3^2}{\beta^2} \right] \zeta \end{pmatrix} \vec{x} \quad \text{其中} \vec{x} \text{是变换前矢量, } \vec{y} \text{是变换后矢量}$$

$\zeta = (1 + (a\gamma\beta_1 + b\gamma\beta_2 + c\gamma\beta_3)^2)^{-1/2}$ 这是个含 a, b, c 的number，说明拟向变换是非线性变换

$$\vec{m} \equiv \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & \left[1 + \frac{(\gamma-1)\beta_1^2}{\beta^2} \right] & \frac{(\gamma-1)\beta_1\beta_2}{\beta^2} & \frac{(\gamma-1)\beta_1\beta_3}{\beta^2} \\ 0 & \frac{(\gamma-1)\beta_1\beta_2}{\beta^2} & \left[1 + \frac{(\gamma-1)\beta_2^2}{\beta^2} \right] & \frac{(\gamma-1)\beta_2\beta_3}{\beta^2} \\ 0 & \frac{(\gamma-1)\beta_1\beta_3}{\beta^2} & \frac{(\gamma-1)\beta_2\beta_3}{\beta^2} & \left[1 + \frac{(\gamma-1)\beta_3^2}{\beta^2} \right] \end{pmatrix} \vec{n} \equiv \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

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$$\therefore \vec{y} = [\zeta \vec{m} + \vec{n}] \vec{x}$$

$$\vec{k} \equiv (0, \gamma\beta_1, \gamma\beta_2, \gamma\beta_3)^T$$

$$\therefore \zeta = \left(1 + (\vec{k}^T \vec{x})^2\right)^{-1/2}$$

$$\therefore \vec{y} = \left[\left(1 + (\vec{k}^T \vec{x})^2\right)^{-1/2} \vec{m} + \vec{n}\right] \vec{x}$$

为了研究拟向变换的Thomas进动性质，依照 $\vec{x} \xrightarrow{\text{L.T.1}} \vec{y} \xrightarrow{\text{L.T.2}} \vec{y}'$ 顺序构造连续不共线拟向变换。

$$\vec{y}'_{\text{IxIy}} = \left[\left(1 + \left(\vec{k}'^T \left[\left(1 + (\vec{k}^T \vec{x})^2\right)^{-1/2} \vec{m} + \vec{n}\right] \vec{x}\right)^2\right)^{-1/2} \vec{m}' + \vec{n}\right] \left[\left(1 + (\vec{k}^T \vec{x})^2\right)^{-1/2} \vec{m} + \vec{n}\right] \vec{x}$$

$$\because \vec{k}'^T \vec{n} = 0$$

$$\therefore \vec{y}'_{\text{IxIy}} = \left[\left(1 + \left(\vec{k}'^T \left[\left(1 + (\vec{k}^T \vec{x})^2\right)^{-1/2} \vec{m}\right] \vec{x}\right)^2\right)^{-1/2} \vec{m}' + \vec{n}\right] \left[\left(1 + (\vec{k}^T \vec{x})^2\right)^{-1/2} \vec{m} + \vec{n}\right] \vec{x}$$

$$\chi \equiv \left(1 + (\vec{k}^T \vec{x})^2\right)^{-1/2}$$

$$\lambda \equiv \left(1 + \chi^2 (\vec{k}'^T \vec{m} \vec{x})^2\right)^{-1/2}$$

$$\because \vec{m} \vec{n} = \vec{n} \vec{m} = \vec{m}' \vec{n} = \vec{n} \vec{m}' = 0, \quad \vec{n} \vec{n} = \vec{n}$$

$$\therefore \vec{y}'_{\text{IxIy}} = [\lambda \vec{m}' + \vec{n}] [\chi \vec{m} + \vec{n}] \vec{x}$$

$$= \lambda \chi \vec{m}' \vec{m} \vec{x} + \vec{n} \vec{x}$$

$$\because \lambda \chi = \left(1 + \left(1 + (\vec{k}^T \vec{x})^2\right)^{-1} (\vec{k}'^T \vec{m} \vec{x})^2\right)^{-1/2} \left(1 + (\vec{k}^T \vec{x})^2\right)^{-1/2}$$

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$$= \left[1 + (\vec{k}^T \vec{x})^2 + \left(1 + (\vec{k}^T \vec{x})^2 \right) \left(1 + (\vec{k}^T \vec{x})^2 \right)^{-1} (\vec{k}'^T \vec{m} \vec{x})^2 \right]^{-1/2}$$

$$= \left[1 + (\vec{k}^T \vec{x})^2 + (\vec{k}'^T \vec{m} \vec{x})^2 \right]^{-1/2}$$

$$\therefore \vec{y}'_{\text{Ixy}} = \left[1 + (\vec{k}^T \vec{x})^2 + (\vec{k}'^T \vec{m} \vec{x})^2 \right]^{-1/2} \vec{m}' \vec{m} \vec{x} + \vec{n} \vec{x}$$

Dirac方程解耦的自旋矢量分量要求对任意矢量 $(1, a, b, c)^T$ ，满足 $a^2 + b^2 + c^2 = 1$ 。

经过拟向变换和Lorentz变换后分别满足：

$$\begin{pmatrix} 1 \\ \left\{ -\gamma\beta_1 + a \left[1 + \frac{(\gamma-1)\beta_1^2}{\beta^2} \right] + b \frac{(\gamma-1)\beta_1\beta_2}{\beta^2} + c \frac{(\gamma-1)\beta_1\beta_3}{\beta^2} \right\} \zeta \\ \left\{ -\gamma\beta_2 + a \frac{(\gamma-1)\beta_1\beta_2}{\beta^2} + b \left[1 + \frac{(\gamma-1)\beta_2^2}{\beta^2} \right] + c \frac{(\gamma-1)\beta_2\beta_3}{\beta^2} \right\} \zeta \\ \left\{ -\gamma\beta_3 + a \frac{(\gamma-1)\beta_1\beta_3}{\beta^2} + b \frac{(\gamma-1)\beta_2\beta_3}{\beta^2} + c \left[1 + \frac{(\gamma-1)\beta_3^2}{\beta^2} \right] \right\} \zeta \end{pmatrix} \xleftarrow{\text{I.T.}} \begin{pmatrix} 1 \\ a \\ b \\ c \end{pmatrix} \xrightarrow{\text{L.T.}} \begin{pmatrix} \gamma - a\gamma\beta_1 - b\gamma\beta_2 - c\gamma\beta_3 \\ -\gamma\beta_1 + a \left[1 + \frac{(\gamma-1)\beta_1^2}{\beta^2} \right] + b \frac{(\gamma-1)\beta_1\beta_2}{\beta^2} + c \frac{(\gamma-1)\beta_1\beta_3}{\beta^2} \\ -\gamma\beta_2 + a \frac{(\gamma-1)\beta_1\beta_2}{\beta^2} + b \left[1 + \frac{(\gamma-1)\beta_2^2}{\beta^2} \right] + c \frac{(\gamma-1)\beta_2\beta_3}{\beta^2} \\ -\gamma\beta_3 + a \frac{(\gamma-1)\beta_1\beta_3}{\beta^2} + b \frac{(\gamma-1)\beta_2\beta_3}{\beta^2} + c \left[1 + \frac{(\gamma-1)\beta_3^2}{\beta^2} \right] \end{pmatrix}$$

$$\therefore \vec{y} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ -\gamma\beta_1\zeta & \left[1 + \frac{(\gamma-1)\beta_1^2}{\beta^2} \right] \zeta & \frac{(\gamma-1)\beta_1\beta_2}{\beta^2} \zeta & \frac{(\gamma-1)\beta_1\beta_3}{\beta^2} \zeta \\ -\gamma\beta_2\zeta & \frac{(\gamma-1)\beta_1\beta_2}{\beta^2} \zeta & \left[1 + \frac{(\gamma-1)\beta_2^2}{\beta^2} \right] \zeta & \frac{(\gamma-1)\beta_2\beta_3}{\beta^2} \zeta \\ -\gamma\beta_3\zeta & \frac{(\gamma-1)\beta_1\beta_3}{\beta^2} \zeta & \frac{(\gamma-1)\beta_2\beta_3}{\beta^2} \zeta & \left[1 + \frac{(\gamma-1)\beta_3^2}{\beta^2} \right] \zeta \end{pmatrix} \vec{x}$$

其中 $\zeta = \gamma - a\gamma\beta_1 - b\gamma\beta_2 - c\gamma\beta_3$

$\vec{k} \equiv (\gamma, -\gamma\beta_1, -\gamma\beta_2, -\gamma\beta_3)^T$

$\therefore \zeta = \vec{k}^T \vec{x}$

拟向变换自旋赝偶极矢量

(Imitative Transfer Pseudodipole Vector, ITPDV)

$$\vec{m} \equiv \begin{pmatrix} 0 & 0 & 0 & 0 \\ -\gamma\beta_1 \left[1 + \frac{(\gamma-1)\beta_1^2}{\beta^2}\right] & \frac{(\gamma-1)\beta_1\beta_2}{\beta^2} & \frac{(\gamma-1)\beta_1\beta_3}{\beta^2} \\ -\gamma\beta_2 & \frac{(\gamma-1)\beta_1\beta_2}{\beta^2} & \left[1 + \frac{(\gamma-1)\beta_2^2}{\beta^2}\right] & \frac{(\gamma-1)\beta_2\beta_3}{\beta^2} \\ -\gamma\beta_3 & \frac{(\gamma-1)\beta_1\beta_3}{\beta^2} & \frac{(\gamma-1)\beta_2\beta_3}{\beta^2} & \left[1 + \frac{(\gamma-1)\beta_3^2}{\beta^2}\right] \end{pmatrix} \quad \vec{n} \equiv \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\therefore \vec{y} = \left(\frac{1}{\vec{k}^T \vec{x}} \vec{m} + \vec{n} \right) \vec{x}$$

为了研究拟向变换的Thomas进动性质，同样也依照 $\vec{x} \xrightarrow{\text{L.T.1}} \vec{y} \xrightarrow{\text{L.T.2}} \vec{y}'$ 顺序构造连续不共

线拟向变换。

$$\vec{y}'_{\text{IxIy}} = \left[\frac{1}{\vec{k}'^T \left(\frac{1}{\vec{k}^T \vec{x}} \vec{m} + \vec{n} \right) \vec{x}} \vec{m}' + \vec{n} \right] \left(\frac{1}{\vec{k}^T \vec{x}} \vec{m} + \vec{n} \right) \vec{x}$$

$$\therefore \vec{k}^T \vec{n} = \gamma \vec{n}^T$$

$$\text{其中 } \vec{n} = (1, 0, 0, 0)^T \quad \vec{n}^T \vec{x} = 1$$

$$\therefore \vec{y}'_{\text{IxIy}} = \left[\frac{1}{\frac{1}{\vec{k}^T \vec{x}} \vec{k}'^T \vec{m} \vec{x} + \gamma' \vec{n}^T \vec{x}} \vec{m}' + \vec{n} \right] \left(\frac{1}{\vec{k}^T \vec{x}} \vec{m} + \vec{n} \right) \vec{x}$$

$$\lambda \equiv \frac{1}{\vec{k}^T \vec{x}} \quad \chi \equiv \frac{1}{\lambda \vec{k}'^T \vec{m} \vec{x} + \gamma' \vec{n}^T \vec{x}}$$

拟向变换自旋赝偶极矢量 (Imitative Transfer Pseudodipole Vector, ITPDV)

$$\because \vec{n}\vec{m} = \vec{n}\vec{m}' = 0, \quad \vec{n}\vec{n} = \vec{n}$$

$$\vec{M} \equiv \vec{m}\vec{n} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ -\gamma\beta_1 & 0 & 0 & 0 \\ -\gamma\beta_2 & 0 & 0 & 0 \\ -\gamma\beta_3 & 0 & 0 & 0 \end{pmatrix} \quad \vec{M}' \equiv \vec{m}'\vec{n} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ -\gamma'\beta'_1 & 0 & 0 & 0 \\ -\gamma'\beta'_2 & 0 & 0 & 0 \\ -\gamma'\beta'_3 & 0 & 0 & 0 \end{pmatrix}$$

$$\begin{aligned} \because \vec{y}'_{\text{IxIy}} &= (\chi\vec{m}' + \vec{n})(\lambda\vec{m} + \vec{n})\vec{x} \\ &= \lambda\chi\vec{m}'\vec{m}\vec{x} + \chi\vec{M}'\vec{x} + \vec{n}\vec{x} \end{aligned}$$

$$\because \lambda\chi = \frac{1}{\lambda\vec{k}'^T\vec{m}\vec{x} + \gamma'\vec{n}^T\vec{x}} \lambda = \frac{1}{\vec{k}'^T\vec{m}\vec{x} + \gamma'\vec{k}^T\vec{x}}$$

$$\because \vec{y}'_{\text{IxIy}} = \frac{1}{\vec{k}'^T\vec{m}\vec{x} + \gamma'\vec{k}^T\vec{x}} \vec{m}'\vec{m}\vec{x} + \frac{\vec{k}^T\vec{x}}{\vec{k}'^T\vec{m}\vec{x} + \gamma'\vec{k}^T\vec{x}} \vec{M}'\vec{x} + \vec{n}\vec{x}$$

考虑对任意自旋矢量，沿xoy面上任意方向拟向变换，再沿zoy面任意方向进行微元拟向变换，将上式带入：

$$\beta_3 = 0, \beta_1 \neq 0, \beta_2 \neq 0$$

$$\gamma = (1 - \beta_1^2 - \beta_2^2)^{-1/2}$$

$$\beta'_1 = 0, \beta'_3 \rightarrow 0, \beta'_2 \rightarrow 0$$

$$\gamma' = 1$$

$$\therefore \vec{m} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ -\gamma\beta_1 & \left[1 + \frac{\gamma^2}{\gamma+1}\beta_1^2\right] & \frac{\gamma^2}{\gamma+1}\beta_1\beta_2 & 0 \\ -\gamma\beta_2 & \frac{\gamma^2}{\gamma+1}\beta_1\beta_2 & \left[1 + \frac{\gamma^2}{\gamma+1}\beta_2^2\right] & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad \vec{k} = \begin{pmatrix} \gamma \\ -\gamma\beta_1 \\ -\gamma\beta_2 \\ 0 \end{pmatrix}$$

拟向变换自旋赝偶极矢量 (Imitative Transfer Pseudodipole Vector, ITPDV)

$$\begin{aligned}
 \vec{m}' &= \begin{pmatrix} 0 & 0 & 0 & 0 \\ -\gamma'\beta'_1 \left[1 + \frac{\gamma'^2}{\gamma'+1}\beta'^2_1\right] & \frac{\gamma'^2}{\gamma'+1}\beta'_1\beta'_2 & \frac{\gamma'^2}{\gamma'+1}\beta'_1\beta'_3 \\ -\gamma'\beta'_2 & \frac{\gamma'^2}{\gamma'+1}\beta'_1\beta'_2 & \left[1 + \frac{\gamma'^2}{\gamma'+1}\beta'^2_2\right] & \frac{\gamma'^2}{\gamma'+1}\beta'_2\beta'_3 \\ -\gamma'\beta'_3 & \frac{\gamma'^2}{\gamma'+1}\beta'_1\beta'_3 & \frac{\gamma'^2}{\gamma'+1}\beta'_2\beta'_3 & \left[1 + \frac{\gamma'^2}{\gamma'+1}\beta'^2_3\right] \end{pmatrix} \xrightarrow{\gamma' \rightarrow 1} \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ -\beta'_2 & 0 & 1 & \frac{\beta'_2\beta'_3}{2} \\ -\beta'_3 & 0 & \frac{\beta'_2\beta'_3}{2} & 1 \end{pmatrix} \vec{k}' = \begin{pmatrix} 1 \\ 0 \\ -\beta'_2 \\ -\beta'_3 \end{pmatrix} \\
 \therefore \vec{m}'\vec{m}'\vec{x} &= \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ -\beta'_2 & 0 & 1 & \frac{\beta'_2\beta'_3}{2} \\ -\beta'_3 & 0 & \frac{\beta'_2\beta'_3}{2} & 1 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 & 0 \\ -\gamma\beta_1 \left[1 + \frac{\gamma^2}{\gamma+1}\beta^2_1\right] & \frac{\gamma^2}{\gamma+1}\beta_1\beta_2 & 0 \\ -\gamma\beta_2 & \frac{\gamma^2}{\gamma+1}\beta_1\beta_2 & \left[1 + \frac{\gamma^2}{\gamma+1}\beta^2_2\right] & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ a \\ b \\ c \end{pmatrix} \\
 &= \begin{pmatrix} 0 & 0 & 0 & 0 \\ -\gamma\beta_1 \left[1 + \frac{\gamma^2}{\gamma+1}\beta^2_1\right] & \frac{\gamma^2}{\gamma+1}\beta_1\beta_2 & 0 \\ -\gamma\beta_2 & \frac{\gamma^2}{\gamma+1}\beta_1\beta_2 & \left[1 + \frac{\gamma^2}{\gamma+1}\beta^2_2\right] & \frac{\beta'_2\beta'_3}{2} \\ -\gamma\beta_2 \frac{\beta'_2\beta'_3}{2} & \frac{\gamma^2}{\gamma+1}\beta_1\beta_2 \frac{\beta'_2\beta'_3}{2} & \left[1 + \frac{\gamma^2}{\gamma+1}\beta^2_2\right] \frac{\beta'_2\beta'_3}{2} & 1 \end{pmatrix} \begin{pmatrix} 1 \\ a \\ b \\ c \end{pmatrix} \\
 &= \begin{pmatrix} 0 \\ -\gamma\beta_1 + a\left[1 + \frac{\gamma^2}{\gamma+1}\beta^2_1\right] + b\frac{\gamma^2}{\gamma+1}\beta_1\beta_2 \\ -\gamma\beta_2 + a\frac{\gamma^2}{\gamma+1}\beta_1\beta_2 + b\left[1 + \frac{\gamma^2}{\gamma+1}\beta^2_2\right] + c\frac{\beta'_2\beta'_3}{2} \\ -\gamma\beta_2 \frac{\beta'_2\beta'_3}{2} + a\frac{\gamma^2}{\gamma+1}\beta_1\beta_2 \frac{\beta'_2\beta'_3}{2} + b\left[1 + \frac{\gamma^2}{\gamma+1}\beta^2_2\right] \frac{\beta'_2\beta'_3}{2} + c \end{pmatrix}
 \end{aligned}$$

拟向变换自旋赝偶极矢量

(Imitative Transfer Pseudodipole Vector, ITPDV)

$$\vec{k}^T \vec{x} \vec{M}' \vec{x} = (\gamma \quad -\gamma\beta_1 \quad -\gamma\beta_2 \quad 0) \begin{pmatrix} 1 \\ a \\ b \\ c \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ -\beta'_2 & 0 & 0 & 0 \\ -\beta'_3 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ a \\ b \\ c \end{pmatrix} = (\gamma - a\gamma\beta_1 - b\gamma\beta_2) \begin{pmatrix} 0 \\ 0 \\ -\beta'_2 \\ -\beta'_3 \end{pmatrix}$$

故变换后方向为

$$\begin{pmatrix} -\gamma\beta_1 + a \left[1 + \frac{\gamma^2}{\gamma+1} \beta_1^2 \right] + b \frac{\gamma^2}{\gamma+1} \beta_1 \beta_2 \\ -\gamma\beta_2 + a \frac{\gamma^2}{\gamma+1} \beta_1 \beta_2 + b \left[1 + \frac{\gamma^2}{\gamma+1} \beta_2^2 \right] + c \frac{\beta'_2 \beta'_3}{2} - (\gamma - a\gamma\beta_1 - b\gamma\beta_2) \beta'_2 \\ \left\{ -\gamma\beta_2 + a \frac{\gamma^2}{\gamma+1} \beta_1 \beta_2 + b \left[1 + \frac{\gamma^2}{\gamma+1} \beta_2^2 \right] \right\} \frac{\beta'_2 \beta'_3}{2} + c - (\gamma - a\gamma\beta_1 - b\gamma\beta_2) \beta'_3 \end{pmatrix}$$

$$\vec{y}'_{\text{LxLy}} = \begin{pmatrix} 1 & 0 & -\beta'_2 & -\beta'_3 \\ 0 & 1 & 0 & 0 \\ -\beta'_2 & 0 & 1 & \frac{1}{2} \beta'_2 \beta'_3 \\ -\beta'_3 & 0 & \frac{1}{2} \beta'_2 \beta'_3 & 1 \end{pmatrix} \begin{pmatrix} \gamma & -\gamma\beta_1 & -\gamma\beta_2 & 0 \\ -\gamma\beta_1 \left[1 + \frac{\gamma^2}{\gamma+1} \beta_1^2 \right] & \frac{\gamma^2}{\gamma+1} \beta_1 \beta_2 & 0 \\ -\gamma\beta_2 & \frac{\gamma^2}{\gamma+1} \beta_1 \beta_2 & \left[1 + \frac{\gamma^2}{\gamma+1} \beta_2^2 \right] & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ a \\ b \\ c \end{pmatrix}$$

$$= \begin{pmatrix} \gamma + \gamma\beta'_2 \beta_2 & -\gamma\beta_1 - \frac{\gamma^2}{\gamma+1} \beta_1 \beta_2 \beta'_2 & -\gamma\beta_2 - \beta'_2 \left[1 + \frac{\gamma^2}{\gamma+1} \beta_2^2 \right] & -\beta'_3 \\ -\gamma\beta_1 & \left[1 + \frac{\gamma^2}{\gamma+1} \beta_1^2 \right] & \frac{\gamma^2}{\gamma+1} \beta_1 \beta_2 & 0 \\ -\gamma\beta'_2 - \gamma\beta_2 & \gamma\beta_1 \beta'_2 + \frac{\gamma^2}{\gamma+1} \beta_1 \beta_2 & \gamma\beta'_2 \beta_2 + \left[1 + \frac{\gamma^2}{\gamma+1} \beta_2^2 \right] & \frac{1}{2} \beta'_2 \beta'_3 \\ -\gamma\beta'_3 - \frac{1}{2} \gamma\beta_2 \beta'_2 \beta'_3 & \gamma\beta'_3 \beta_1 + \frac{1}{2} \frac{\gamma^2}{\gamma+1} \beta'_2 \beta'_3 \beta_1 \beta_2 & \gamma\beta'_3 \beta_2 + \frac{1}{2} \beta'_2 \beta'_3 \left[1 + \frac{\gamma^2}{\gamma+1} \beta_2^2 \right] & 1 \end{pmatrix} \begin{pmatrix} 1 \\ a \\ b \\ c \end{pmatrix}$$

拟向变换自旋赝偶极矢量

(Imitative Transfer Pseudodipole Vector, ITPDV)

$$\text{方向: } \begin{pmatrix} -\gamma\beta_1 + a\left[1 + \frac{\gamma^2}{\gamma+1}\beta_1^2\right] + b\frac{\gamma^2}{\gamma+1}\beta_1\beta_2 \\ -\gamma\beta'_2 - \gamma\beta_2 + a\left(\gamma\beta_1\beta'_2 + \frac{\gamma^2}{\gamma+1}\beta_1\beta_2\right) + b\left(\gamma\beta'_2\beta_2 + \left[1 + \frac{\gamma^2}{\gamma+1}\beta_2^2\right]\right) + \frac{1}{2}c\beta'_2\beta'_3 \\ -\gamma\beta'_3 - \frac{1}{2}\gamma\beta_2\beta'_2\beta'_3 + a\left(\gamma\beta'_3\beta_1 + \frac{1}{2}\frac{\gamma^2}{\gamma+1}\beta'_2\beta'_3\beta_1\beta_2\right) + b\left(\gamma\beta'_3\beta_2 + \frac{1}{2}\beta'_2\beta'_3\left[1 + \frac{\gamma^2}{\gamma+1}\beta_2^2\right]\right) + c \end{pmatrix}$$

在一般情况下, 连续不共线Lorentz变换与连续不共线拟向变换方向相同

综上所述, 总拟向变换后矢量的方向一定与总Lorentz变换后矢量的方向一致。并且由于拟向变换的保空间内积特性, 两次连续不共线拟向变换后的矢量与总拟向变换后的矢量的空间范数一定相等(时间部分也都等于1), 二者一定可以通过空间旋转变换相联系。且因连续不共线拟向变换和连续不共线Lorentz变换后方向相同, 所以空间旋转变换是线性的, 角速度等于一般Thomas进动角速度。

浅析一下连续不共线的拟向变换(非线性)和Lorentz变换(线性)的Thomas进动一致的原因, 我认为有两点比较关键:

- 不同向拟向变换(基于空间旋转)的生成元对易关系与不同向Lorentz变换生成元对易关系一致;

$$[S_i, S_j] = \varepsilon_{ijk} S_k$$

拟向变换自旋赝偶极矢量

(Imitative Transfer Pseudodipole Vector, ITPDV)

$$[L_i, L_j] = -\varepsilon_{ijk} S_k$$

➤ Thomas进动的第二帧变换是一个微元过程，这个过程的非线性和其他一些细微的差异被抹平。

$$\because L_{\tau+d\tau} R_{d\tau}(x^\mu) = L_{d\tau} L_\tau(x^\mu)$$

$$\because R_{d\tau}^{-1} L_{\tau+d\tau}^{-1}(x^\mu) = L_\tau^{-1} L_{d\tau}^{-1}(x^\mu)$$

$$\therefore \frac{dL^{-1}}{d\tau} X^\mu = \frac{dL^{-1}}{d\tau} (Lx^\mu) = \frac{L_{\tau+d\tau}^{-1} - L_\tau^{-1}}{d\tau} (Lx^\mu)$$

$$= \frac{L_{\tau+d\tau}^{-1} - L_\tau^{-1}}{d\tau} L_\tau x_\tau^\mu$$

$$= \frac{R_{d\tau} L_\tau^{-1} L_{d\tau}^{-1} - L_\tau^{-1}}{d\tau} L_\tau x_\tau^\mu$$

$$\because L_{d\tau}^{-1} L_\tau - L_\tau L_{d\tau}^{-1} = \begin{pmatrix} 0 & 0 & \beta_2 - \beta_2 \gamma & 0 \\ 0 & 0 & \beta \beta_2 \gamma & 0 \\ -\beta_2 + \beta_2 \gamma & -\beta \beta_2 \gamma & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \rightarrow 0$$

$$\therefore \text{原式} = \frac{R_{d\tau} L_{d\tau}^{-1} - I_4}{d\tau} x_\tau^\mu, \text{分子的 } L_{d\tau}^{-1} \text{ 是纯 Lorentz 转动, 作用到 } x_\tau^\mu \text{ 后得到②式的 } x_0 \text{ 项,}$$

而空间 $\vec{x}$ 项则完全由 Thomas 进动 $R_{d\tau}$ 得到，解释了 Thomas 动力学与 Thomas 运动学的一致性

现在要考虑拟向变换是否也有类似的动力学－运动学一致

$$\because L_{I, \tau+d\tau} [R_{d\tau}(x^\mu)] = L_{I, d\tau} [L_{I, \tau}(x^\mu)]$$

$$\because R_{d\tau}^{-1} L_{I, \tau+d\tau}^{-1}(x^\mu) = L_{I, \tau}^{-1} [L_{I, d\tau}^{-1}(x^\mu)]$$

$$\therefore \frac{dL_I^{-1}}{d\tau} X^\mu = \frac{dL_I^{-1}}{d\tau} [L_I(x^\mu)] = \frac{L_{I, \tau+d\tau}^{-1} - L_{I, \tau}^{-1}}{d\tau} (L_I x^\mu)$$

拟向变换自旋赝偶极矢量 (Imitative Transfer Pseudodipole Vector, ITPDV)

$$= \frac{R_{d\tau} L_{I,\tau}^{-1} [\hat{L}_{I,d\tau}^{-1}] - \hat{L}_{I,\tau}^{-1}}{d\tau} (L_I x^\mu)$$

$$= \frac{R_{d\tau} L_{I,\tau}^{-1} [L_{I,d\tau}^{-1} (L_{I,\tau} x^\mu)] - x^\mu}{d\tau}$$

以 $\vec{k} = (\gamma, -\gamma\beta_1, 0, 0)^\top$ $\vec{k}' = (1, 0, -\beta'_2, 0)^\top$ 为例

$$\vec{m} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ -\gamma\beta_1 & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad \vec{m}' = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ -\beta'_2 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$L_{I,\tau} [L_{I,d\tau} (x^\mu)] - L_{I,d\tau} [L_{I,\tau} (x^\mu)]$$

$$= \left(\frac{1}{\vec{k}^\top \left[\left(\frac{1}{\vec{k}'^\top \vec{x}} \vec{m} + \vec{n} \right) \vec{x} \right]} \vec{m} + \vec{n} \right) \left(\frac{1}{\vec{k}'^\top \vec{x}} \vec{m} + \vec{n} \right) \vec{x} - \left(\frac{1}{\vec{k}'^\top \left[\left(\frac{1}{\vec{k}^\top \vec{x}} \vec{m} + \vec{n} \right) \vec{x} \right]} \vec{m} + \vec{n} \right) \left(\frac{1}{\vec{k}^\top \vec{x}} \vec{m} + \vec{n} \right) \vec{x}$$

$$= \frac{1}{\vec{k}^\top \vec{m}' \vec{x} + \gamma \vec{k}'^\top \vec{x}} \vec{m} \vec{m}' \vec{x} + \frac{\vec{k}'^\top \vec{x}}{\vec{k}^\top \vec{m}' \vec{x} + \gamma \vec{k}'^\top \vec{x}} \vec{M} \vec{x} + \vec{n} \vec{x} - \frac{1}{\vec{k}'^\top \vec{m} \vec{x} + \gamma' \vec{k}^\top \vec{x}} \vec{m}' \vec{m} \vec{x} - \frac{\vec{k}^\top \vec{x}}{\vec{k}'^\top \vec{m} \vec{x} + \gamma' \vec{k}^\top \vec{x}} \vec{M}' \vec{x} - \vec{n} \vec{x}$$

$$= \frac{1}{-a\gamma\beta_1 + \gamma - b\gamma\beta'_2} \begin{pmatrix} 0 & 0 & 0 & 0 \\ -(1-b\beta'_2)\gamma\beta_1 & \gamma & 0 & 0 \\ -\beta'_2 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \vec{x} - \frac{1}{-a\gamma\beta_1 + \gamma - b\beta'_2} \begin{pmatrix} 0 & 0 & 0 & 0 \\ -\gamma\beta_1 & \gamma & 0 & 0 \\ -(\gamma - a\gamma\beta_1)\beta'_2 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \vec{x}$$

由于 $-a\gamma\beta_1 + \gamma > 0$ 恒成立，故省略两项分母的 β'_2 项：

$$\therefore L_{I,\tau} [L_{I,d\tau} (x^\mu)] - L_{I,d\tau} [L_{I,\tau} (x^\mu)] = \frac{1}{-a\gamma\beta_1 + \gamma} \begin{pmatrix} 0 & 0 & 0 & 0 \\ b\gamma\beta_1\beta'_2 & 0 & 0 & 0 \\ (\gamma - a\gamma\beta_1 - 1)\beta'_2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \vec{x} \rightarrow 0$$

拟向变换自旋赝偶极矢量 (Imitative Transfer Pseudodipole Vector, ITPDV)

$$\therefore \frac{dL_I^{-1}}{d\tau} X^\mu = \frac{R_{d\tau} L_{I,d\tau}^{-1} - I_4}{d\tau} x^\mu$$

故拟向变换也有类似Lorentz变换的动力学－运动学一致性

$$\begin{aligned} \dot{\vec{X}} \Big|_{\vec{x}} &= \left(-\frac{1}{(\vec{k}^T \vec{x})^2} (\dot{\vec{k}}^T \vec{x}) \vec{m} + \frac{1}{\vec{k}^T \vec{x}} \dot{\vec{m}} \right) \vec{x} \\ &= \left(-\frac{1}{\left(\vec{k}^T \left(\frac{1}{\vec{k}^T \vec{X}} \vec{m}_- + \vec{n} \right) \vec{X} \right)^2} \left(\dot{\vec{k}}^T \left(\frac{1}{\vec{k}^T \vec{X}} \vec{m}_- + \vec{n} \right) \vec{X} \right) \vec{m} + \frac{1}{\vec{k}^T \left(\frac{1}{\vec{k}^T \vec{X}} \vec{m}_- + \vec{n} \right) \vec{X}} \dot{\vec{m}} \right) \left(\frac{1}{\vec{k}^T \vec{X}} \vec{m}_- + \vec{n} \right) \vec{X} \end{aligned}$$

现在只关注 X_0 对 $\dot{\vec{X}} \Big|_{\vec{x}}$ 空间分量的贡献，如果作线性近似（认为 X_0 对 $\dot{\vec{X}} \Big|_{\vec{x}}$ 的贡献仅与 X_0 有关），设 $\vec{X} = (1, 0, 0, 0)^T$ ，代入上式得：

$$\dot{\vec{X}} \Big|_{\vec{x}} = 4\gamma^6 (\vec{\beta} \cdot \dot{\vec{\beta}}) \vec{\beta} - \gamma \dot{\vec{\beta}} - \frac{\gamma^3}{\gamma+1} (\vec{\beta} \cdot \dot{\vec{\beta}}) \vec{\beta}, \text{ 其中蓝色部分与②式 } X_0 \text{ 贡献一致，红色部分则是拟向变换特有的。}$$

如果令最前面括号里 $\vec{X} = (1, X_1, X_2, X_3)^T$ 其余不变，重复以上操作就得到了考虑一阶非线性近似的 X_0 对 $\dot{\vec{X}} \Big|_{\vec{x}}$ 贡献：

$$\begin{aligned} \left(\frac{1}{\vec{k}^T \vec{X}} \vec{m}_- + \vec{n} \right) \vec{X} &= \frac{\vec{X} - \gamma \vec{\beta} + \frac{(\gamma-1)}{\beta^2} (\vec{\beta} \cdot \vec{X}) \vec{\beta}}{\gamma(1 + \vec{\beta} \cdot \vec{X})} + \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} \\ \dot{\vec{k}}^T \left(\frac{1}{\vec{k}^T \vec{X}} \vec{m}_- + \vec{n} \right) \vec{X} &= \frac{2\gamma^3 (\vec{\beta} \cdot \dot{\vec{\beta}}) - (\dot{\vec{\beta}} \cdot \vec{X}) - \frac{\gamma^2}{\gamma+1} (\vec{\beta} \cdot \dot{\vec{\beta}}) (\vec{\beta} \cdot \vec{X})}{(1 + \vec{\beta} \cdot \vec{X})} \\ \vec{k}^T \left(\frac{1}{\vec{k}^T \vec{X}} \vec{m}_- + \vec{n} \right) \vec{X} &= \frac{2\gamma^2 (\vec{\beta} \cdot \vec{X}) + 1}{\gamma(1 + \vec{\beta} \cdot \vec{X})} \end{aligned}$$

拟向变换自旋赝偶极矢量

(Imitative Transfer Pseudodipole Vector, ITPDV)

$$\therefore \dot{\vec{X}} \Big|_{\vec{x}} = \frac{2\gamma^3(1+\vec{\beta} \cdot \vec{X})}{(2\gamma^2(\vec{\beta} \cdot \vec{X})+1)^2} \left[2\gamma^3(\vec{\beta} \cdot \dot{\vec{\beta}}) - (\dot{\vec{\beta}} \cdot \vec{X}) - \frac{\gamma^2}{\gamma+1} (\vec{\beta} \cdot \dot{\vec{\beta}})(\vec{\beta} \cdot \vec{X}) \right] \vec{\beta} - \frac{\gamma(1+\vec{\beta} \cdot \vec{X})}{2\gamma^2(\vec{\beta} \cdot \vec{X})+1} \dot{\vec{\beta}} - \frac{\gamma(1+\vec{\beta} \cdot \vec{X})}{2\gamma^2(\vec{\beta} \cdot \vec{X})+1} \frac{\gamma^2}{\gamma+1} (\vec{\beta} \cdot \dot{\vec{\beta}}) \vec{\beta}$$

至此已完成对P41提出的拟向变换的动力学探究。但是，F-W变换表明自旋4-vector的时间分量在任何参考系下都为0，所以应遵循P44提出的拟向变换。修改前面排版太麻烦，所以我直接在下面补充。

对含时Dirac方程的演化算符作F – W变换：

$$e^{i\hat{S}} i\hbar \frac{\partial}{\partial t} e^{-i\hat{S}} \\ = i\hbar \frac{\partial}{\partial t} + i \left[\hat{S}, i\hbar \frac{\partial}{\partial t} \right] + \frac{i^2}{2!} \left[\hat{S}, \left[\hat{S}, i\hbar \frac{\partial}{\partial t} \right] \right] + \dots$$

其中：

$$i \left[\hat{S}, i\hbar \frac{\partial}{\partial t} \right] = \frac{i\hbar}{2mc^2} \left[\hat{\beta} \hat{O}, \frac{\partial}{\partial t} \right] = \frac{\hbar^2}{2mc} \hat{\beta} \hat{\alpha} \cdot \left(\hat{\nabla} \frac{\partial}{\partial t} \right) - \frac{\hbar^2}{2mc} \hat{\beta} \hat{\alpha} \cdot \left(\frac{\partial}{\partial t} \hat{\nabla} \right) = 0$$

$$\frac{i^2}{2!} \left[\hat{S}, \left[\hat{S}, i\hbar \frac{\partial}{\partial t} \right] \right] = \frac{i\hbar}{8m^2c^2} \left[\hat{\beta} \hat{\alpha} \cdot \hat{p}, \left[\hat{\beta} \hat{\alpha} \cdot \hat{p}, \frac{\partial}{\partial t} \right] \right] = 0$$

⋮

$$\therefore e^{i\hat{S}} i\hbar \frac{\partial}{\partial t} e^{-i\hat{S}} = i\hbar \frac{\partial}{\partial t}$$

拟向变换自旋赝偶极矢量

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演化算符可视为 $i\hbar \frac{\partial}{\partial t} = \hat{\mathbf{I}}_4 \cdot \hat{\mathbf{E}}$, 将其与 $\hat{\alpha} \cdot \hat{\mathbf{p}}$ 的 $F-W$ 变换结果对比不难发现:

$$e^{i\hat{S}} \hat{\mathbf{I}}_4 \cdot \hat{\mathbf{E}} e^{-i\hat{S}} = \hat{\mathbf{E}}$$

$$e^{i\hat{S}} c\hat{\alpha} \cdot \hat{\mathbf{p}} e^{-i\hat{S}} = -c\hat{\alpha} \cdot \hat{\mathbf{p}} + \frac{\hat{\mathbf{p}}^2}{2m} - \frac{\hat{\mathbf{p}}^4}{8m^3c^2} + \frac{1}{8m^2c^2} \hat{\nabla}^2 \hat{V} + \frac{1}{4m^2c^2} \hat{\Sigma} \cdot (\hat{\nabla} \hat{V} \times \hat{\mathbf{p}}) + \dots$$

所以从 4-momentum 的角度, 自旋 4-vector 空间分量以 $\hat{\Sigma}$ 算符与轨道角动量耦合, 时间分量以 $\hat{O}$ (零矩阵) 算符与轨道角能量耦合。

$$\therefore \langle \sigma_t \rangle = \langle \psi | \hat{O}_4 | \psi \rangle = 0$$

接着 P46 的结果, 对任意自旋矢量沿 xoy 面上任意方向拟向变换, 再沿 zoy 面任意方向进行微元拟向变换:

$$\beta_3 = 0, \beta_1 \neq 0, \beta_2 \neq 0$$

$$\gamma = (1 - \beta_1^2 - \beta_2^2)^{-1/2}$$

$$\beta'_1 = 0, \beta'_3 \rightarrow 0, \beta'_2 \rightarrow 0$$

$$\gamma' = 1$$

$$\therefore \vec{m} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & \left[1 + \frac{\gamma^2}{\gamma+1}\beta_1^2\right] & \frac{\gamma^2}{\gamma+1}\beta_1\beta_2 & 0 \\ 0 & \frac{\gamma^2}{\gamma+1}\beta_1\beta_2 & \left[1 + \frac{\gamma^2}{\gamma+1}\beta_2^2\right] & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad \vec{k} = \begin{pmatrix} 0 \\ \gamma\beta_1 \\ \gamma\beta_2 \\ 0 \end{pmatrix}$$

拟向变换自旋赝偶极矢量 (Imitative Transfer Pseudodipole Vector, ITPDV)

$$\begin{aligned}
 \vec{m}' &= \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & \left[1 + \frac{\gamma'^2}{\gamma' + 1} \beta_1'^2\right] & \frac{\gamma'^2}{\gamma' + 1} \beta_1' \beta_2' & \frac{\gamma'^2}{\gamma' + 1} \beta_1' \beta_3' \\ 0 & \frac{\gamma'^2}{\gamma' + 1} \beta_1' \beta_2' & \left[1 + \frac{\gamma'^2}{\gamma' + 1} \beta_2'^2\right] & \frac{\gamma'^2}{\gamma' + 1} \beta_2' \beta_3' \\ 0 & \frac{\gamma'^2}{\gamma' + 1} \beta_1' \beta_3' & \frac{\gamma'^2}{\gamma' + 1} \beta_2' \beta_3' & \left[1 + \frac{\gamma'^2}{\gamma' + 1} \beta_3'^2\right] \end{pmatrix} \xrightarrow{\gamma' \rightarrow 1} \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & \frac{\beta_2' \beta_3'}{2} \\ 0 & 0 & \frac{\beta_2' \beta_3'}{2} & 1 \end{pmatrix} \quad \vec{k}' = \begin{pmatrix} 0 \\ 0 \\ \beta_2' \\ \beta_3' \end{pmatrix} \\
 \therefore \vec{m}' \vec{m} \vec{x} &= \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & \frac{\beta_2' \beta_3'}{2} \\ 0 & 0 & \frac{\beta_2' \beta_3'}{2} & 1 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & \left[1 + \frac{\gamma^2}{\gamma + 1} \beta_1^2\right] & \frac{\gamma^2}{\gamma + 1} \beta_1 \beta_2 & 0 \\ 0 & \frac{\gamma^2}{\gamma + 1} \beta_1 \beta_2 & \left[1 + \frac{\gamma^2}{\gamma + 1} \beta_2^2\right] & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ a \\ b \\ c \end{pmatrix} \\
 &= \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & \left[1 + \frac{\gamma^2}{\gamma + 1} \beta_1^2\right] & \frac{\gamma^2}{\gamma + 1} \beta_1 \beta_2 & 0 \\ 0 & \frac{\gamma^2}{\gamma + 1} \beta_1 \beta_2 & \left[1 + \frac{\gamma^2}{\gamma + 1} \beta_2^2\right] & \frac{\beta_2' \beta_3'}{2} \\ 0 & \frac{\gamma^2}{\gamma + 1} \beta_1 \beta_2 \frac{\beta_2' \beta_3'}{2} & \left[1 + \frac{\gamma^2}{\gamma + 1} \beta_2^2\right] \frac{\beta_2' \beta_3'}{2} & 1 \end{pmatrix} \begin{pmatrix} 0 \\ a \\ b \\ c \end{pmatrix} \\
 &= \begin{pmatrix} 0 \\ a \left[1 + \frac{\gamma^2}{\gamma + 1} \beta_1^2\right] + b \frac{\gamma^2}{\gamma + 1} \beta_1 \beta_2 \\ a \frac{\gamma^2}{\gamma + 1} \beta_1 \beta_2 + b \left[1 + \frac{\gamma^2}{\gamma + 1} \beta_2^2\right] + c \frac{\beta_2' \beta_3'}{2} \\ a \frac{\gamma^2}{\gamma + 1} \beta_1 \beta_2 \frac{\beta_2' \beta_3'}{2} + b \left[1 + \frac{\gamma^2}{\gamma + 1} \beta_2^2\right] \frac{\beta_2' \beta_3'}{2} + c \end{pmatrix}
 \end{aligned}$$

拟向变换自旋赝偶极矢量 (Imitative Transfer Pseudodipole Vector, ITPDV)

$$\vec{k}'^T \vec{m} \vec{x} = \begin{pmatrix} 0 & 0 & \beta'_2 & \beta'_3 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & \left[1 + \frac{\gamma^2}{\gamma+1} \beta_1^2\right] & \frac{\gamma^2}{\gamma+1} \beta_1 \beta_2 & 0 \\ 0 & \frac{\gamma^2}{\gamma+1} \beta_1 \beta_2 & \left[1 + \frac{\gamma^2}{\gamma+1} \beta_2^2\right] & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ a \\ b \\ c \end{pmatrix} = a \frac{\gamma^2}{\gamma+1} \beta'_2 \beta_1 \beta_2 + b \beta'_2 \left[1 + \frac{\gamma^2}{\gamma+1} \beta_2^2\right] + c \beta'_3$$

故变换后方向为

$$\begin{pmatrix} a \left[1 + \frac{\gamma^2}{\gamma+1} \beta_1^2\right] + b \frac{\gamma^2}{\gamma+1} \beta_1 \beta_2 \\ a \frac{\gamma^2}{\gamma+1} \beta_1 \beta_2 + b \left[1 + \frac{\gamma^2}{\gamma+1} \beta_2^2\right] + c \frac{\beta'_2 \beta'_3}{2} \\ a \frac{\gamma^2}{\gamma+1} \beta_1 \beta_2 \frac{\beta'_2 \beta'_3}{2} + b \left[1 + \frac{\gamma^2}{\gamma+1} \beta_2^2\right] \frac{\beta'_2 \beta'_3}{2} + c \end{pmatrix}$$

$$\vec{y}'_{\text{LxLy}} = \begin{pmatrix} 1 & 0 & -\beta'_2 & -\beta'_3 \\ 0 & 1 & 0 & 0 \\ -\beta'_2 & 0 & 1 & \frac{1}{2} \beta'_2 \beta'_3 \\ -\beta'_3 & 0 & \frac{1}{2} \beta'_2 \beta'_3 & 1 \end{pmatrix} \begin{pmatrix} \gamma & -\gamma \beta_1 & -\gamma \beta_2 & 0 \\ -\gamma \beta_1 \left[1 + \frac{\gamma^2}{\gamma+1} \beta_1^2\right] & \frac{\gamma^2}{\gamma+1} \beta_1 \beta_2 & 0 & 0 \\ -\gamma \beta_2 & \frac{\gamma^2}{\gamma+1} \beta_1 \beta_2 & \left[1 + \frac{\gamma^2}{\gamma+1} \beta_2^2\right] & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ a \\ b \\ c \end{pmatrix}$$

$$= \begin{pmatrix} \gamma + \gamma \beta'_2 \beta_2 & -\gamma \beta_1 - \frac{\gamma^2}{\gamma+1} \beta_1 \beta_2 \beta'_2 & -\gamma \beta_2 - \beta'_2 \left[1 + \frac{\gamma^2}{\gamma+1} \beta_2^2\right] & -\beta'_3 \\ -\gamma \beta_1 & \left[1 + \frac{\gamma^2}{\gamma+1} \beta_1^2\right] & \frac{\gamma^2}{\gamma+1} \beta_1 \beta_2 & 0 \\ -\gamma \beta'_2 - \gamma \beta_2 & \gamma \beta_1 \beta'_2 + \frac{\gamma^2}{\gamma+1} \beta_1 \beta_2 & \gamma \beta'_2 \beta_2 + \left[1 + \frac{\gamma^2}{\gamma+1} \beta_2^2\right] & \frac{1}{2} \beta'_2 \beta'_3 \\ -\gamma \beta'_3 - \frac{1}{2} \gamma \beta_2 \beta'_2 \beta'_3 & \gamma \beta'_3 \beta_1 + \frac{1}{2} \frac{\gamma^2}{\gamma+1} \beta'_2 \beta'_3 \beta_1 \beta_2 & \gamma \beta'_3 \beta_2 + \frac{1}{2} \beta'_2 \beta'_3 \left[1 + \frac{\gamma^2}{\gamma+1} \beta_2^2\right] & 1 \end{pmatrix} \begin{pmatrix} 0 \\ a \\ b \\ c \end{pmatrix}$$

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$$\text{方向: } \begin{pmatrix} a \left[1 + \frac{\gamma^2}{\gamma+1} \beta_1^2 \right] + b \frac{\gamma^2}{\gamma+1} \beta_1 \beta_2 \\ a \gamma \beta_1 \beta'_2 + b \gamma \beta'_2 \beta_2 + a \frac{\gamma^2}{\gamma+1} \beta_1 \beta_2 + b \left[1 + \frac{\gamma^2}{\gamma+1} \beta_2^2 \right] + c \frac{\beta'_2 \beta'_3}{2} \\ a \gamma \beta'_3 \beta_1 + b \gamma \beta'_3 \beta_2 + a \frac{\gamma^2}{\gamma+1} \beta_1 \beta_2 \frac{\beta'_2 \beta'_3}{2} + b \left[1 + \frac{\gamma^2}{\gamma+1} \beta_2^2 \right] \frac{\beta'_2 \beta'_3}{2} + c \end{pmatrix}$$

可见 $\vec{y}'_{\text{LxLy}}$ 较 $\vec{y}'_{\text{LxLy}}$ (上式蓝色项) 多出一个少量, 这点与之前探讨的拟向变换不同。

由于自旋矢量经历单次拟向变换后与单次Lorentz变换后空间方向一致

$$\therefore \begin{pmatrix} \left[1 + \frac{\gamma^2}{\gamma+1} \beta_1^2 \right] & \frac{\gamma^2}{\gamma+1} \beta_1 \beta_2 & \frac{\gamma^2}{\gamma+1} \beta_1 \beta'_3 \\ \frac{\gamma^2}{\gamma+1} \beta_1 \beta_2 & \left[1 + \frac{\gamma^2}{\gamma+1} \beta_2^2 \right] & \frac{\gamma^2}{\gamma+1} \beta_2 \beta'_3 \\ \frac{\gamma^2}{\gamma+1} \beta_1 \beta'_3 & \frac{\gamma^2}{\gamma+1} \beta_2 \beta'_3 & 1 \end{pmatrix} \hat{R} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} a \left[1 + \frac{\gamma^2}{\gamma+1} \beta_1^2 \right] + b \frac{\gamma^2}{\gamma+1} \beta_1 \beta_2 \\ a \frac{\gamma^2}{\gamma+1} \beta_1 \beta_2 + b \left[1 + \frac{\gamma^2}{\gamma+1} \beta_2^2 \right] + c \frac{\beta'_2 \beta'_3}{2} \\ a \frac{\gamma^2}{\gamma+1} \beta_1 \beta_2 \frac{\beta'_2 \beta'_3}{2} + b \left[1 + \frac{\gamma^2}{\gamma+1} \beta_2^2 \right] \frac{\beta'_2 \beta'_3}{2} + c \end{pmatrix}$$

简单起见令 $\beta_2 = 0, \beta'_3 = 0$

$$\therefore \begin{pmatrix} \gamma & \frac{\gamma-1}{\beta_1} \beta'_2 & 0 \\ \frac{\gamma-1}{\beta_1} \beta'_2 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & -\delta\varphi & 0 \\ \delta\varphi & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} a\gamma \\ b \\ c \end{pmatrix} \quad \dots (11)$$

$$\therefore \begin{cases} a \frac{\gamma-1}{\beta_1} \beta'_2 \delta\varphi + b \left(\frac{\gamma-1}{\beta_1} \beta'_2 - \gamma \delta\varphi \right) = 0 \\ a \left(\frac{\gamma-1}{\beta_1} \beta'_2 + \delta\varphi \right) - b \frac{\gamma-1}{\beta_1} \beta'_2 \delta\varphi = 0 \end{cases} \quad \dots (12)$$

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$$-\frac{\left(\frac{\gamma-1}{\beta_1}\beta'_2 - \gamma\delta\varphi\right)}{\frac{\gamma-1}{\beta_1}\beta'_2\delta\varphi} = \frac{\frac{\gamma-1}{\beta_1}\beta'_2\delta\varphi}{\frac{\gamma-1}{\beta_1}\beta'_2 + \delta\varphi}$$

做一个线性近似:

$$\frac{(\gamma-1)^2}{\beta_1^2}\beta'^2_2 - \frac{(\gamma-1)^2}{\beta_1}\beta'_2\delta\varphi - \gamma(\delta\varphi)^2 = 0$$

$$\therefore \delta\varphi = \frac{\frac{(\gamma-1)^2}{\beta_1}\beta'_2 \pm \sqrt{\frac{(\gamma-1)^4}{\beta_1^2}\beta'^2_2 + 4\gamma\frac{(\gamma-1)^2}{\beta_1^2}\beta'^2_2}}{-2\gamma}$$

$$\text{其中 } \frac{(\gamma-1)^4}{\beta_1^2}\beta'^2_2 + 4\gamma\frac{(\gamma-1)^2}{\beta_1^2}\beta'^2_2 = (\gamma^2-1)^2\frac{\beta'^2_2}{\beta_1^2} > 0 \text{ 恒成立}$$

$$\therefore \begin{cases} \delta\varphi_1 = -\frac{(\gamma-1)}{\beta_1}\beta'_2 \\ \delta\varphi_2 = \frac{(\gamma-1)}{\beta_1\gamma}\beta'_2 \end{cases}$$

这个结果基于线性近似，与Lorentz-Thomas进动一致。另外，这里 $\beta''_2 = \beta'_2$ ，因为 β'_2 最后对应的加速度用的是坐标时， $\beta''_2 = \frac{\beta'_2}{\gamma}$ 的情况用的是固有时(即 β'_2 为 Σ_2 参考系下的微元速度)，这两种方案都转换为固有时后Thomas进动表达形式完全一致。

若考虑非线性效应，从(11)式出发，并用 $\hat{R}$ 代替旋转矩阵:

$$\hat{R} = \frac{1}{\gamma - \beta'^2_2 \frac{(\gamma-1)^2}{\beta_1^2}} \begin{pmatrix} 1 & \frac{\beta'_2}{\beta_1}(1-\gamma) & 0 \\ \frac{\beta'_2}{\beta_1}(1-\gamma) & \gamma & 0 \\ 0 & 0 & \gamma \end{pmatrix} \begin{pmatrix} \gamma & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & \frac{\beta'_2}{\beta_1\gamma}(1-\gamma) & 0 \\ \frac{\beta'_2}{\beta_1}(1-\gamma) & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad \cdots(13)$$

拟向变换自旋赝偶极矢量

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一个小插曲：现在的(13) $\hat{R}$ 矩阵相当于将 $\vec{\beta}$ 与 $\vec{\beta}$ 方向对换，然后对 $\vec{\beta}$ 和 $\vec{\beta} \times \vec{\beta}$ 方向作线性拟向变换。

但(13)式有一个问题，在于(11)式左侧和右侧的归一化系数差了一个小量，我原本是想忽略它，

后来发现由于后面要减一个单位阵，导致一阶小量被暴露出来，所以一阶小量不能忽略。（高阶小量与dt的微商仍为小量）

修改后的(11)式为：

$$\frac{1}{\sqrt{1+\gamma^2(a\beta_1+b\beta'_2)^2}} \begin{pmatrix} \gamma & \frac{\gamma-1}{\beta_1}\beta'_2 & 0 \\ \frac{\gamma-1}{\beta_1}\beta'_2 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \hat{R} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \frac{1}{\sqrt{1+\gamma^2\beta_1^2 a^2 + \beta'^2_2 b^2}} \begin{pmatrix} a\gamma \\ b \\ c \end{pmatrix} \quad \dots(11')$$

$$\therefore \hat{R} = \sqrt{\frac{1+\gamma^2(a\beta_1+b\beta'_2)^2}{1+\gamma^2\beta_1^2 a^2 + \beta'^2_2 b^2}} \frac{1}{\gamma - \beta'^2_2 \frac{(\gamma-1)^2}{\beta_1^2}} \begin{pmatrix} 1 & \frac{\beta'_2}{\beta_1}(1-\gamma) & 0 \\ \frac{\beta'_2}{\beta_1}(1-\gamma) & \gamma & 0 \\ 0 & 0 & \gamma \end{pmatrix} \begin{pmatrix} \gamma & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\underline{\underline{\text{系数保留一阶小量}}} \begin{pmatrix} 1 + \frac{\gamma^2 a \beta_1 b \beta'_2}{1 + \gamma^2 \beta_1^2 a^2} & \frac{\beta'_2}{\beta_1 \gamma} (1 - \gamma) & 0 \\ \frac{\beta'_2}{\beta_1} (1 - \gamma) & 1 + \frac{\gamma^2 a \beta_1 b \beta'_2}{1 + \gamma^2 \beta_1^2 a^2} & 0 \\ 0 & 0 & 1 + \frac{\gamma^2 a \beta_1 b \beta'_2}{1 + \gamma^2 \beta_1^2 a^2} \end{pmatrix} \quad \dots(13')$$

若连续Lorentz变换结果计算，则(11)式变为：

$$\begin{pmatrix} \gamma & \frac{\gamma-1}{\beta_1}\beta'_2 & 0 \\ \frac{\gamma-1}{\beta_1}\beta'_2 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & -\delta\varphi & 0 \\ \delta\varphi & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} a\gamma \\ a\gamma\beta_1\beta'_2 + b \\ c \end{pmatrix}$$

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$$\therefore \begin{cases} a \frac{\gamma-1}{\gamma\beta_1} \beta'_2 \delta\varphi + b \left(\frac{\gamma-1}{\beta_1} \beta'_2 - \gamma \delta\varphi \right) = 0 \\ a \left(\delta\varphi - \frac{\gamma-1}{\gamma\beta_1} \beta'_2 \right) - b \frac{\gamma-1}{\gamma\beta_1} \beta'_2 \delta\varphi = 0 \end{cases}$$

两个方程在 $\delta\varphi = \frac{\gamma-1}{\gamma\beta_1} \beta'_2$ 时同时成立，正确导出了 Lorentz - Thomas 进动。

导致连续 Lorentz 变换与连续拟向变换不同的原因在红色项，更进一步讲是因为 $\vec{y}'_{LxLy}$ 中第一步变换后时间项非零，通过第二步变换 involve 到空间项；而 $\vec{m}'\vec{m}\vec{x}$ 则没有这种现象。

下面探讨下拟向变换的动力学 - 运动学一致性

$$\begin{aligned} & L_{I,\tau}[L_{I,d\tau}(x^\mu)] - L_{I,d\tau}[L_{I,\tau}(x^\mu)] \\ &= \left[1 + (\vec{k}^T \vec{x})^2 + (\vec{k}'^T \vec{m}\vec{x})^2 \right]^{-1/2} \vec{m}'\vec{m}\vec{x} + \vec{n}\vec{x} - \left[1 + (\vec{k}'^T \vec{x})^2 + (\vec{k}^T \vec{m}'\vec{x})^2 \right]^{-1/2} \vec{m}\vec{m}'\vec{x} - \vec{n}\vec{x} \\ &= \left[1 + (a\gamma\beta_1)^2 + (b\beta'_2)^2 \right]^{-1/2} \begin{pmatrix} 0 \\ a\gamma \\ b \\ c \end{pmatrix} - \left[1 + (b\beta'_2)^2 + (a\gamma\beta_1)^2 \right]^{-1/2} \begin{pmatrix} 0 \\ a\gamma \\ b \\ c \end{pmatrix} = 0 \end{aligned}$$

可见，在线性近似的前提下拟向变换严格满足动力学 - 运动学一致性。

并且，由于 $X_0 = 0$ 始终严格成立，故不存在类似 P55 中的额外项。

下一步要推导拟向变换自旋赝偶极矢量 Thomas-Hamiltonian，方法与前面推导经典

Thomas-Hamiltonian 类似，详细的过程我给放在附录2里了。这个推导里面最关键的问题是确定进动发生在什么空间，它不像经典 Thomas 进动那么显然。带入原子单位制 $\hbar$ 自旋

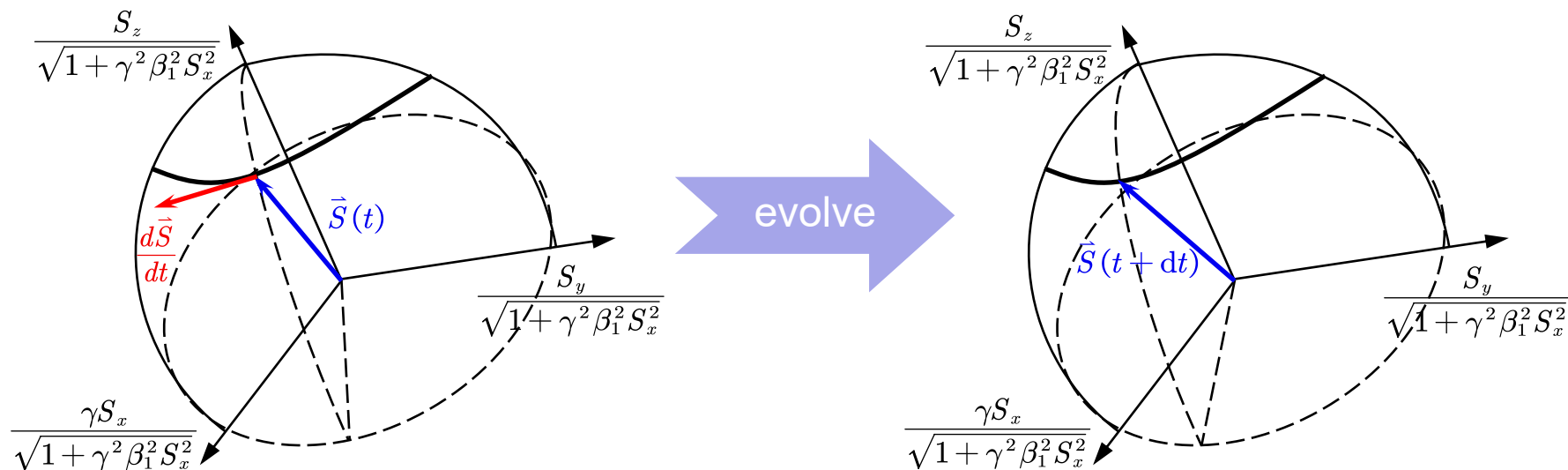
拟向变换自旋赝偶极矢量

(Imitative Transfer Pseudodipole Vector, ITPDV)

绝对值 $1/2$ ；经推导，拟向变换自旋赝偶极矢量Thomas进动的Hamiltonian表达式为：

$$H = \frac{\gamma}{2\beta^2} \vec{S}_\gamma \cdot (\dot{\vec{\beta}} \times \vec{\beta}) \frac{\frac{4}{3}(\vec{\beta} \cdot \vec{S})^2}{1 - \beta^2 + \frac{4}{3}(\vec{\beta} \cdot \vec{S})^2}, \quad S_{\gamma,i} = \frac{1}{\sqrt{1 + \frac{4}{3}\gamma^2(\vec{\beta} \cdot \vec{S})^2}} \frac{1}{\sqrt{1 - \beta_i^2}} S_i \quad \dots(14)$$

上式在 $\beta \ll 1$ 下近似成立，进动的物理图像如下。



拟向变换的理论基础是 $\hat{\vec{S}} = \hbar/2 \hat{\vec{\sigma}}$ 中角动量大小 $\hbar/2$ 在任意参考系下数值恒定，这使自旋角动量大小是Lorentz标量；狭义相对论运动学、动力学也是建立在光速 c 在任意参考系下数值恒定的基础假设上。可见，二者的理论有相通之处：都是基于物理常数在参考系变换下不变；但自旋 $(0, \vec{S})$ 、光速 $(0, \vec{c})$ 两个物理量的描述对象、描述性质截然不同，导致两个理论的数学形式、适用范围不同。

Lorentz 变换自旋赝偶极张量

(Lorentz Transfer Pseudodipole Tensor, LTPDT)

基于LTPDT的自旋动力学与基于ITPDV的自旋动力学完全不相干。LTPDT理论认为电子具有磁偶极矩和电偶极矩，并可以用二阶4-Tensor描述之。与ITPDV相比，LTPDT理论的优势主要有两点：

- 能够描述Dirac方程F-W变换的结果中电场旋轨耦合项；
- 能够将自旋偶极项引入Maxwell方程组，全面描述外场与电子(自旋)的相互作用，并可以解释自旋辐射现象；

仿照ITPDV定义拟向变换自旋赝偶极张量是可行的，但我并不打算这么做；因为这样定义的拟向变换不仅十分复杂，而且还将失去LTPDT的第二点优势(Maxwell方程Lorentz协变)。有相当多的针对方程的研究。

早在1971年，Fisher^[1]就已提出自旋磁偶极、电偶极随参考系变换互化的观点。最近的研究中，Silenko^[2-4]提出了带静电偶极和静磁偶极的Dirac $\frac{1}{2}$ 旋量的自旋动力学，并利用

[1] Rębilas, Krzysztof. "Subtleties of the Thomas precession." European Journal of Physics 36.4 (2015): 045007.

Lorentz 变换自旋赝偶极张量

(Lorentz Transfer Pseudodipole Tensor, LTPDT)

精确F-W变换关联了量子Dirac方程和经典自旋偶极张量动力学的结果，虽然与我们现在讨论的问题不完全一致，但仍有巨大的参考意义：

- Dirac方程解耦项中包含了电子在闵式时空背景中运动的Thomas动力学项，这对后续研究非惯性系下的二阶张量Thomas动力学有借鉴意义；
- 研究自旋辐射应当以自旋偶极密度为基本量，它未在Pauli方程中直接表达，下文中没有特殊说明时， $\vec{S}$ 代表实验室参考系下的自旋偶极密度；

依据前面对二阶张量Thomas动力学的研究成果：

$$\vec{S}_\mu \stackrel{\text{def}}{=} \gamma \mu_B \vec{\sigma}$$

$$\vec{S}_\eta \stackrel{\text{def}}{=} \gamma \mu_B \vec{\sigma} \times \vec{\beta}$$

$$= \vec{S}_\mu \times \vec{\beta}$$

$$\therefore S^{\mu\nu} \equiv (\vec{S}_\eta, -\vec{S}_\mu) = \begin{pmatrix} 0 & -S_\eta^1 & -S_\eta^2 & -S_\eta^3 \\ S_\eta^1 & 0 & S_\mu^3 & -S_\mu^2 \\ S_\eta^2 & -S_\mu^3 & 0 & S_\mu^1 \\ S_\eta^3 & S_\mu^2 & -S_\mu^1 & 0 \end{pmatrix}$$

$$= (\gamma \mu_B \vec{\sigma} \times \vec{\beta}, -\gamma \mu_B \vec{\sigma})$$

$$= (\vec{S}_\mu \times \vec{\beta}, -\vec{S}_\mu)$$

Lorentz 变换自旋赝偶极张量

(Lorentz Transfer Pseudodipole Tensor, LTPDT)

$$\begin{cases} \frac{\partial}{\partial t} \vec{S}_\eta \Big|_{\text{Thomas}} = \gamma^2 \vec{S}_\mu \times \frac{\partial \vec{\beta}}{\partial t} + \frac{\gamma^3}{\gamma+1} \vec{S}_\mu \times \left(\vec{\beta} \times \left(\vec{\beta} \times \frac{\partial \vec{\beta}}{\partial t} \right) \right) + \frac{\gamma^2}{\gamma+1} \left(\vec{\beta} \times \frac{\partial \vec{\beta}}{\partial t} \cdot \vec{S}_\mu \right) \vec{\beta} \\ \frac{\partial}{\partial t} \vec{S}_\mu \Big|_{\text{Thomas}} = -\gamma^2 (\vec{S}_\mu \times \vec{\beta}) \times \frac{\partial \vec{\beta}}{\partial t} - \frac{\gamma^3}{\gamma+1} (\vec{S}_\mu \times \vec{\beta}) \times \left(\vec{\beta} \times \left(\vec{\beta} \times \frac{\partial \vec{\beta}}{\partial t} \right) \right) + \frac{\gamma^2}{\gamma+1} \vec{S}_\mu \times \left(\vec{\beta} \times \frac{\partial \vec{\beta}}{\partial t} \right) \end{cases} \quad \dots(15)$$

化简到最后的結果是：

$$\begin{cases} \frac{\partial}{\partial t} \vec{S}_\eta \Big|_{\text{Thomas}} = -\gamma^2 \left(\frac{\partial \vec{\beta}}{\partial t} \times \vec{S}_\mu \Big|_{\perp} \right) - \gamma \left(\frac{\partial \vec{\beta}}{\partial t} \times \vec{S}_\mu \Big|_{\parallel} \right) + \gamma^2 \left(\vec{\beta} \cdot \left(\frac{\partial \vec{\beta}}{\partial t} \times \vec{S}_\mu \right) \right) \vec{\beta} \\ \frac{\partial}{\partial t} \vec{S}_\mu \Big|_{\text{Thomas}} = -\gamma^2 \left(\frac{\partial \vec{\beta}}{\partial t} \Big|_{\parallel} \times (\vec{\beta} \times \vec{S}_\mu) \right) - \gamma \left(\frac{\partial \vec{\beta}}{\partial t} \Big|_{\perp} \times (\vec{\beta} \times \vec{S}_\mu) \right) + \frac{\gamma^2}{\gamma+1} \vec{S}_\mu \times \left(\vec{\beta} \times \frac{\partial \vec{\beta}}{\partial t} \right) \end{cases} \quad \dots(16)$$

其中 $\vec{S}_\mu \Big|_{\perp}, \frac{\partial \vec{\beta}}{\partial t} \Big|_{\perp}$ 表示垂直于 $\vec{\beta}$ 方向的分量； $\vec{S}_\mu \Big|_{\parallel}, \frac{\partial \vec{\beta}}{\partial t} \Big|_{\parallel}$ 表示平行于 $\vec{\beta}$ 方向的分量。

仿照拟向变换Thomas动力学的处理办法，上式整合化作：

$$\begin{cases} \frac{\partial}{\partial t} \vec{S}_\eta \Big|_{\text{Thomas}} = -\gamma \left(\frac{\partial \vec{\beta}}{\partial t} \times \vec{S}_{\mu\gamma\perp} \right) + \gamma^2 \left(\vec{\beta} \cdot \left(\frac{\partial \vec{\beta}}{\partial t} \times \vec{S}_\mu \right) \right) \vec{\beta} \\ \frac{\partial}{\partial t} \vec{S}_\mu \Big|_{\text{Thomas}} = -\gamma \left(\frac{\partial \vec{\beta}}{\partial t} \Big|_{\gamma\parallel} \times (\vec{\beta} \times \vec{S}_\mu) \right) + \frac{\gamma^2}{\gamma+1} \vec{S}_\mu \times \left(\vec{\beta} \times \frac{\partial \vec{\beta}}{\partial t} \right) \end{cases}$$

Lorentz变换自旋赝偶极张量

(Lorentz Transfer Pseudodipole Tensor, LTPDT)

$$\text{其中 } \vec{S}_{\mu\gamma\perp} = \left(1 - \frac{\beta^2 S_\mu^2 - (\vec{\beta} \cdot \vec{S}_\mu)^2}{S_\mu^2} \right)^{-1/2} \vec{S}_\mu, \quad \left. \frac{\partial \vec{\beta}}{\partial t} \right|_{\gamma\parallel} = \left(1 - \frac{\left(\vec{\beta} \cdot \frac{\partial \vec{\beta}}{\partial t} \right)^2}{\beta^2} \right)^{-1/2} \frac{\partial \vec{\beta}}{\partial t}$$

但这是典型的非线性变换，不符合(15)式与(16)式的线性特征。对于拟向变换，(13')式定义了一个关于 S_γ 的线性空间，态矢量在该线性空间的投影发生Thomas进动从而导致拟向变换Thomas进动是一个关于 $\vec{S}$ 的非线性变换；而对于Lorentz变换自旋赝偶极张量的Thomas进动，(15)式的定义是确定的、离散的，并且没有定义类似的连续线性空间。这二者的区别类似量子力学的相干态与非相干态。总之，(16)式

不可被整合，只能将 $\vec{S}_\mu$ 、 $\frac{\partial \vec{\beta}}{\partial t}$ 的垂直分量与平行分量分开定义。

附录1 一般二阶张量Thomas动力学表达式化简

第一项=

$$\begin{aligned}
 & \left\{ \frac{(2\gamma+4)\gamma^6(\vec{\beta} \cdot \dot{\vec{\beta}})}{(\gamma+1)^3} \vec{\beta} \vec{\beta}^T (\vec{\beta} \vec{\beta}^T)^{ip} + \frac{\gamma^4}{(\gamma+1)^2} (\vec{\beta} \dot{\vec{\beta}}^T) (\vec{\beta} \vec{\beta}^T)^{ip} + \frac{\gamma^4}{(\gamma+1)^2} \vec{\beta} \vec{\beta}^T (\vec{\beta} \dot{\vec{\beta}}^T)^{ip} \right\} \\
 & \left\{ \vec{x} + \frac{\gamma^4}{(\gamma+1)^2} \vec{\beta} \vec{\beta}^T [(\vec{\beta} \vec{\beta}^T) \cdot \vec{x}] - \frac{\gamma^3}{\gamma+1} \vec{\beta} \vec{\beta}^T (\vec{\beta} \cdot \vec{x}_1) + \gamma^2 \vec{\beta} \vec{\beta}^T x^{11} + \frac{\gamma^2}{\gamma+1} (\vec{\beta} \vec{\beta}^T) \vec{x}^s - \gamma \vec{\beta} \vec{x}_{lim1}^T - \gamma \vec{x}_{col1} \vec{\beta}^T \right\} \\
 & = \frac{(2\gamma+4)\gamma^6(\vec{\beta} \cdot \dot{\vec{\beta}})}{(\gamma+1)^3} \vec{\beta} \vec{\beta}^T [(\vec{\beta} \vec{\beta}^T) \cdot \vec{x}] + \frac{(2\gamma+4)\gamma^{10}(\vec{\beta} \cdot \dot{\vec{\beta}})}{(\gamma+1)^5} [(\vec{\beta} \vec{\beta}^T) \cdot \vec{x}] [(\vec{\beta} \vec{\beta}^T) \cdot (\vec{\beta} \vec{\beta}^T)] \vec{\beta} \vec{\beta}^T \\
 & - \frac{(2\gamma+4)\gamma^9(\vec{\beta} \cdot \dot{\vec{\beta}})}{(\gamma+1)^4} [(\vec{\beta} \vec{\beta}^T) \cdot (\vec{\beta} \vec{\beta}^T)] (\vec{\beta} \cdot \vec{x}_1) \vec{\beta} \vec{\beta}^T \\
 & + \frac{(2\gamma+4)\gamma^8(\vec{\beta} \cdot \dot{\vec{\beta}})}{(\gamma+1)^3} [(\vec{\beta} \vec{\beta}^T) \cdot (\vec{\beta} \vec{\beta}^T)] x^{11} \vec{\beta} \vec{\beta}^T + \frac{(2\gamma+4)\gamma^8(\vec{\beta} \cdot \dot{\vec{\beta}})}{(\gamma+1)^4} [(\vec{\beta} \vec{\beta}^T) \cdot (\vec{\beta} \vec{\beta}^T \vec{x}^s)] \vec{\beta} \vec{\beta}^T \\
 & - \frac{(2\gamma+4)\gamma^7(\vec{\beta} \cdot \dot{\vec{\beta}})}{(\gamma+1)^3} [(\vec{\beta} \vec{\beta}^T) \cdot (\vec{\beta} \vec{x}_{lim1}^T)] \vec{\beta} \vec{\beta}^T - \frac{(2\gamma+4)\gamma^7(\vec{\beta} \cdot \dot{\vec{\beta}})}{(\gamma+1)^3} [(\vec{\beta} \vec{\beta}^T) \cdot (\vec{x}_{col1} \vec{\beta}^T)] \vec{\beta} \vec{\beta}^T \\
 & + \frac{\gamma^4}{(\gamma+1)^2} [(\vec{\beta} \vec{\beta}^T) \cdot \vec{x}] (\vec{\beta} \dot{\vec{\beta}}^T) + \frac{\gamma^8}{(\gamma+1)^4} [(\vec{\beta} \vec{\beta}^T) \cdot \vec{x}] [(\vec{\beta} \vec{\beta}^T) \cdot (\vec{\beta} \vec{\beta}^T)] (\vec{\beta} \dot{\vec{\beta}}^T) - \frac{\gamma^7}{(\gamma+1)^3} [(\vec{\beta} \vec{\beta}^T) \cdot (\vec{\beta} \vec{\beta}^T)] (\vec{\beta} \cdot \vec{x}_1) (\vec{\beta} \dot{\vec{\beta}}^T) \\
 & + \frac{\gamma^6}{(\gamma+1)^2} [(\vec{\beta} \vec{\beta}^T) \cdot (\vec{\beta} \vec{\beta}^T)] x^{11} (\vec{\beta} \dot{\vec{\beta}}^T) + \frac{\gamma^6}{(\gamma+1)^3} [(\vec{\beta} \vec{\beta}^T) \cdot (\vec{\beta} \vec{\beta}^T \vec{x}^s)] (\vec{\beta} \dot{\vec{\beta}}^T) \\
 & - \frac{\gamma^5}{(\gamma+1)^2} [(\vec{\beta} \vec{\beta}^T) \cdot (\vec{\beta} \vec{x}_{lim1}^T)] (\vec{\beta} \dot{\vec{\beta}}^T) - \frac{\gamma^5}{(\gamma+1)^2} [(\vec{\beta} \vec{\beta}^T) \cdot (\vec{x}_{col1} \vec{\beta}^T)] (\vec{\beta} \dot{\vec{\beta}}^T) \\
 & + \frac{\gamma^4}{(\gamma+1)^2} [(\vec{\beta} \dot{\vec{\beta}}^T) \cdot \vec{x}] \vec{\beta} \vec{\beta}^T + \frac{\gamma^8}{(\gamma+1)^4} [(\vec{\beta} \vec{\beta}^T) \cdot \vec{x}] [(\vec{\beta} \dot{\vec{\beta}}^T) \cdot (\vec{\beta} \vec{\beta}^T)] \vec{\beta} \vec{\beta}^T - \frac{\gamma^7}{(\gamma+1)^3} [(\vec{\beta} \dot{\vec{\beta}}^T) \cdot (\vec{\beta} \vec{\beta}^T)] (\vec{\beta} \cdot \vec{x}_1) \vec{\beta} \vec{\beta}^T \\
 & + \frac{\gamma^6}{(\gamma+1)^2} [(\vec{\beta} \dot{\vec{\beta}}^T) \cdot (\vec{\beta} \vec{\beta}^T)] x^{11} \vec{\beta} \vec{\beta}^T + \frac{\gamma^6}{(\gamma+1)^3} [(\vec{\beta} \dot{\vec{\beta}}^T) \cdot (\vec{\beta} \vec{\beta}^T \vec{x}^s)] \vec{\beta} \vec{\beta}^T \\
 & - \frac{\gamma^5}{(\gamma+1)^2} [(\vec{\beta} \dot{\vec{\beta}}^T) \cdot (\vec{\beta} \vec{x}_{lim1}^T)] \vec{\beta} \vec{\beta}^T - \frac{\gamma^5}{(\gamma+1)^2} [(\vec{\beta} \dot{\vec{\beta}}^T) \cdot (\vec{x}_{col1} \vec{\beta}^T)] \vec{\beta} \vec{\beta}^T
 \end{aligned}$$

附录1 一般二阶张量Thomas动力学表达式化简

作部分项代换

$$\left[(\vec{\beta} \vec{\beta}^T) \cdot (\vec{\beta} \vec{\beta}^T) \right] = \sum_{ij} (\beta_i \beta_j)^2 = \sum_i \beta_i^2 \sum_j \beta_j^2 = \beta^2 \beta^2 = \beta^4 = \frac{1}{\gamma^4} (\gamma + 1)^2 (\gamma - 1)^2$$

$$\left[(\vec{\beta} \dot{\vec{\beta}}^T) \cdot (\vec{\beta} \vec{\beta}^T) \right] = 2 \sum_{ij} \dot{\beta}_i \beta_j \beta_i \beta_j = \sum_i (\dot{\beta}_i^2) \sum_j \beta_j^2 = 2 \dot{\beta} \beta^3 = 2 \left(\vec{\beta} \cdot \dot{\vec{\beta}} \right) \frac{1}{\gamma^2} (\gamma + 1) (\gamma - 1)$$

$$\left[(\vec{\beta} \vec{\beta}^T) \cdot (\vec{\beta} \vec{\beta}^T \vec{x}^S) \right] = \beta^2 \sum_{ij} \beta_i \beta_j x_{ij}^S = \beta^2 \left[(\vec{\beta} \vec{\beta}^T) \cdot \vec{x}^S \right] = \frac{1}{\gamma^2} (\gamma + 1) (\gamma - 1) \left[(\vec{\beta} \vec{\beta}^T) \cdot \vec{x}^S \right]$$

$$\left[(\vec{\beta} \dot{\vec{\beta}}^T) \cdot (\vec{\beta} \vec{\beta}^T \vec{x}^S) \right] = \beta \dot{\beta} \sum_{ij} \beta_i \beta_j x_{ij}^S + \beta^2 \sum_{ij} \beta_i \dot{\beta}_j x_{ij}^S = \beta \dot{\beta} \left[(\vec{\beta} \vec{\beta}^T) \cdot \vec{x}^S \right] + \frac{1}{\gamma^2} (\gamma + 1) (\gamma - 1) \left[(\vec{\beta} \dot{\vec{\beta}}^T) \cdot \vec{x}^S \right]$$

$$\left[(\vec{\beta} \vec{\beta}^T) \cdot (\vec{\beta} \vec{x}_{lin1}^T) \right] = \sum_{ij} \beta_i \beta_j \beta_i x_{lin1,j} = \sum_i \beta_i^2 \sum_j \beta_j x_{lin1,j} = \frac{1}{\gamma^2} (\gamma + 1) (\gamma - 1) (\vec{\beta} \cdot \vec{x}_{lin1})$$

$$\left[(\vec{\beta} \dot{\vec{\beta}}^T) \cdot (\vec{\beta} \vec{x}_{lin1}^T) \right] = \sum_i \dot{\beta}_i \beta_i \sum_j \beta_j x_{lin1,j} + \sum_i \beta_i^2 \sum_j \dot{\beta}_j x_{lin1,j} = \left(\vec{\beta} \cdot \dot{\vec{\beta}} \right) (\vec{\beta} \cdot \vec{x}_{lin1}) + \frac{1}{\gamma^2} (\gamma + 1) (\gamma - 1) \left(\dot{\vec{\beta}} \cdot \vec{x}_{lin1} \right)$$

$$\left[(\vec{\beta} \vec{\beta}^T) \cdot (\vec{x}_{col1} \vec{\beta}^T) \right] = \sum_{ij} \beta_i \beta_j x_{col1,i} \beta_j = \sum_j \beta_j^2 \sum_i \beta_i x_{col1,i} = \frac{1}{\gamma^2} (\gamma + 1) (\gamma - 1) (\vec{\beta} \cdot \vec{x}_{col1})$$

$$\left[(\vec{\beta} \dot{\vec{\beta}}^T) \cdot (\vec{x}_{col1} \vec{\beta}^T) \right] = \sum_j \dot{\beta}_j \beta_j \sum_i \beta_i x_{col1,i} + \sum_j \beta_j^2 \sum_i \dot{\beta}_i x_{col1,i} = \left(\vec{\beta} \cdot \dot{\vec{\beta}} \right) (\vec{\beta} \cdot \vec{x}_{col1}) + \frac{1}{\gamma^2} (\gamma + 1) (\gamma - 1) \left(\dot{\vec{\beta}} \cdot \vec{x}_{col1} \right)$$

上式考虑到

$$\beta^2 = \frac{1}{\gamma^2} (\gamma^2 - 1) = \frac{1}{\gamma^2} (\gamma + 1) (\gamma - 1)$$

$$\beta^4 = \frac{1}{\gamma^2} (\gamma^2 - 1) = \frac{1}{\gamma^4} (\gamma + 1)^2 (\gamma - 1)^2$$

附录1 一般二阶张量Thomas动力学表达式化简

第一项

$$\begin{aligned}
 &= \frac{(2\gamma^3 - 2\gamma + 6)\gamma^6 \left(\vec{\beta} \cdot \dot{\vec{\beta}} \right)}{(\gamma + 1)^3} \left[(\vec{\beta} \vec{\beta}^T) \cdot \vec{x} \right] (\vec{\beta} \vec{\beta}^T) + \frac{\gamma^4}{(\gamma + 1)^2} \left[(\vec{\beta} \dot{\vec{\beta}}^T) \cdot \vec{x} \right] (\vec{\beta} \vec{\beta}^T) + \frac{(\gamma^2 - 2\gamma + 2)\gamma^4}{(\gamma + 1)^2} \left[(\vec{\beta} \vec{\beta}^T) \cdot \vec{x} \right] (\vec{\beta} \dot{\vec{\beta}}^T) \\
 &+ \frac{(2\gamma^2 + 2\gamma - 3)\gamma^6 \left(\vec{\beta} \cdot \dot{\vec{\beta}} \right)}{(\gamma + 1)^3} \left[(\vec{\beta} \vec{\beta}^T) \cdot \vec{x}^S \right] (\vec{\beta} \vec{\beta}^T) + \frac{(\gamma - 1)\gamma^4}{2(\gamma + 1)^2} \left[(\vec{\beta} \dot{\vec{\beta}}^T) \cdot \vec{x}^S \right] (\vec{\beta} \vec{\beta}^T) + \frac{(\gamma - 1)\gamma^4}{(\gamma + 1)^2} \left[(\vec{\beta} \vec{\beta}^T) \cdot \vec{x}^S \right] (\vec{\beta} \dot{\vec{\beta}}^T) \\
 &- \frac{(2\gamma^3 + 2\gamma^2 - 2\gamma - 1)\gamma^5 \left(\vec{\beta} \cdot \dot{\vec{\beta}} \right)}{(\gamma + 1)^2} (\vec{\beta} \cdot \vec{x}_1) (\vec{\beta} \vec{\beta}^T) - \frac{(\gamma - 1)\gamma^3}{\gamma + 1} (\vec{\beta} \cdot \vec{x}_1) (\vec{\beta} \dot{\vec{\beta}}^T) - \frac{(\gamma^2 - \gamma)\gamma^3}{\gamma + 1} (\vec{\beta} \cdot \vec{x}_1) (\vec{\beta} \dot{\vec{\beta}}^T) \\
 &+ \frac{(2\gamma^3 - 4\gamma + 2)\gamma^4 \left(\vec{\beta} \cdot \dot{\vec{\beta}} \right)}{\gamma + 1} x^{11} (\vec{\beta} \vec{\beta}^T) + (\gamma - 1)^2 \gamma^2 x^{11} (\vec{\beta} \dot{\vec{\beta}}^T)
 \end{aligned}$$

注意虽然 $\vec{\beta} \dot{\vec{\beta}}^T \neq \dot{\vec{\beta}} \vec{\beta}^T$

但由于 $\vec{x}^S$ 是对称矩阵, 故 $(\vec{\beta} \dot{\vec{\beta}}^T) \cdot \vec{x}^S = \sum_{ij} \beta_i \dot{\beta}_j \vec{x}_{ij}^S = \sum_{ij} \beta_i \dot{\beta}_j \vec{x}_{ji}^S = (\dot{\vec{\beta}} \vec{\beta}^T) \cdot \vec{x}^S = \frac{1}{2} (\vec{\beta} \dot{\vec{\beta}}^T) \cdot \vec{x}^S$

附录1 一般二阶张量Thomas动力学表达式化简

第二项 =

$$\begin{aligned}
 & \left[\frac{(2\gamma+3)\gamma^5(\vec{\beta} \cdot \dot{\vec{\beta}})}{(\gamma+1)^2} (\vec{\beta}\vec{\beta}^T) \vec{\beta}^T + \frac{\gamma^3}{\gamma+1} (\vec{\beta}\dot{\vec{\beta}}^T) \vec{\beta}^T + \frac{\gamma^3}{\gamma+1} (\vec{\beta}\vec{\beta}^T) \dot{\vec{\beta}}^T \right] \\
 & \left\{ -\frac{2\gamma^3}{\gamma+1} \vec{\beta} [(\vec{\beta}\vec{\beta}^T) \cdot \vec{x}] - \gamma \vec{x}^s \vec{\beta} + \frac{\gamma^3}{\gamma+1} \vec{\beta} (\vec{\beta} \cdot \vec{x}_1) + \gamma \vec{x}_1 + \gamma^2 \vec{\beta} (\vec{\beta} \cdot \vec{x}_1 - 2x^{11}) \right\} \\
 & = -\frac{(4\gamma+6)\gamma^8(\vec{\beta} \cdot \dot{\vec{\beta}})}{(\gamma+1)^3} (\vec{\beta}\vec{\beta}^T) (\vec{\beta}^T \vec{\beta}) [(\vec{\beta}\vec{\beta}^T) \cdot \vec{x}] - \frac{(2\gamma+3)\gamma^6(\vec{\beta} \cdot \dot{\vec{\beta}})}{(\gamma+1)^2} (\vec{\beta}\vec{\beta}^T) [\vec{\beta}^T (\vec{x}^s \vec{\beta})] + \frac{(2\gamma+3)\gamma^8(\vec{\beta} \cdot \dot{\vec{\beta}})}{(\gamma+1)^3} (\vec{\beta}\vec{\beta}^T) (\vec{\beta}^T \vec{\beta}) (\vec{\beta} \cdot \vec{x}_1) \\
 & + \frac{(2\gamma+3)\gamma^6(\vec{\beta} \cdot \dot{\vec{\beta}})}{(\gamma+1)^2} (\vec{\beta}\vec{\beta}^T) (\vec{\beta}^T \vec{x}_1) + \frac{(2\gamma+3)\gamma^7(\vec{\beta} \cdot \dot{\vec{\beta}})}{(\gamma+1)^2} (\vec{\beta}\vec{\beta}^T) (\vec{\beta}^T \vec{\beta}) (\vec{\beta} \cdot \vec{x}_1 - 2x^{11}) \\
 & - \frac{2\gamma^6}{(\gamma+1)^2} (\vec{\beta}\dot{\vec{\beta}}^T) (\vec{\beta}^T \vec{\beta}) [(\vec{\beta}\vec{\beta}^T) \cdot \vec{x}] - \frac{\gamma^4}{\gamma+1} (\vec{\beta}\dot{\vec{\beta}}^T) [\vec{\beta}^T (\vec{x}^s \vec{\beta})] + \frac{\gamma^6}{(\gamma+1)^2} (\vec{\beta}\dot{\vec{\beta}}^T) (\vec{\beta}^T \vec{\beta}) (\vec{\beta} \cdot \vec{x}_1) + \frac{\gamma^4}{\gamma+1} (\vec{\beta}\dot{\vec{\beta}}^T) (\vec{\beta}^T \vec{x}_1) \\
 & + \frac{\gamma^5}{\gamma+1} (\vec{\beta}\dot{\vec{\beta}}^T) (\vec{\beta}^T \vec{\beta}) (\vec{\beta} \cdot \vec{x}_1 - 2x^{11}) \\
 & - \frac{2\gamma^6}{(\gamma+1)^2} (\vec{\beta}\vec{\beta}^T) (\dot{\vec{\beta}}^T \vec{\beta}) [(\vec{\beta}\vec{\beta}^T) \cdot \vec{x}] - \frac{\gamma^4}{\gamma+1} (\vec{\beta}\vec{\beta}^T) [\dot{\vec{\beta}}^T (\vec{x}^s \vec{\beta})] + \frac{\gamma^6}{(\gamma+1)^2} (\vec{\beta}\vec{\beta}^T) (\dot{\vec{\beta}}^T \vec{\beta}) (\vec{\beta} \cdot \vec{x}_1) + \frac{\gamma^4}{\gamma+1} (\vec{\beta}\vec{\beta}^T) (\dot{\vec{\beta}}^T \vec{x}_1) \\
 & + \frac{\gamma^5}{\gamma+1} (\vec{\beta}\vec{\beta}^T) (\dot{\vec{\beta}}^T \vec{\beta}) (\vec{\beta} \cdot \vec{x}_1 - 2x^{11})
 \end{aligned}$$

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第二项

$$\begin{aligned}
 &= -\frac{2(2\gamma^2 + \gamma - 2)\gamma^6(\vec{\beta} \cdot \dot{\vec{\beta}})}{(\gamma + 1)^2} [(\vec{\beta}\vec{\beta}^T) \cdot \vec{x}] (\vec{\beta}\vec{\beta}^T) - \frac{2(\gamma - 1)\gamma^4}{\gamma + 1} [(\vec{\beta}\vec{\beta}^T) \cdot \vec{x}] (\vec{\beta}\dot{\vec{\beta}}^T) \\
 &- \frac{(2\gamma + 3)\gamma^6(\vec{\beta} \cdot \dot{\vec{\beta}})}{(\gamma + 1)^2} [(\vec{\beta}\vec{\beta}^T) \cdot \vec{x}^s] (\vec{\beta}\vec{\beta}^T) - \frac{\gamma^4}{2(\gamma + 1)} \left[(\vec{\beta}\dot{\vec{\beta}}^T) \cdot \vec{x}^s \right] (\vec{\beta}\vec{\beta}^T) - \frac{\gamma^4}{\gamma + 1} [(\vec{\beta}\vec{\beta}^T) \cdot \vec{x}^s] (\vec{\beta}\dot{\vec{\beta}}^T) \\
 &+ \frac{2(2\gamma^3 + 3\gamma^2 - 1)\gamma^5(\vec{\beta} \cdot \dot{\vec{\beta}})}{(\gamma + 1)^2} (\vec{\beta} \cdot \vec{x}_1) (\vec{\beta}\vec{\beta}^T) + \frac{\gamma^4}{\gamma + 1} (\dot{\vec{\beta}} \cdot \vec{x}_1) (\vec{\beta}\vec{\beta}^T) + \frac{(2\gamma^2 - 1)\gamma^3}{\gamma + 1} (\vec{\beta} \cdot \vec{x}_1) (\vec{\beta}\dot{\vec{\beta}}^T) \\
 &- \frac{2(2\gamma^2 + \gamma - 2)\gamma^5(\vec{\beta} \cdot \dot{\vec{\beta}})}{\gamma + 1} x^{11} (\vec{\beta}\vec{\beta}^T) - 2(\gamma - 1)\gamma^3 x^{11} (\vec{\beta}\dot{\vec{\beta}}^T)
 \end{aligned}$$

上式考虑到:

$$(\vec{\beta}\vec{\beta}^T) \cdot \vec{x}^s = \vec{\beta} \cdot (\vec{x}^s \vec{\beta})$$

$$\dot{\vec{\beta}} \cdot (\vec{x}^s \vec{\beta}) = (\dot{\vec{\beta}}\vec{\beta}^T) \cdot \vec{x}^s = \frac{1}{2} (\vec{\beta}\dot{\vec{\beta}}^T) \cdot \vec{x}^s$$

第三项

$$\begin{aligned}
 &= \left[2\gamma^4 (\vec{\beta} \cdot \dot{\vec{\beta}}) \vec{\beta}\vec{\beta}^T + \gamma^2 (\vec{\beta}\dot{\vec{\beta}}^T) \right] \left\{ \gamma^2 [(\vec{\beta}\vec{\beta}^T) \cdot \vec{x}] - \gamma^2 (\vec{\beta} \cdot \vec{x}_1) + \gamma^2 x^{11} \right\} \\
 &= 2\gamma^4 (\vec{\beta} \cdot \dot{\vec{\beta}}) \vec{\beta}\vec{\beta}^T \gamma^2 [(\vec{\beta}\vec{\beta}^T) \cdot \vec{x}] - 2\gamma^4 (\vec{\beta} \cdot \dot{\vec{\beta}}) \vec{\beta}\vec{\beta}^T \gamma^2 (\vec{\beta} \cdot \vec{x}_1) + 2\gamma^4 (\vec{\beta} \cdot \dot{\vec{\beta}}) \vec{\beta}\vec{\beta}^T \gamma^2 x^{11} + \gamma^2 (\vec{\beta}\dot{\vec{\beta}}^T) \gamma^2 [(\vec{\beta}\vec{\beta}^T) \cdot \vec{x}] - \gamma^2 (\vec{\beta}\dot{\vec{\beta}}^T) \gamma^2 (\vec{\beta} \cdot \vec{x}_1) + \gamma^2 (\vec{\beta}\dot{\vec{\beta}}^T) \gamma^2 x^{11} \\
 &= 2\gamma^6 (\vec{\beta} \cdot \dot{\vec{\beta}}) [(\vec{\beta}\vec{\beta}^T) \cdot \vec{x}] \vec{\beta}\vec{\beta}^T - 2\gamma^6 (\vec{\beta} \cdot \dot{\vec{\beta}}) (\vec{\beta} \cdot \vec{x}_1) \vec{\beta}\vec{\beta}^T + 2\gamma^6 (\vec{\beta} \cdot \dot{\vec{\beta}}) x^{11} \vec{\beta}\vec{\beta}^T + \gamma^4 [(\vec{\beta}\dot{\vec{\beta}}^T) \cdot \vec{x}] (\vec{\beta}\dot{\vec{\beta}}^T) - \gamma^4 (\vec{\beta} \cdot \vec{x}_1) (\vec{\beta}\dot{\vec{\beta}}^T) + \gamma^4 x^{11} (\vec{\beta}\dot{\vec{\beta}}^T)
 \end{aligned}$$

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第一加第二项

$$\begin{aligned} &= \frac{(-2\gamma^3 - 6\gamma^2 + 10)\gamma^6 (\vec{\beta} \cdot \dot{\vec{\beta}})}{(\gamma + 1)^3} [(\vec{\beta} \vec{\beta}^T) \cdot \vec{x}] (\vec{\beta} \vec{\beta}^T) + \frac{\gamma^4}{(\gamma + 1)^2} [(\vec{\beta} \dot{\vec{\beta}}^T) \cdot \vec{x}] (\vec{\beta} \vec{\beta}^T) + \frac{(-\gamma^2 - 2\gamma + 4)\gamma^4}{(\gamma + 1)^2} [(\vec{\beta} \vec{\beta}^T) \cdot \vec{x}] (\vec{\beta} \dot{\vec{\beta}}^T) \\ &- \frac{3(\gamma + 2)\gamma^6 (\vec{\beta} \cdot \dot{\vec{\beta}})}{(\gamma + 1)^3} [(\vec{\beta} \vec{\beta}^T) \cdot \vec{x}^s] (\vec{\beta} \vec{\beta}^T) - \frac{\gamma^4}{(\gamma + 1)^2} [(\vec{\beta} \dot{\vec{\beta}}^T) \cdot \vec{x}^s] (\vec{\beta} \vec{\beta}^T) - \frac{2\gamma^4}{(\gamma + 1)^2} [(\vec{\beta} \vec{\beta}^T) \cdot \vec{x}^s] (\vec{\beta} \dot{\vec{\beta}}^T) \\ &+ \frac{(2\gamma^3 + 4\gamma^2 + 2\gamma - 1)\gamma^5 (\vec{\beta} \cdot \dot{\vec{\beta}})}{(\gamma + 1)^2} (\vec{\beta} \cdot \vec{x}_1) (\vec{\beta} \vec{\beta}^T) + \frac{\gamma^3}{\gamma + 1} (\vec{\beta} \cdot \vec{x}_1) (\vec{\beta} \dot{\vec{\beta}}^T) + \frac{(\gamma^2 + \gamma - 1)\gamma^3}{\gamma + 1} (\vec{\beta} \cdot \vec{x}_1) (\vec{\beta} \dot{\vec{\beta}}^T) \\ &+ \frac{(-2\gamma^3 - 2\gamma^2 + 2)\gamma^4 (\vec{\beta} \cdot \dot{\vec{\beta}})}{\gamma + 1} x^{11} (\vec{\beta} \vec{\beta}^T) + (1 - \gamma^2)\gamma^2 x^{11} (\vec{\beta} \dot{\vec{\beta}}^T) \end{aligned}$$

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第四项

$$\begin{aligned}
 &= \left[\frac{(\gamma+2)\gamma^4(\vec{\beta} \cdot \dot{\vec{\beta}})}{(\gamma+1)^2} (\vec{\beta}\vec{\beta}^T) + \frac{\gamma^2}{\gamma+1} (\vec{\beta}\dot{\vec{\beta}}^T) \right] \\
 &\left\{ \vec{x}^s + \frac{\gamma^4}{(\gamma+1)^2} \vec{\beta}\vec{\beta}^T [(\vec{\beta}\vec{\beta}^T) \cdot \vec{x}^s] - \frac{2\gamma^3}{\gamma+1} \vec{\beta}\vec{\beta}^T (\vec{\beta} \cdot \vec{x}_1) + 2\gamma^2 \vec{\beta}\vec{\beta}^T x^{11} + \frac{\gamma^2}{\gamma+1} [\vec{\beta}\vec{\beta}^T, \vec{x}^s]_+ - \gamma(\vec{\beta}\vec{x}_1^T + \vec{x}_1\vec{\beta}^T) \right\} \\
 &= \frac{(\gamma+2)\gamma^4(\vec{\beta} \cdot \dot{\vec{\beta}})}{(\gamma+1)^2} (\vec{\beta}\vec{\beta}^T) \vec{x}^s + \frac{(\gamma+2)\gamma^8(\vec{\beta} \cdot \dot{\vec{\beta}})}{(\gamma+1)^4} [(\vec{\beta}\vec{\beta}^T) \cdot \vec{x}^s] (\vec{\beta}\vec{\beta}^T)^2 - \frac{2(\gamma+2)\gamma^7(\vec{\beta} \cdot \dot{\vec{\beta}})}{(\gamma+1)^3} (\vec{\beta} \cdot \vec{x}_1) (\vec{\beta}\vec{\beta}^T)^2 \\
 &+ \frac{2(\gamma+2)\gamma^6(\vec{\beta} \cdot \dot{\vec{\beta}})}{(\gamma+1)^2} x^{11} (\vec{\beta}\vec{\beta}^T)^2 + \frac{(\gamma+2)\gamma^6(\vec{\beta} \cdot \dot{\vec{\beta}})}{(\gamma+1)^3} (\vec{\beta}\vec{\beta}^T) [\vec{\beta}\vec{\beta}^T, \vec{x}^s]_+ - \frac{(\gamma+2)\gamma^5(\vec{\beta} \cdot \dot{\vec{\beta}})}{(\gamma+1)^2} (\vec{\beta}\vec{\beta}^T) (\vec{\beta}\vec{x}_1^T + \vec{x}_1\vec{\beta}^T) \\
 &+ \frac{\gamma^2}{\gamma+1} (\vec{\beta}\dot{\vec{\beta}}^T) \vec{x}^s + \frac{\gamma^6}{(\gamma+1)^3} [(\vec{\beta}\vec{\beta}^T) \cdot \vec{x}^s] (\vec{\beta}\dot{\vec{\beta}}^T) \vec{\beta}\vec{\beta}^T - \frac{2\gamma^5}{(\gamma+1)^2} (\vec{\beta} \cdot \vec{x}_1) (\vec{\beta}\dot{\vec{\beta}}^T) \vec{\beta}\vec{\beta}^T + \frac{2\gamma^4}{\gamma+1} x^{11} (\vec{\beta}\dot{\vec{\beta}}^T) \vec{\beta}\vec{\beta}^T \\
 &+ \frac{\gamma^4}{(\gamma+1)^2} (\vec{\beta}\dot{\vec{\beta}}^T) [\vec{\beta}\vec{\beta}^T, \vec{x}^s]_+ - \frac{\gamma^3}{\gamma+1} (\vec{\beta}\dot{\vec{\beta}}^T) (\vec{\beta}\vec{x}_1^T + \vec{x}_1\vec{\beta}^T)
 \end{aligned}$$

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作部分项代换

$$(\vec{\beta}\vec{\beta}^T)^2 = \beta_i\beta_j\beta_1^2 + \beta_i\beta_j\beta_2^2 + \beta_i\beta_j\beta_3^2 = \beta^2(\vec{\beta}\vec{\beta}^T)$$

$$(\vec{\beta}\dot{\vec{\beta}}^T)\vec{\beta}\vec{\beta}^T = \beta^2(\dot{\vec{\beta}}\vec{\beta}^T) + (\vec{\beta} \cdot \dot{\vec{\beta}})(\vec{\beta}\vec{\beta}^T)$$

$$(\vec{\beta}\vec{\beta}^T)\vec{\beta}\vec{x}_1^T = \beta_i x_{1,j} \beta_1^2 + \beta_i x_{1,j} \beta_2^2 + \beta_i x_{1,j} \beta_3^2 = \beta^2(\vec{\beta}\vec{x}_1^T)$$

$$(\vec{\beta}\vec{\beta}^T)\vec{x}_1\vec{\beta}^T = \beta_i\beta_j\beta_1 x_{1,1} + \beta_i\beta_j\beta_2 x_{1,2} + \beta_i\beta_j\beta_3 x_{1,3} = (\vec{\beta} \cdot \vec{x}_1)(\vec{\beta}\vec{\beta}^T)$$

$$(\vec{\beta}\dot{\vec{\beta}}^T)\vec{\beta}\vec{x}_1^T = \beta^2(\dot{\vec{\beta}}\vec{x}_1^T) + (\vec{\beta} \cdot \dot{\vec{\beta}})(\vec{\beta}\vec{x}_1^T)$$

$$(\vec{\beta}\dot{\vec{\beta}}^T)\vec{x}_1\vec{\beta}^T = (\vec{\beta} \cdot \vec{x}_1)(\dot{\vec{\beta}}\vec{\beta}^T) + (\dot{\vec{\beta}} \cdot \vec{x}_1)(\vec{\beta}\vec{\beta}^T)$$

$$(\vec{\beta}\vec{\beta}^T)(\vec{\beta}\vec{\beta}^T)\vec{x}^s = \beta^2(\vec{\beta}\vec{\beta}^T)\vec{x}^s$$

$$(\vec{\beta}\vec{\beta}^T)\vec{x}^s(\vec{\beta}\vec{\beta}^T) = \beta_\mu\beta_\nu \sum_{ij} \beta_i\beta_j x_{ij}^s = (\vec{\beta}\vec{\beta}^T)[(\vec{\beta}\vec{\beta}^T) \cdot \vec{x}^s]$$

$$(\vec{\beta}\dot{\vec{\beta}}^T)(\vec{\beta}\vec{\beta}^T)\vec{x}^s = \beta^2(\dot{\vec{\beta}}\vec{\beta}^T)\vec{x}^s + (\vec{\beta} \cdot \dot{\vec{\beta}})(\vec{\beta}\vec{\beta}^T)\vec{x}^s$$

$$(\vec{\beta}\dot{\vec{\beta}}^T)\vec{x}^s(\vec{\beta}\vec{\beta}^T) = \dot{\beta}_\mu\beta_\nu \sum_{ij} \beta_i\beta_j x_{ij}^s + \beta_\mu\beta_\nu \sum_{ij} \dot{\beta}_i\beta_j x_{ij}^s = (\dot{\vec{\beta}}\vec{\beta}^T)[(\vec{\beta}\vec{\beta}^T) \cdot \vec{x}^s] + (\vec{\beta}\vec{\beta}^T)[(\dot{\vec{\beta}}\vec{\beta}^T) \cdot \vec{x}^s]$$

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第四项

$$\begin{aligned}
 &= \frac{\gamma^6 (\vec{\beta} \cdot \dot{\vec{\beta}})}{\gamma+1} [(\vec{\beta} \vec{\beta}^T) \cdot \vec{x}^s] (\vec{\beta} \vec{\beta}^T) + \frac{\gamma^4}{2(\gamma+1)^2} [(\dot{\vec{\beta}} \vec{\beta}^T) \cdot \vec{x}^s] (\vec{\beta} \vec{\beta}^T) + \frac{\gamma^5}{(\gamma+1)^2} [(\vec{\beta} \vec{\beta}^T) \cdot \vec{x}^s] (\dot{\vec{\beta}} \vec{\beta}^T) \\
 &\gamma^4 (\vec{\beta} \cdot \dot{\vec{\beta}}) (\vec{\beta} \vec{\beta}^T) \vec{x}^s + \frac{\gamma^2}{\gamma+1} (\dot{\vec{\beta}} \vec{\beta}^T) \vec{x}^s + \frac{\gamma^3}{\gamma+1} (\dot{\vec{\beta}} \vec{\beta}^T) \vec{x}^s \\
 &- \frac{(2\gamma+3)\gamma^6 (\vec{\beta} \cdot \dot{\vec{\beta}})}{(\gamma+1)^2} (\vec{\beta} \cdot \vec{x}_1) (\vec{\beta} \vec{\beta}^T) - \frac{(2\gamma-1)\gamma^3}{\gamma+1} (\vec{\beta} \cdot \vec{x}_1) (\dot{\vec{\beta}} \vec{\beta}^T) - \frac{\gamma^3}{\gamma+1} (\dot{\vec{\beta}} \cdot \vec{x}_1) (\vec{\beta} \vec{\beta}^T) \\
 &- \frac{(\gamma^2 + \gamma - 1)\gamma^3 (\vec{\beta} \cdot \dot{\vec{\beta}})}{\gamma+1} (\vec{\beta} \vec{x}_1^T) - \gamma(\gamma-1) (\dot{\vec{\beta}} \vec{x}_1^T) \\
 &+ \frac{2(\gamma^2 + \gamma - 1)\gamma^4 (\vec{\beta} \cdot \dot{\vec{\beta}})}{\gamma+1} x^{11} (\vec{\beta} \vec{\beta}^T) + 2(\gamma-1)\gamma^2 x^{11} (\dot{\vec{\beta}} \vec{\beta}^T)
 \end{aligned}$$

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第三加第四项

$$\begin{aligned}
 &= 2\gamma^6 (\vec{\beta} \cdot \dot{\vec{\beta}}) [(\vec{\beta} \vec{\beta}^T) \cdot \vec{x}] (\vec{\beta} \vec{\beta}^T) + \gamma^4 [(\vec{\beta} \vec{\beta}^T) \cdot \vec{x}] (\dot{\vec{\beta}} \vec{\beta}^T) + \gamma^4 [(\vec{\beta} \vec{\beta}^T) \cdot \vec{x}] (\vec{\beta} \dot{\vec{\beta}}^T) \\
 &+ \frac{\gamma^6 (\vec{\beta} \cdot \dot{\vec{\beta}})}{\gamma+1} [(\vec{\beta} \vec{\beta}^T) \cdot \vec{x}^s] (\vec{\beta} \vec{\beta}^T) + \frac{\gamma^4}{2(\gamma+1)^2} [(\dot{\vec{\beta}} \vec{\beta}^T) \cdot \vec{x}^s] (\vec{\beta} \vec{\beta}^T) + \frac{\gamma^5}{(\gamma+1)^2} [(\vec{\beta} \vec{\beta}^T) \cdot \vec{x}^s] (\dot{\vec{\beta}} \vec{\beta}^T) \\
 &+ \gamma^4 (\vec{\beta} \cdot \dot{\vec{\beta}}) (\vec{\beta} \vec{\beta}^T) \vec{x}^s + \frac{\gamma^2}{\gamma+1} (\vec{\beta} \dot{\vec{\beta}}^T) \vec{x}^s + \frac{\gamma^3}{\gamma+1} (\dot{\vec{\beta}} \vec{\beta}^T) \vec{x}^s \\
 &- \frac{(2\gamma^2 + 6\gamma + 5)}{(\gamma+1)^2} \gamma^6 (\vec{\beta} \cdot \dot{\vec{\beta}}) (\vec{\beta} \cdot \vec{x}_1) (\vec{\beta} \vec{\beta}^T) - \frac{\gamma^3}{\gamma+1} (\dot{\vec{\beta}} \cdot \vec{x}_1) (\vec{\beta} \vec{\beta}^T) - \frac{\gamma^2 + 3\gamma - 1}{\gamma+1} \gamma^3 (\vec{\beta} \cdot \vec{x}_1) (\dot{\vec{\beta}} \vec{\beta}^T) - \gamma^4 (\vec{\beta} \cdot \vec{x}_1) (\vec{\beta} \dot{\vec{\beta}}^T) \\
 &- \frac{(\gamma^2 + \gamma - 1)\gamma^3 (\vec{\beta} \cdot \dot{\vec{\beta}})}{\gamma+1} (\vec{\beta} \vec{x}_1^T) - \gamma(\gamma-1) (\dot{\vec{\beta}} \vec{x}_1^T) \\
 &+ \frac{2\gamma^3 + 4\gamma^2 + 2\gamma - 2}{\gamma+1} \gamma^4 (\vec{\beta} \cdot \dot{\vec{\beta}}) x^{11} (\vec{\beta} \vec{\beta}^T) + (\gamma^2 + 2\gamma - 2) \gamma^2 x^{11} (\dot{\vec{\beta}} \vec{\beta}^T) + \gamma^4 x^{11} (\vec{\beta} \dot{\vec{\beta}}^T)
 \end{aligned}$$

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第五加第六项

$$\begin{aligned}
 &= \left[\gamma^3 (\vec{\beta} \cdot \dot{\vec{\beta}}) \vec{\beta} + \gamma \dot{\vec{\beta}} \right] \left\{ -\frac{\gamma^3}{\gamma+1} \vec{\beta}^T [(\vec{\beta} \vec{\beta}^T) \cdot \vec{x}] - \gamma \vec{\beta}^T \vec{x} + \frac{\gamma^3}{\gamma+1} \vec{\beta}^T (\vec{\beta} \cdot \vec{x}_{lin1}) + \gamma \vec{x}_{lin1}^T + \gamma^2 \vec{\beta}^T (\vec{\beta} \cdot \vec{x}_{col1} - x^{11}) \right\} \\
 &+ \gamma^3 (\vec{\beta} \cdot \dot{\vec{\beta}}) \left[-\frac{\gamma^3}{\gamma+1} \vec{\beta} [(\vec{\beta} \vec{\beta}^T) \cdot \vec{x}] - \gamma \vec{x} \vec{\beta} + \frac{\gamma^3}{\gamma+1} \vec{\beta} (\vec{\beta} \cdot \vec{x}_{col1}) + \gamma \vec{x}_{col1} + \gamma^2 \vec{\beta} (\vec{\beta} \cdot \vec{x}_{lin1} - x^{11}) \right] \vec{\beta}^T \\
 &+ \gamma \left[-\frac{\gamma^3}{\gamma+1} \vec{\beta} [(\vec{\beta} \vec{\beta}^T) \cdot \vec{x}] - \gamma \vec{x} \vec{\beta} + \frac{\gamma^3}{\gamma+1} \vec{\beta} (\vec{\beta} \cdot \vec{x}_{col1}) + \gamma \vec{x}_{col1} + \gamma^2 \vec{\beta} (\vec{\beta} \cdot \vec{x}_{lin1} - x^{11}) \right] \dot{\vec{\beta}}^T \\
 &= -\frac{\gamma^6}{\gamma+1} (\vec{\beta} \cdot \dot{\vec{\beta}}) \vec{\beta} \vec{\beta}^T [(\vec{\beta} \vec{\beta}^T) \cdot \vec{x}] - \gamma^4 (\vec{\beta} \cdot \dot{\vec{\beta}}) \vec{\beta} \vec{\beta}^T \vec{x} + \frac{\gamma^6}{\gamma+1} (\vec{\beta} \cdot \dot{\vec{\beta}}) \vec{\beta} \vec{\beta}^T (\vec{\beta} \cdot \vec{x}_{lin1}) + \gamma^4 (\vec{\beta} \cdot \dot{\vec{\beta}}) \vec{\beta} \vec{x}_{lin1}^T + \gamma^5 (\vec{\beta} \cdot \dot{\vec{\beta}}) \vec{\beta} \vec{\beta}^T (\vec{\beta} \cdot \vec{x}_{col1} - x^{11}) \\
 &- \frac{\gamma^4}{\gamma+1} \dot{\vec{\beta}} \vec{\beta}^T [(\vec{\beta} \vec{\beta}^T) \cdot \vec{x}] - \gamma^2 (\dot{\vec{\beta}} \vec{\beta}^T) \vec{x} + \frac{\gamma^4}{\gamma+1} (\vec{\beta} \cdot \vec{x}_{lin1}) \dot{\vec{\beta}} \vec{\beta}^T + \gamma^2 \dot{\vec{\beta}} \vec{x}_{lin1}^T + \gamma^3 \dot{\vec{\beta}} \vec{\beta}^T (\vec{\beta} \cdot \vec{x}_{col1} - x^{11}) \\
 &- \frac{\gamma^6}{\gamma+1} (\vec{\beta} \cdot \dot{\vec{\beta}}) \vec{\beta} \vec{\beta}^T [(\vec{\beta} \vec{\beta}^T) \cdot \vec{x}] - \gamma^4 (\vec{\beta} \cdot \dot{\vec{\beta}}) \vec{x} (\vec{\beta} \vec{\beta}^T) + \frac{\gamma^6}{\gamma+1} (\vec{\beta} \cdot \dot{\vec{\beta}}) (\vec{\beta} \cdot \vec{x}_{col1}) \vec{\beta} \vec{\beta}^T + \gamma^4 (\vec{\beta} \cdot \dot{\vec{\beta}}) \vec{x}_{col1} \vec{\beta}^T + \gamma^5 (\vec{\beta} \cdot \dot{\vec{\beta}}) (\vec{\beta} \cdot \vec{x}_{lin1} - x^{11}) \vec{\beta} \vec{\beta}^T \\
 &- \frac{\gamma^4}{\gamma+1} [(\vec{\beta} \vec{\beta}^T) \cdot \vec{x}] \vec{\beta} \dot{\vec{\beta}}^T - \gamma^2 \vec{x} (\dot{\vec{\beta}} \vec{\beta}^T) + \frac{\gamma^4}{\gamma+1} (\vec{\beta} \cdot \vec{x}_{col1}) \vec{\beta} \dot{\vec{\beta}}^T + \gamma^2 \vec{x}_{col1} \dot{\vec{\beta}}^T + \gamma^3 (\vec{\beta} \cdot \vec{x}_{lin1} - x^{11}) \vec{\beta} \dot{\vec{\beta}}^T \\
 &= -\frac{2\gamma^6}{\gamma+1} (\vec{\beta} \cdot \dot{\vec{\beta}}) [(\vec{\beta} \vec{\beta}^T) \cdot \vec{x}] (\vec{\beta} \vec{\beta}^T) - \gamma^4 (\vec{\beta} \cdot \dot{\vec{\beta}}) (\vec{\beta} \vec{\beta}^T) \vec{x} - \gamma^4 (\vec{\beta} \cdot \dot{\vec{\beta}}) \vec{x} (\vec{\beta} \vec{\beta}^T) + \frac{\gamma^6}{\gamma+1} (\vec{\beta} \cdot \dot{\vec{\beta}}) (\vec{\beta} \cdot \vec{x}_1) \vec{\beta} \vec{\beta}^T \\
 &+ \gamma^4 (\vec{\beta} \cdot \dot{\vec{\beta}}) \vec{\beta} \vec{x}_{lin1}^T + \gamma^4 (\vec{\beta} \cdot \dot{\vec{\beta}}) \vec{x}_{col1} \vec{\beta}^T + \gamma^5 (\vec{\beta} \cdot \dot{\vec{\beta}}) (\vec{\beta} \cdot \vec{x}_1) (\vec{\beta} \vec{\beta}^T) - 2\gamma^5 (\vec{\beta} \cdot \dot{\vec{\beta}}) x^{11} (\vec{\beta} \vec{\beta}^T) \\
 &- \frac{\gamma^4}{\gamma+1} [(\vec{\beta} \vec{\beta}^T) \cdot \vec{x}] (\dot{\vec{\beta}} \vec{\beta}^T) + \frac{\gamma^4}{\gamma+1} (\vec{\beta} \cdot \vec{x}_1) (\dot{\vec{\beta}} \vec{\beta}^T) - \gamma^2 (\dot{\vec{\beta}} \vec{\beta}^T) \vec{x} - \gamma^2 \vec{x} (\dot{\vec{\beta}} \vec{\beta}^T) + \gamma^2 \dot{\vec{\beta}} \vec{x}_{lin1}^T + \gamma^2 \vec{x}_{col1} \dot{\vec{\beta}}^T \\
 &+ \frac{\gamma^3}{\gamma+1} (\vec{\beta} \cdot \vec{x}_{col1}) (\dot{\vec{\beta}} \vec{\beta}^T) + \frac{\gamma^3}{\gamma+1} (\vec{\beta} \cdot \vec{x}_{lin1}) (\dot{\vec{\beta}} \vec{\beta}^T) - \gamma^3 x^{11} (\dot{\vec{\beta}} \vec{\beta}^T)
 \end{aligned}$$

附录1 一般二阶张量Thomas动力学表达式化简

第三加第四加第五加第六项

$$\begin{aligned}
 &= \frac{2\gamma^7}{\gamma+1} (\vec{\beta} \cdot \dot{\vec{\beta}}) [(\vec{\beta}\vec{\beta}^T) \cdot \vec{x}] (\vec{\beta}\vec{\beta}^T) + \frac{\gamma^5}{\gamma+1} [(\vec{\beta}\vec{\beta}^T) \cdot \vec{x}] (\dot{\vec{\beta}}\vec{\beta}^T) \\
 &+ \frac{\gamma^6 (\vec{\beta} \cdot \dot{\vec{\beta}})}{\gamma+1} [(\vec{\beta}\vec{\beta}^T) \cdot \vec{x}^s] (\vec{\beta}\vec{\beta}^T) + \frac{\gamma^4}{2(\gamma+1)^2} [(\dot{\vec{\beta}}\vec{\beta}^T) \cdot \vec{x}^s] (\vec{\beta}\vec{\beta}^T) + \frac{\gamma^5}{(\gamma+1)^2} [(\vec{\beta}\vec{\beta}^T) \cdot \vec{x}^s] (\dot{\vec{\beta}}\vec{\beta}^T) \\
 &+ \gamma^4 (\vec{\beta} \cdot \dot{\vec{\beta}}) (\vec{\beta}\vec{\beta}^T) \vec{x}^T + \frac{\gamma^2}{\gamma+1} (\dot{\vec{\beta}}\vec{\beta}^T) \vec{x}^s + \frac{\gamma^3}{\gamma+1} (\dot{\vec{\beta}}\vec{\beta}^T) \vec{x}^s \\
 &- \frac{(2\gamma^3 + 4\gamma^2 + 2\gamma - 1)}{(\gamma+1)^2} \gamma^5 (\vec{\beta} \cdot \dot{\vec{\beta}}) (\vec{\beta} \cdot \vec{x}_1) (\vec{\beta}\vec{\beta}^T) - \frac{\gamma^3}{\gamma+1} (\dot{\vec{\beta}} \cdot \vec{x}_1) (\vec{\beta}\vec{\beta}^T) - \frac{\gamma^2 + 2\gamma - 1}{\gamma+1} \gamma^3 (\vec{\beta} \cdot \vec{x}_1) (\dot{\vec{\beta}}\vec{\beta}^T) - \frac{\gamma^5}{\gamma+1} (\vec{\beta} \cdot \vec{x}_1) (\dot{\vec{\beta}}\vec{\beta}^T) \\
 &- \frac{(\gamma^2 + \gamma - 1)\gamma^3 (\vec{\beta} \cdot \dot{\vec{\beta}})}{\gamma+1} (\vec{\beta}\vec{x}_1^T) - \gamma(\gamma-1) (\dot{\vec{\beta}}\vec{x}_1^T) \\
 &+ \frac{2\gamma^3 + 2\gamma^2 - 2}{\gamma+1} \gamma^4 (\vec{\beta} \cdot \dot{\vec{\beta}}) x^{11} (\vec{\beta}\vec{\beta}^T) + (\gamma^2 + \gamma - 2) \gamma^2 x^{11} (\dot{\vec{\beta}}\vec{\beta}^T) + \gamma^3 (\gamma-1) x^{11} (\dot{\vec{\beta}}\vec{\beta}^T) \\
 &- \gamma^4 (\vec{\beta} \cdot \dot{\vec{\beta}}) \vec{x} (\vec{\beta}\vec{\beta}^T) + \gamma^4 (\vec{\beta} \cdot \dot{\vec{\beta}}) \vec{\beta} \vec{x}_{lin1}^T + \gamma^4 (\vec{\beta} \cdot \dot{\vec{\beta}}) \vec{x}_{col1} \vec{\beta}^T \\
 &- \gamma^2 (\dot{\vec{\beta}}\vec{\beta}^T) \vec{x} - \gamma^2 \vec{x} (\dot{\vec{\beta}}\vec{\beta}^T) + \gamma^2 \dot{\vec{\beta}} \vec{x}_{lin1}^T + \gamma^2 \vec{x}_{col1} \dot{\vec{\beta}}^T + \frac{\gamma^3}{\gamma+1} (\vec{\beta} \cdot \vec{x}_{col1}) (\dot{\vec{\beta}}\vec{\beta}^T) + \frac{\gamma^3}{\gamma+1} (\vec{\beta} \cdot \vec{x}_{lin1}) (\dot{\vec{\beta}}\vec{\beta}^T)
 \end{aligned}$$

附录1 一般二阶张量Thomas动力学表达式化简

总和

$$\begin{aligned}
 &= \frac{\gamma^5}{(\gamma+1)^2} [(\vec{\beta}\vec{\beta}^T) \cdot \vec{x}] \dot{\vec{\beta}}\vec{\beta}^T - \frac{\gamma^5}{(\gamma+1)^2} [(\vec{\beta}\vec{\beta}^T) \cdot \vec{x}] \vec{\beta}\dot{\vec{\beta}}^T + \gamma^4 (\vec{\beta} \cdot \dot{\vec{\beta}}) (\vec{\beta}\vec{\beta}^T) \vec{x}^T - \gamma^4 (\vec{\beta} \cdot \dot{\vec{\beta}}) \vec{x} (\vec{\beta}\vec{\beta}^T) \\
 &+ \frac{\gamma^3}{\gamma+1} (\dot{\vec{\beta}}\vec{\beta}^T) \vec{x}^T - \frac{\gamma^3}{\gamma+1} \vec{x} (\vec{\beta}\dot{\vec{\beta}}^T) + \frac{\gamma^2}{\gamma+1} (\vec{\beta}\dot{\vec{\beta}}^T) \vec{x}^T - \frac{\gamma^2}{\gamma+1} (\vec{\beta}\dot{\vec{\beta}}^T) \vec{x} + \frac{\gamma^2}{\gamma+1} (\vec{\beta}\dot{\vec{\beta}}^T) \vec{x} - \frac{\gamma^2}{\gamma+1} \vec{x} (\vec{\beta}\dot{\vec{\beta}}^T) \\
 &+ \frac{\gamma^4}{\gamma+1} (\vec{\beta} \cdot \vec{x}_1) \vec{\beta}\dot{\vec{\beta}}^T - \frac{\gamma^4}{\gamma+1} (\vec{\beta} \cdot \vec{x}_1) \dot{\vec{\beta}}\vec{\beta}^T + \frac{\gamma^3}{\gamma+1} (\vec{\beta} \cdot \vec{x}_{col1}) (\vec{\beta}\dot{\vec{\beta}}^T) - \frac{\gamma^3}{\gamma+1} (\vec{\beta} \cdot \vec{x}_{col1}) (\dot{\vec{\beta}}\vec{\beta}^T) \\
 &+ \gamma^2 (\gamma-1) x^{11} (\dot{\vec{\beta}}\vec{\beta}^T) - \gamma^2 (\gamma-1) x^{11} (\vec{\beta}\dot{\vec{\beta}}^T) + \gamma^2 \vec{x}_{col1} \dot{\vec{\beta}}^T - \gamma^2 (\dot{\vec{\beta}}\vec{x}_{col1}^T) + \gamma^4 (\vec{\beta} \cdot \dot{\vec{\beta}}) \vec{x}_{col1} \vec{\beta}^T - \gamma^4 (\vec{\beta} \cdot \dot{\vec{\beta}}) (\vec{\beta}\vec{x}_{col1}^T) \\
 &+ \frac{\gamma^3 (\vec{\beta} \cdot \dot{\vec{\beta}})}{\gamma+1} (\vec{\beta}\vec{x}_1^T) + \gamma (\dot{\vec{\beta}}\vec{x}_1^T)
 \end{aligned}$$

上式考虑到

$$(\vec{\beta}\vec{\beta}^T) \cdot \vec{x} = \sum_{ij} \beta_i \beta_j x_{ij} = \sum_{ij} \beta_j \beta_i x_{ij} = (\vec{\beta}\vec{\beta}^T) \cdot \vec{x}^T$$

$$(\dot{\vec{\beta}}\vec{\beta}^T) \cdot \vec{x} = \sum_{ij} \dot{\beta}_i \beta_j x_{ij} + \sum_{ij} \beta_i \dot{\beta}_j x_{ij} = \sum_{ij} \beta_j \dot{\beta}_i x_{ij} + \sum_{ij} \dot{\beta}_j \beta_i x_{ij} = (\dot{\vec{\beta}}\vec{\beta}^T) \cdot \vec{x}^T$$

附录2 拟向变换自旋赝偶极矢量Thomas-Hamiltonian推导

基于(13'):

$$\frac{\partial U}{\partial \theta} \begin{pmatrix} -\sin \varphi \\ \cos \varphi \\ 0 \end{pmatrix} - \frac{\partial U}{\partial \varphi} \begin{pmatrix} \cot \theta \cos \varphi \\ \cot \theta \sin \varphi \\ -1 \end{pmatrix} = \begin{pmatrix} \frac{\gamma^2 S_x \beta_1 S_y \dot{\beta}_2}{1 + \gamma^2 \beta_1^2 S_x^2} & \frac{\dot{\beta}_2}{\beta_1 \gamma} (1 - \gamma) & 0 \\ \frac{\dot{\beta}_2}{\beta_1} (1 - \gamma) & \frac{\gamma^2 S_x \beta_1 S_y \dot{\beta}_2}{1 + \gamma^2 \beta_1^2 S_x^2} & 0 \\ 0 & 0 & \frac{\gamma^2 S_x \beta_1 S_y \dot{\beta}_2}{1 + \gamma^2 \beta_1^2 S_x^2} \end{pmatrix} \begin{pmatrix} S_x \\ S_y \\ S_z \end{pmatrix}$$

$$\therefore \begin{cases} -\frac{\partial U}{\partial \theta} \sin \varphi - \frac{\partial U}{\partial \varphi} \cot \theta \cos \varphi = \frac{\gamma^2 S_x \beta_1 S_y \dot{\beta}_2}{1 + \gamma^2 \beta_1^2 S_x^2} S_x + \frac{\dot{\beta}_2}{\beta_1 \gamma} (1 - \gamma) S_y \\ \frac{\partial U}{\partial \theta} \cos \varphi - \frac{\partial U}{\partial \varphi} \cot \theta \sin \varphi = \frac{\dot{\beta}_2}{\beta_1} (1 - \gamma) S_x + \frac{\gamma^2 S_x \beta_1 S_y \dot{\beta}_2}{1 + \gamma^2 \beta_1^2 S_x^2} S_y \\ \frac{\partial U}{\partial \varphi} = \frac{\gamma^2 S_x \beta_1 S_y \dot{\beta}_2}{1 + \gamma^2 \beta_1^2 S_x^2} S_z \end{cases}$$

$$\begin{cases} -\frac{\partial U}{\partial \theta} \sin \varphi - \frac{\gamma^2}{\gamma^2 \beta_1^2 S_x^2 + 1} \beta_1 \dot{\beta}_2 S_x S_y S_z \cot \theta \cos \varphi = \frac{\gamma^2}{\gamma^2 \beta_1^2 S_x^2 + 1} \beta_1 \dot{\beta}_2 S_y S_x^2 - \frac{\gamma}{\gamma + 1} \beta_1 \dot{\beta}_2 S_y \\ \frac{\partial U}{\partial \theta} \cos \varphi - \frac{\gamma^2}{\gamma^2 \beta_1^2 S_x^2 + 1} \beta_1 \dot{\beta}_2 S_x S_y S_z \cot \theta \sin \varphi = -\frac{\gamma^2}{\gamma + 1} \beta_1 \dot{\beta}_2 S_x + \frac{\gamma^2}{\gamma^2 \beta_1^2 S_x^2 + 1} \beta_1 \dot{\beta}_2 S_x S_y^2 \end{cases}$$

$$\begin{cases} -\frac{\partial U}{\partial \theta} \sin \varphi = \frac{\gamma^2}{\gamma^2 \beta_1^2 S_x^2 + 1} \beta_1 \dot{\beta}_2 \left[S_x S_y (S_z \cot \theta \cos \varphi + S_x) - \frac{1}{\gamma(\gamma + 1)} (\gamma^2 \beta_1^2 S_x^2 S_y + S_y) \right] \\ \frac{\partial U}{\partial \theta} \cos \varphi = \frac{\gamma^2}{\gamma^2 \beta_1^2 S_x^2 + 1} \beta_1 \dot{\beta}_2 \left[S_x S_y (S_z \cot \theta \sin \varphi + S_y) - \frac{1}{\gamma + 1} (\gamma^2 \beta_1^2 S_x^2 S_x + S_x) \right] \end{cases}$$

$$\tan \varphi = - \frac{\left[(S_z \cot \theta \cos \varphi + S_x) - \frac{1}{\gamma(\gamma + 1) S_x} (\gamma^2 \beta_1^2 S_x^2 + 1) \right]}{\left[(S_z \cot \theta \sin \varphi + S_y) - \frac{1}{(\gamma + 1) S_y} (\gamma^2 \beta_1^2 S_x^2 + 1) \right]}$$

附录2 拟向变换自旋赭偶极矢量Thomas-Hamiltonian推导

$$\begin{aligned} \therefore \tan \varphi &= -\frac{\left[S_z \cot \theta \cos \varphi + S_x - \frac{1}{\gamma(\gamma+1)S_x}(\gamma^2 \beta_1^2 S_x^2 + 1)\right]}{\left[S_z \cot \theta \sin \varphi + S_y - \frac{1}{(\gamma+1)S_y}(\gamma^2 \beta_1^2 S_x^2 + 1)\right]} \\ \therefore S_z \cot \theta + S_y \sin \varphi - \frac{1}{(\gamma+1)S_y}(\gamma^2 \beta_1^2 S_x^2 + 1) \sin \varphi &= -S_x \cos \varphi + \frac{1}{\gamma(\gamma+1)S_x}(\gamma^2 \beta_1^2 S_x^2 + 1) \cos \varphi \\ \therefore S_z \cot \theta + S_y \sin \varphi + S_x \cos \varphi &= \left((\gamma-1)S_x^2 + \frac{1}{\gamma+1}\right) \left(\frac{\sin \varphi}{S_y} + \frac{\cos \varphi}{\gamma S_x}\right) \\ \therefore S_z \cot \theta + \frac{1}{\gamma} S_x \cos \varphi + S_y \sin \varphi &= \left[\frac{1}{\gamma+1} + (\gamma-1)S_x^2\right] \frac{\sin \varphi}{S_y} + \frac{1}{\gamma+1} \frac{\cos \varphi}{\gamma S_x} \\ \therefore (\gamma+1)S_z S_y \cot \theta + \left[(\gamma+1)S_x S_y - \frac{S_y}{S_x}\right] \frac{\cos \varphi}{\gamma} &= [\gamma^2 S_x^2 - \gamma S_y^2 + S_z^2] \sin \varphi \\ \therefore (\gamma+1)S_z S_y \bar{S}_z &= \left(\frac{S_y}{\gamma S_x} - \frac{\gamma+1}{\gamma} S_x S_y\right) \bar{S}_x + ((\gamma+1)((\gamma-1)S_x^2 - S_y^2) + 1) \bar{S}_y \quad \cdots (14) \end{aligned}$$

满足上式的 $\bar{S}_z$ 、 $\bar{S}_x$ 、 $\bar{S}_y$ 有无穷组，但同时也要满足 $\bar{S}_x^2 + \bar{S}_y^2 + \bar{S}_z^2$ 为定值。

设 $\bar{S}_z \neq 0, \bar{S}_x \neq 0, \bar{S}_y \neq 0$;

$$\text{则可令 } \bar{S}_z = \frac{2\zeta}{(\gamma+1)S_z S_y}, \quad \bar{S}_x = \frac{\zeta\gamma}{\frac{S_y}{S_x} - (\gamma+1)S_x S_y}, \quad \bar{S}_y = \frac{\zeta}{\gamma^2 S_x^2 - \gamma S_y^2 + S_z^2}$$

$$\text{为保障径向梯度为零, 令 } \frac{4\zeta^2}{(\gamma+1)^2 S_z^2 S_y^2} + \frac{\zeta^2 \gamma^2}{\left(\frac{S_y}{S_x} - (\gamma+1)S_x S_y\right)^2} + \frac{\zeta^2}{(\gamma^2 S_x^2 - \gamma S_y^2 + S_z^2)^2} = 1$$

$$\therefore \zeta^2 = \frac{(\gamma+1)^2 S_z^2 S_y^2 \left(\frac{S_y}{S_x} - (\gamma+1)S_x S_y\right)^2 (\gamma^2 S_x^2 - \gamma S_y^2 + S_z^2)^2}{4\left(\frac{S_y}{S_x} - (\gamma+1)S_x S_y\right)^2 (\gamma^2 S_x^2 - \gamma S_y^2 + S_z^2)^2 + \gamma^2 (\gamma+1)^2 S_z^2 S_y^2 (\gamma^2 S_x^2 - \gamma S_y^2 + S_z^2)^2 + (\gamma+1)^2 S_z^2 S_y^2 \left(\frac{S_y}{S_x} - (\gamma+1)S_x S_y\right)^2}$$

这个方程太复杂，我们对(14)式进行一下近似，认为蓝色项里 $\gamma-1 \rightarrow 0$

附录2 拟向变换自旋赝偶极矢量Thomas-Hamiltonian推导

$$(\gamma+1)S_z S_y \bar{S}_z = \left(\frac{S_y}{\gamma S_x} - \frac{\gamma+1}{\gamma} S_x S_y \right) \bar{S}_x + (1 - (\gamma+1)S_y^2) \bar{S}_y \quad \dots(14')$$

显然可设 $\bar{S}_z = \frac{S_z}{S_y} \zeta$; $\frac{\bar{S}_x}{\gamma} = \frac{S_x}{S_y} \zeta$; $\bar{S}_y = \zeta$; 则(14')式为:

$$(\gamma+1)S_z^2 \zeta = (1 - (\gamma+1)S_x^2) \zeta + (1 - (\gamma+1)S_y^2) \zeta$$

$\therefore (\gamma+1)(S_z^2 + S_x^2 + S_y^2) = \gamma+1 = 2$, 此式在 $\vec{\beta} \ll 1$ 时近似成立。

为保障径向梯度为零, 令 $\frac{S_z^2}{S_y^2} \zeta^2 + \gamma^2 \frac{S_x^2}{S_y^2} \zeta^2 + \zeta^2 = 1$

$$\text{解得: } \bar{S}_z = \frac{S_z}{\sqrt{1 + \gamma^2 \beta_1^2 S_x^2}}; \bar{S}_x = \frac{\gamma S_x}{\sqrt{1 + \gamma^2 \beta_1^2 S_x^2}}; \bar{S}_y = \frac{S_y}{\sqrt{1 + \gamma^2 \beta_1^2 S_x^2}};$$

$$\text{且 } \frac{\bar{S}_x}{\gamma(1 - \beta_1^2 \bar{S}_x^2)} = S_x$$

$$\therefore \frac{\partial U}{\partial \varphi} = \frac{\gamma^2 S_x \beta_1 S_y \dot{\beta}_2}{1 + \gamma^2 \beta_1^2 S_x^2} S_z = \bar{S}_z \bar{S}_y \bar{S}_x \beta_1 \dot{\beta}_2 \frac{\gamma}{(1 - \beta_1^2 \bar{S}_x^2)} \rightarrow \gamma \beta_1 \dot{\beta}_2 \bar{S}_z \bar{S}_y \bar{S}_x$$

$$\therefore \frac{\partial U}{\partial \varphi} = \gamma \beta_1 \dot{\beta}_2 \sin^2 \theta \cos \theta \sin \varphi \cos \varphi$$

$$\therefore U = \frac{1}{2} \gamma \beta_1 \dot{\beta}_2 \sin^2 \theta \cos \theta \sin^2 \varphi$$

$$\begin{aligned} \frac{\partial U}{\partial \theta} \cos \varphi &= \frac{\gamma^2}{\gamma^2 \beta_1^2 S_x^2 + 1} \beta_1 \dot{\beta}_2 \left[S_x S_y (S_z \cot \theta \sin \varphi + S_y) - \frac{1}{\gamma+1} (\gamma^2 \beta_1^2 S_x^2 S_x + S_x) \right] \\ &= \frac{\gamma^2}{\gamma^2 \beta_1^2 S_x^2 + 1} \beta_1 \dot{\beta}_2 S_x S_y S_z \cot \theta \sin \varphi + \frac{\gamma^2}{\gamma^2 \beta_1^2 S_x^2 + 1} \beta_1 \dot{\beta}_2 S_x S_y S_y - \frac{\gamma^2}{\gamma^2 \beta_1^2 S_x^2 + 1} \beta_1 \dot{\beta}_2 \frac{\gamma^2}{\gamma+1} \beta_1^2 S_x^2 S_x - \frac{\gamma^2}{\gamma^2 \beta_1^2 S_x^2 + 1} \beta_1 \dot{\beta}_2 \frac{1}{\gamma+1} S_x \\ &= \beta_1 \dot{\beta}_2 \bar{S}_z \bar{S}_y \bar{S}_x \cot \theta \sin \varphi \frac{\gamma}{(1 - \beta_1^2 \bar{S}_x^2)} + \bar{S}_y \bar{S}_y \bar{S}_x \beta_1 \dot{\beta}_2 \frac{\gamma}{(1 - \beta_1^2 \bar{S}_x^2)} - \beta_1 \dot{\beta}_2 \bar{S}_x \bar{S}_x \bar{S}_x \frac{(\gamma-1)}{\gamma(1 - \beta_1^2 \bar{S}_x^2)} - \beta_1 \dot{\beta}_2 \frac{\gamma}{\gamma+1} (1 - \beta_1^2 \bar{S}_x^2) \bar{S}_x \end{aligned}$$

附录2 拟向变换自旋赭偶极矢量Thomas-Hamiltonian推导

$$\begin{aligned}
 &= \beta_1 \dot{\beta}_2 \bar{S}_z^2 \bar{S}_y^2 \bar{S}_x \frac{1}{\bar{S}_x^2 + \bar{S}_y^2} \frac{\gamma}{(1 - \beta_1^2 \bar{S}_x^2)} + \bar{S}_y \bar{S}_y \bar{S}_x \beta_1 \dot{\beta}_2 \frac{\gamma}{(1 - \beta_1^2 \bar{S}_x^2)} - \beta_1 \dot{\beta}_2 \bar{S}_x^3 \frac{(\gamma - 1)}{\gamma(1 - \beta_1^2 \bar{S}_x^2)} - \beta_1 \dot{\beta}_2 \frac{\gamma}{\gamma + 1} (1 - \beta_1^2 \bar{S}_x^2) \bar{S}_x \\
 &\rightarrow \gamma \beta_1 \dot{\beta}_2 \bar{S}_z^2 \bar{S}_y^2 \bar{S}_x \frac{1}{\bar{S}_x^2 + \bar{S}_y^2} + \gamma \beta_1 \dot{\beta}_2 \bar{S}_y^2 \bar{S}_x - \beta_1 \dot{\beta}_2 \bar{S}_x^3 \frac{\gamma - 1}{\gamma} - \beta_1 \dot{\beta}_2 \frac{\gamma}{\gamma + 1} \bar{S}_x \\
 &\xrightarrow{\text{忽略蓝色项}} \gamma \beta_1 \dot{\beta}_2 \bar{S}_y^2 \bar{S}_x \frac{1}{\bar{S}_x^2 + \bar{S}_y^2} - \beta_1 \dot{\beta}_2 \frac{\gamma}{\gamma + 1} \bar{S}_x \\
 &= \gamma \beta_1 \dot{\beta}_2 \sin \theta \cos \varphi \sin^2 \varphi - \beta_1 \dot{\beta}_2 \frac{\gamma}{\gamma + 1} \sin \theta \cos \varphi
 \end{aligned}$$

$$\therefore \frac{\partial U}{\partial \theta} = \gamma \beta_1 \dot{\beta}_2 \sin \theta \sin^2 \varphi - \beta_1 \dot{\beta}_2 \frac{\gamma}{\gamma + 1} \sin \theta$$

$$\therefore U = \gamma \beta_1 \dot{\beta}_2 \left(\frac{1}{\gamma + 1} - \sin^2 \varphi \right) \cos \theta, \text{ 与之前 } U = \frac{1}{2} \gamma \beta_1 \dot{\beta}_2 \sin^2 \theta \cos \theta \sin^2 \varphi \text{ 不一致, 原因是引入了一些近似。}$$

下面看看能不能少用一些近似:

$$\frac{\partial U}{\partial \varphi} = \frac{\gamma^2 S_x \beta_1 S_y \dot{\beta}_2}{1 + \gamma^2 \beta_1^2 S_x^2} S_z = \bar{S}_z \bar{S}_y \bar{S}_x \beta_1 \dot{\beta}_2 \frac{\gamma}{(1 - \beta_1^2 \bar{S}_x^2)} = \gamma \beta_1 \dot{\beta}_2 \sin^2 \theta \cos \theta \frac{\sin \varphi \cos \varphi}{1 - \beta_1^2 \sin^2 \theta \cos^2 \varphi}$$

$$\therefore U = \gamma \dot{\beta}_2 \frac{\cos \theta}{2 \beta_1} \ln(1 - \beta_1^2 \sin^2 \theta \cos^2 \varphi) \xrightarrow{\beta_1 = 0 \text{ 一阶Taylor展开}} -\frac{1}{2} \gamma \beta_1 \dot{\beta}_2 \sin^2 \theta \cos^2 \varphi \cos \theta$$

$$\begin{aligned}
 \frac{\partial U}{\partial \theta} \cos \varphi &= \beta_1 \dot{\beta}_2 \bar{S}_z^2 \bar{S}_y^2 \bar{S}_x \frac{1}{\bar{S}_x^2 + \bar{S}_y^2} \frac{\gamma}{(1 - \beta_1^2 \bar{S}_x^2)} + \bar{S}_y \bar{S}_y \bar{S}_x \beta_1 \dot{\beta}_2 \frac{\gamma}{(1 - \beta_1^2 \bar{S}_x^2)} - \beta_1 \dot{\beta}_2 \bar{S}_x^3 \frac{(\gamma - 1)}{\gamma(1 - \beta_1^2 \bar{S}_x^2)} - \beta_1 \dot{\beta}_2 \frac{\gamma}{\gamma + 1} (1 - \beta_1^2 \bar{S}_x^2) \bar{S}_x \\
 &= \frac{1}{1 - \beta_1^2 \sin^2 \theta \cos^2 \varphi} \left[\gamma \beta_1 \dot{\beta}_2 \sin \theta \cos \varphi \sin^2 \varphi - \frac{\gamma - 1}{\gamma} \beta_1 \dot{\beta}_2 \sin^3 \theta \cos^3 \varphi \right] - \beta_1 \dot{\beta}_2 \frac{\gamma}{\gamma + 1} \sin \theta \cos \varphi + \beta_1 \dot{\beta}_2 \frac{\gamma - 1}{\gamma} \sin^3 \theta \cos^3 \varphi \\
 &= \frac{\gamma \beta_1 \dot{\beta}_2 \sin \theta \cos \varphi \sin^2 \varphi}{1 - \beta_1^2 \sin^2 \theta \cos^2 \varphi} - \beta_1 \dot{\beta}_2 \frac{\gamma}{\gamma + 1} \sin \theta \cos \varphi - \frac{\beta_1^2 \sin^2 \theta \cos^2 \varphi}{1 - \beta_1^2 \sin^2 \theta \cos^2 \varphi} \frac{\gamma - 1}{\gamma} \beta_1 \dot{\beta}_2 \sin^3 \theta \cos^3 \varphi \\
 &\rightarrow \frac{\gamma \beta_1 \dot{\beta}_2 \sin \theta \cos \varphi \sin^2 \varphi}{1 - \beta_1^2 \sin^2 \theta \cos^2 \varphi} - \beta_1 \dot{\beta}_2 \frac{\gamma}{\gamma + 1} \sin \theta \cos \varphi
 \end{aligned}$$

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$$\begin{aligned}
 \therefore \frac{\partial U}{\partial \theta} &= \frac{\gamma \beta_1 \dot{\beta}_2 \sin \theta \sin^2 \varphi}{1 - \beta_1^2 \sin^2 \theta \cos^2 \varphi} - \frac{\gamma - 1}{\gamma} \frac{\beta_1 \dot{\beta}_2 \beta_1^2 \sin^5 \theta \cos^3 \varphi}{1 - \beta_1^2 \sin^2 \theta \cos^2 \varphi} - \beta_1 \dot{\beta}_2 \frac{\gamma}{\gamma + 1} \sin \theta \\
 &\rightarrow \frac{\gamma \beta_1 \dot{\beta}_2 \sin \theta \sin^2 \varphi}{1 - \beta_1^2 \sin^2 \theta \cos^2 \varphi} - \beta_1 \dot{\beta}_2 \frac{\gamma}{\gamma + 1} \sin \theta \\
 \therefore U &= \int d\theta \frac{\gamma \beta_1 \dot{\beta}_2 \sin^2 \varphi \sin \theta - \frac{\gamma - 1}{\gamma} \beta_1 \dot{\beta}_2 \beta_1^2 \cos^3 \varphi \sin^5 \theta}{1 - \beta_1^2 \cos^2 \varphi \sin^2 \theta} + \beta_1 \dot{\beta}_2 \frac{\gamma}{\gamma + 1} \cos \theta \\
 &\rightarrow - \frac{\gamma \beta_1 \dot{\beta}_2 \sin^2 \varphi \left(\arctan \left(\frac{\beta_1 \cos \varphi - \tan \frac{\theta}{2}}{\sqrt{1 - \beta_1^2 \cos^2 \varphi}} \right) + \arctan \left(\frac{\beta_1 \cos \varphi + \tan \frac{\theta}{2}}{\sqrt{1 - \beta_1^2 \cos^2 \varphi}} \right) \right)}{\beta_1 \cos \varphi \sqrt{1 - \beta_1^2 \cos^2 \varphi}} + \beta_1 \dot{\beta}_2 \frac{\gamma}{\gamma + 1} \cos \theta \\
 &\xrightarrow{\text{零点一阶Taylor展开}} - \frac{\gamma \beta_1 \dot{\beta}_2 \sin^2 \varphi \left(-\frac{\theta}{2} + \frac{\beta_1 \cos \varphi}{1 + \tan^2 \frac{\theta}{2}} + \frac{\theta}{2} + \frac{\beta_1 \cos \varphi}{1 + \tan^2 \frac{\theta}{2}} \right)}{\beta_1 \cos \varphi \sqrt{1 - \beta_1^2 \cos^2 \varphi}} + \beta_1 \dot{\beta}_2 \frac{\gamma}{\gamma + 1} \cos \theta \\
 &= \frac{\gamma \beta_1 \dot{\beta}_2 \sin^2 \varphi \cos \theta}{\sqrt{1 - \beta_1^2 \cos^2 \varphi}} - \frac{\gamma \beta_1 \dot{\beta}_2 \sin^2 \varphi}{\sqrt{1 - \beta_1^2 \cos^2 \varphi}} + \beta_1 \dot{\beta}_2 \frac{\gamma}{\gamma + 1} \cos \theta
 \end{aligned}$$

看来除非不做任何近似，否则无法统一 $\frac{\partial U}{\partial \theta}$ 和 $\frac{\partial U}{\partial \varphi}$ 推导的结果。我们选择保留蓝色项的结果，即：

$$\begin{aligned}
 U &= -\frac{\gamma}{2} \beta_1 \dot{\beta}_2 \sin^2 \theta \cos^2 \varphi \cos \theta \\
 &= -\frac{\gamma}{2} \vec{S}_\gamma \cdot (\vec{\beta} \times \dot{\vec{\beta}}) \vec{S}_x^2 \\
 &= \frac{\gamma}{2\beta^2} \vec{S}_\gamma \cdot (\dot{\vec{\beta}} \times \vec{\beta}) \frac{(\vec{\beta} \cdot \vec{S})^2}{1 - \beta^2 + (\vec{\beta} \cdot \vec{S})^2}
 \end{aligned}$$

其中 $S_{\gamma,i} = \frac{1}{\sqrt{1 + \gamma^2 (\vec{\beta} \cdot \vec{S})^2}} \frac{1}{\sqrt{1 - \beta_i^2}} S_i$, 前提条件 $|\vec{S}| = 1$ 。

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若 $|\vec{S}| = s \neq 1$

则(13')式变为:

$$\frac{\partial U}{\partial \theta} \begin{pmatrix} -\sin \varphi \\ \cos \varphi \\ 0 \end{pmatrix} - \frac{\partial U}{\partial \varphi} \begin{pmatrix} \cot \theta \cos \varphi \\ \cot \theta \sin \varphi \\ -1 \end{pmatrix} = \begin{pmatrix} \frac{\gamma^2 S_x \beta_1 S_y \dot{\beta}_2}{s^2 + \gamma^2 \beta_1^2 S_x^2} & \frac{\dot{\beta}_2}{\beta_1 \gamma} (1 - \gamma) & 0 \\ \frac{\dot{\beta}_2}{\beta_1} (1 - \gamma) & \frac{\gamma^2 S_x \beta_1 S_y \dot{\beta}_2}{s^2 + \gamma^2 \beta_1^2 S_x^2} & 0 \\ 0 & 0 & \frac{\gamma^2 S_x \beta_1 S_y \dot{\beta}_2}{s^2 + \gamma^2 \beta_1^2 S_x^2} \end{pmatrix} \begin{pmatrix} S_x \\ S_y \\ S_z \end{pmatrix}$$

据此可推旋转空间的坐标变为:

$$\bar{S}_z = \frac{S_z}{\sqrt{s^2 + \gamma^2 \beta_1^2 S_x^2}}; \bar{S}_x = \frac{\gamma S_x}{\sqrt{s^2 + \gamma^2 \beta_1^2 S_x^2}}; \bar{S}_y = \frac{S_y}{\sqrt{s^2 + \gamma^2 \beta_1^2 S_x^2}};$$

$$\text{且 } \frac{s \bar{S}_x}{\gamma (1 - \beta_1^2 \bar{S}_x^2)} = S_x$$

$$\therefore \frac{\partial U}{\partial \varphi} = \frac{\gamma^2 S_x \beta_1 S_y \dot{\beta}_2}{s^2 + \gamma^2 \beta_1^2 S_x^2} S_z = \bar{S}_z \bar{S}_y \bar{S}_x \beta_1 \dot{\beta}_2 \frac{s \gamma}{(1 - \beta_1^2 \bar{S}_x^2)} \rightarrow s \gamma \beta_1 \dot{\beta}_2 \bar{S}_z \bar{S}_y \bar{S}_x$$

$$\therefore U = s \gamma \beta_2 \frac{\cos \theta}{2 \beta_1} \ln(1 - \beta_1^2 \sin^2 \theta \cos^2 \varphi)$$

$$\xrightarrow{\beta_1=0 \text{—阶 Taylor 展开}} -\frac{s}{2} \gamma \beta_1 \dot{\beta}_2 \sin^2 \theta \cos^2 \varphi \cos \theta$$

$$= \frac{s \gamma}{2 \beta^2} \vec{S}_\gamma \cdot (\dot{\vec{\beta}} \times \vec{\beta}) \frac{(\vec{\beta} \cdot \vec{S})^2}{s^2 (1 - \beta^2) + (\vec{\beta} \cdot \vec{S})^2}$$

$$\text{其中 } S_{\gamma,i} = \frac{1}{\sqrt{s^2 + \gamma^2 (\vec{\beta} \cdot \vec{S})^2}} \frac{1}{\sqrt{1 - \beta_i^2}} S_i$$